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**UEHARA et al.**(10) **Pub. No.: US 2025/0278021 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **RESIST COMPOSITION, RESIST PATTERN  
FORMATION METHOD, AND COMPOUND****7/0382** (2013.01); **G03F 7/0397** (2013.01);  
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**ABSTRACT**(72) Inventors: **Takuya UEHARA**, Kawasaki-shi (JP);  
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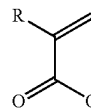
A resist composition including a resin component having a constitutional unit derived from a compound represented by General Formula (a0-1). In the formulas, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms; R<sub>pg</sub> represents an acid dissociable group represented by General Formula (a0-pg); A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton; Ra<sup>01</sup> represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms; m1 represents an integer of 0 to 20; m2 represents an integer of 1 to 20; and \* represents a bonding site

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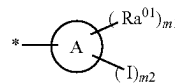
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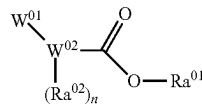
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(a0-1)



(a0-pg)



(a0-m0)

# RESIST COMPOSITION, RESIST PATTERN FORMATION METHOD, AND COMPOUND

## TECHNICAL FIELD

[0001] The present invention relates to a resist composition, a resist pattern formation method, and a compound.

[0002] Priority is claimed on Japanese Patent Application No. 2022-080807, filed May 17, 2022, the content of which is incorporated herein by reference.

## BACKGROUND ART

[0003] In recent years, in the manufacture of semiconductor elements and liquid crystal display elements, advances in lithography technologies have led to rapid progress in the field of pattern miniaturization. These pattern miniaturization techniques typically involve shortening the wavelength (increasing the energy) of the exposure light source.

[0004] Resist materials are required to have lithography characteristics such as sensitivity to these exposure light sources and resolution that enables reproduction of patterns with minute dimensions.

[0005] As a resist material that satisfies these requirements, a chemically amplified resist composition containing a base material component whose solubility in a developing solution is changed by an action of an acid and an acid generator component that generates an acid upon light exposure has been used.

[0006] For example, Patent Document 1 discloses a resist composition containing a sulfonium-based acid generator containing an electron-withdrawing group in a cation moiety. Patent Document 1 discloses that this resist composition is capable of achieving high sensitivity.

## CITATION LIST

Patent Document

[0007] Patent Document 1: Japanese Unexamined Patent Application, First Publication No. 2017-227734

## SUMMARY OF INVENTION

### Technical Problem

[0008] With further advances in lithography technologies, rapid progress in the field of pattern fining is being achieved together with the expansion of application fields. In association with this, in a case of manufacturing a semiconductor element or the like, there is a demand for a technology that enables formation of a fine pattern in a satisfactory shape. For example, the purpose of lithography using EUV or EB is to form a fine pattern with a size of several tens of nanometers.

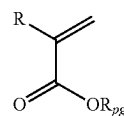
[0009] The present invention has been made in consideration of the above-described circumstances, and an object thereof is to provide a resist composition which can achieve high sensitivity during formation of a resist pattern, a resist pattern formation method using the resist composition, and a compound useful as a raw material of a base material component used in the resist composition.

### Solution to Problem

[0010] In order to solve the above-described problems, the present invention has adopted the following configurations.

[0011] That is, according to a first aspect of the present invention, there is provided a resist composition which generates an acid upon light exposure and whose solubility in a developing solution is changed by an action of the acid, the resist composition including: a resin component (A1) whose solubility in a developing solution is changed by the action of the acid, in which the resin component (A1) has a constitutional unit (a0) derived from a compound represented by General Formula (a0-1).

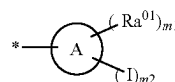
[Chemical Formula 1]



(a0-1)

[0012] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. Rpg represents an acid dissociable group represented by General Formula (a0-pg).]

[Chemical Formula 2]



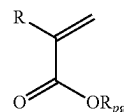
(a0-pg)

[0013] [In the formula, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton.  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms.  $m1$  represents an integer of 0 to 20.  $m2$  represents an integer of 1 to 20. Here,  $m1+m2 \leq$  (a total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied. In a case where  $m1$  represents an integer of 2 or greater, a plurality of  $Ra^{01}$ 's may be the same as or different from each other. \* represents a bonding site.]

[0014] According to a second aspect of the present invention, there is provided a resist pattern formation method including a step of forming a resist film on a support using the resist composition according to the first aspect, a step of exposing the resist film to light, and a step of developing the resist film exposed to light to form a resist pattern.

[0015] According to a third aspect of the present invention, there is provided a compound represented by General Formula (a0-1).

[Chemical Formula 3]

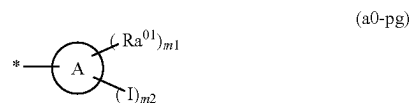


(a0-1)

[0016] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated

alkyl group having 1 to 5 carbon atoms. Rpg represents an acid dissociable group represented by General Formula (a0-pg).]

[Chemical Formula 4]



**[0017]** [In the formula, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton. Ra<sup>01</sup> represents a monovalent hydrocarbon group having 1 to 20 carbon atoms. m1 represents an integer of 0 to 20. m2 represents an integer of 1 to 20. Here, m1+m2≤(a total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied. In a case where m1 represents an integer of 2 or greater, a plurality of Ra<sup>01</sup>'s may be the same as or different from each other. \* represents a bonding site.]

#### Advantageous Effects of Invention

**[0018]** According to the present invention, it is possible to provide a resist composition which can achieve high sensitivity during formation of a resist pattern, a resist pattern formation method using the resist composition, and a compound useful as a raw material of a base material component used in the resist composition.

#### DESCRIPTION OF EMBODIMENTS

**[0019]** In the present specification and the scope of the present claims, the term “aliphatic” is a relative concept used in relation to the term “aromatic”, and defines a group, a compound, or the like that has no aromaticity.

**[0020]** The term “alkyl group” includes a linear, branched, or cyclic monovalent saturated hydrocarbon group unless otherwise specified. The same applies to the alkyl group in an alkoxy group.

**[0021]** The term “alkylene group” includes a linear, branched, or cyclic divalent saturated hydrocarbon group unless otherwise specified.

**[0022]** Examples of “halogen atom” include a fluorine atom, a chlorine atom, a bromine atom, and an iodine atom.

**[0023]** The term “constitutional unit” indicates a monomer unit constituting a polymer compound (a resin, a polymer, or a copolymer).

**[0024]** The expression “may have a substituent” includes both a case where a hydrogen atom (—H) is substituted with a monovalent group and a case where a methylene (—CH<sub>2</sub>—) group is substituted with a divalent group.

**[0025]** The term “light exposure” is a general concept for irradiation with radiation.

**[0026]** The term “acid decomposable group” indicates a group having acid decomposability in which at least a part of a bond in the structure of the acid decomposable group can be cleaved due to the action of an acid.

**[0027]** Examples of the acid decomposable group whose polarity is increased by the action of an acid include groups which are decomposed by the action of an acid to generate a polar group.

**[0028]** Examples of the polar group include a carboxy group, a hydroxyl group, an amino group, and a sulfo group (—SO<sub>3</sub>H).

**[0029]** More specific examples of the acid decomposable group include a group in which the above-described polar group has been protected by an acid dissociable group (such as a group in which a hydrogen atom of the OH-containing polar group has been protected by an acid dissociable group).

**[0030]** Here, the term “acid dissociable group” indicates both a group (i) having an acid dissociation property in which a bond between the acid dissociable group and an atom adjacent to the acid dissociable group can be cleaved due to the action of an acid and a group (ii) in which some bonds are cleaved due to the action of an acid, a decarboxylation reaction occurs, and thus the bond between the acid dissociable group and the atom adjacent to the acid dissociable group can be cleaved.

**[0031]** It is necessary that the acid dissociable group that constitutes the acid decomposable group is a group which exhibits a lower polarity than that of the polar group generated by the dissociation of the acid dissociable group. Thus, in a case where the acid dissociable group is dissociated by the action of an acid, a polar group exhibiting a higher polarity than that of the acid dissociable group is generated so that the polarity is increased. As a result, the polarity of an entire component (A1) is increased. Due to the increase in the polarity, the solubility in a developing solution is relatively changed such that the solubility is increased in a case where the developing solution is an alkali developing solution and the solubility is decreased in a case where the developing solution is an organic developing solution.

**[0032]** The term “base material component” denotes an organic compound having a film-forming ability. Organic compounds used as the base material component are classified into non-polymers and polymers. As the non-polymers, typically non-polymers having a molecular weight of 500 or greater and less than 4000 (hereinafter, referred to as “low-molecular-weight compounds”) are used. Hereinafter, “resin”, “polymer compound”, or “polymer” indicates a polymer having a molecular weight of 1000 or greater. As the molecular weight of the polymer, the weight-average molecular weight in terms of polystyrene according to gel permeation chromatography (GPC) is used.

**[0033]** The expression “constitutional unit to be derived” denotes a constitutional unit formed by cleavage of a multiple bond between carbon atoms, for example, an ethylenic double bond.

**[0034]** In “acrylic acid ester”, the hydrogen atom bonded to the carbon atom at the α-position may be substituted with a substituent. The substituent (R<sup>α</sup>) that substitutes the hydrogen atom bonded to the carbon atom at the α-position is an atom other than the hydrogen atom or a group. Further, the acrylic acid ester includes itaconic acid diester in which the substituent (R<sup>α</sup>) has been substituted with a substituent having an ester bond and α-hydroxyacryl ester in which the substituent (R<sup>α</sup>) has been substituted with a hydroxyalkyl group or a group obtained by modifying a hydroxyl group thereof. Further, the carbon atom at the α-position of acrylic acid ester indicates the carbon atom to which the carbonyl group of acrylic acid is bonded, unless otherwise specified.

**[0035]** Hereinafter, acrylic acid ester in which the hydrogen atom bonded to the carbon atom at the α-position has

been substituted with a substituent is also referred to as  $\alpha$ -substituted acrylic acid ester.

**[0036]** The concept “derivative” includes those obtained by substituting a hydrogen atom at the  $\alpha$ -position of a target compound with another substituent such as an alkyl group or a halogenated alkyl group, and derivatives thereof. Examples of the derivatives thereof include those obtained by substituting a hydrogen atom of a hydroxyl group of a target compound, in which the hydrogen atom at the  $\alpha$ -position may be substituted with a substituent, with an organic group, and those obtained by bonding a substituent other than a hydroxyl group to a target compound in which the hydrogen atom at the  $\alpha$ -position may be substituted with a substituent. Further, the  $\alpha$ -position denotes the first carbon atom adjacent to a functional group unless otherwise specified.

**[0037]** Examples of the substituent that substitutes the hydrogen atom at the  $\alpha$ -position of hydroxystyrene include those for  $R^{xx}$ .

**[0038]** In the present specification and the scope of the present claims, asymmetric carbons may be present and enantiomers or diastereomers may be present depending on the structures of the chemical formulae. In this case, these isomers are represented by one chemical formula. These isomers may be used alone or in the form of a mixture.

#### (Resist Composition)

**[0039]** The resist composition according to the present embodiment is a resist composition which generates an acid upon light exposure and whose solubility in a developing solution is changed by the action of the acid.

**[0040]** The resist composition contains a base material component (A) (hereinafter, also referred to as “component (A)”) whose solubility in a developing solution is changed by the action of the acid.

**[0041]** In the resist composition according to the present embodiment, the component (A) may generate acid upon exposure, or an additive component that is blended separately from the component (A) may generate acid upon exposure.

**[0042]** Specifically, the resist composition according to the present embodiment may (1) further contain an acid generator component (B) (hereinafter, referred to as “component (B)”) that generates an acid upon light exposure; (2) have a component (A) that generates an acid upon light exposure; and (3) have a component (A) that generates an acid upon light exposure and further contains component (B).

**[0043]** That is, in the cases of (2) and (3) described above, the component (A) is “base material component which generates an acid upon light exposure and whose solubility in a developing solution is changed by the action of the acid”. In a case where the component (A) is a base material component which generates an acid upon light exposure and whose solubility in a developing solution is changed by the action of the acid, it is preferable that the component (A1) described below is a resin which generates an acid upon light exposure and whose solubility in a developing solution is changed by the action of the acid. As such a resin, a polymer compound having a constitutional unit that generates an acid upon light exposure can be used. As the constitutional unit that generates an acid upon light exposure, those which have been known can be used.

**[0044]** Among the examples, it is preferable that the resist composition according to the present embodiment corresponds to the case (1). That is, it is preferable that the resist composition according to the present embodiment contains the component (A) and the component (B).

**[0045]** In a case where a resist film is formed using the resist composition according to the present embodiment and the formed resist film is subjected to selective exposure, acid is generated, for example, from the component (B) at exposed portions of the resist film, and the generated acid acts on the component (A) to change the solubility of the component (A) in a developing solution, whereas the solubility of the component (A) in a developing solution is not changed at unexposed portions, thereby that generates the difference in solubility in the developing solution between exposed portions and unexposed portions of the resist film. Therefore, by subjecting the resist film to development, exposed portions of the resist film are dissolved and removed to form a positive-tone resist pattern in a case where the resist composition is a positive-tone type, whereas unexposed portions of the resist film are dissolved and removed to form a negative-tone resist pattern in a case where the resist composition is a negative-tone type.

**[0046]** The resist composition of the present embodiment may be a positive-tone resist composition or a negative-tone resist composition. Further, in the formation of a resist pattern, the resist composition according to the present embodiment can be applied to an alkali developing process using an alkali developing solution in the developing treatment, or a solvent developing process using a developing solution containing an organic solvent (organic developing solution) in the developing treatment.

#### <Component (A)>

**[0047]** In the resist composition according to the present embodiment, the component (A) contains a resin component (A1) (hereinafter, also referred to as “component (A1)”) whose solubility in a developing solution is changed by the action of an acid.

**[0048]** Since the polarity of the base material component before and after the light exposure is changed by using the component (A1), an excellent development contrast can be obtained not only in an alkali developing process but also in a solvent developing process.

**[0049]** As the component (A), another polymeric compound and/or a low molecular weight compound may be used in combination with the component (A1).

**[0050]** In the resist composition according to the present embodiment, the component (A) may be used alone or in a combination of two or more kinds thereof.

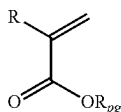
#### In Regard to Component (A1)

**[0051]** The component (A1) is a resin component whose solubility in a developing solution is changed by the action of an acid.

**[0052]** The component (A1) has a constitutional unit (a0) derived from a compound represented by General Formula (a0-1).

**[0053]** Further, the component (A1) may have other constitutional units as necessary in addition to the constitutional unit (a0).

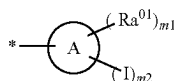
[Chemical Formula 5]



(a0-1)

**[0054]** [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. Rpg represents an acid dissociable group represented by General Formula (a0-pg).]

[Chemical Formula 6]



(a0-pg)

**[0055]** [In the formula, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton.  $\text{Ra}^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms.  $m_1$  represents an integer of 0 to 20.  $m_2$  represents an integer of 1 to 20. Here,  $m_1 + m_2 \leq$  (a total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied. In a case where  $m_1$  represents an integer of 2 or greater, a plurality of Raft's may be the same as or different from each other. \* represents a bonding site.]

**[0056]** In Formula (a0-1), R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms.

**[0057]** R represents preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms, and from the viewpoint of industrial availability, more preferably a hydrogen atom, a methyl group, or trifluoromethyl group, still more preferably a hydrogen atom or a methyl group, and particularly preferably a methyl group.

**[0058]** In Formula (a0-1), Rpg represents an acid dissociable group represented by Formula (a0-pg).

**[0059]** That is, the constitutional unit (a0) contains an acid decomposable group whose polarity is increased by the action of an acid. Specifically, the acid decomposable group Rpg is bonded to the carbonyloxy group in Formula (a0-1) to form an acid decomposable group. Therefore, in a case where the carbon atom ( $\text{C}^4$ ) forming a ring skeleton of the cyclic alkene A, which is bonded to the carbonyloxy group in Formula (a0-1), is a secondary carbon atom, the carbon-carbon unsaturated bond included in the ring skeleton of the cyclic alkene A is formed between a carbon atom ( $\text{C}^\alpha$ ) at a position ( $\alpha$ -position of the carbon atom ( $\text{C}^4$ )) adjacent to the carbon atom ( $\text{C}^4$ ) and a carbon atom ( $\text{C}^\beta$ ) at a position ( $\beta$ -position of the carbon atom ( $\text{C}^4$ )) adjacent to the carbon atom ( $\text{C}^\alpha$ ). In a case where the carbon atom ( $\text{C}^4$ ) is a tertiary carbon atom, the position of the carbon-carbon unsaturated bond included in the carbocyclic skeleton of the cyclic alkene A is not particularly limited. From the viewpoint of improving acid dissociation properties, in a case where the carbon atom ( $\text{C}^4$ ) is a tertiary carbon atom, it is preferable

that the carbon-carbon unsaturated bond included in the ring skeleton of the cyclic alkene A is formed between the carbon atom ( $\text{C}^\alpha$ ) and the carbon atom ( $\text{C}^\beta$ ).

**[0060]** In Formula (a0-pg), A represents a cyclic alkene having only one carbon-carbon unsaturated bond in the ring skeleton. The cyclic alkene as A may be a polycyclic group or a monocyclic group.

**[0061]** As the cyclic alkene which is a monocyclic group, a group obtained by removing one hydrogen atom from a monocycloalkene is preferable. The monocycloalkene preferably has 3 to 6 carbon atoms, and specific examples thereof include cyclopentene and cyclohexene.

**[0062]** A group obtained by removing one hydrogen atom from a polycycloalkene is preferable as the cyclic alkene which is a polycyclic group, and a polycycloalkane having 7 to 12 carbon atoms is preferable as the polycycloalkane, and specific examples thereof include adamantene, norbornene, isobornene, tricyclodecene, and tetracyclododecene.

**[0063]** Among these, as the cyclic alkene represented by A, a monocyclic cyclic alkene is preferable, a group obtained by removing one hydrogen atom from a monocycloalkene is more preferable, and a group obtained by removing one hydrogen atom from cyclopentene or cyclohexene is still more preferable.

**[0064]** In Formula (a0-pg), examples of the monovalent hydrocarbon group having 1 to 20 carbon atoms as  $\text{Ra}^{01}$  include a chain-like alkyl group and a chain-like alkenyl group.

**[0065]** The chain-like alkyl group as  $\text{Ra}^{01}$  may be linear or branched.

**[0066]** The linear alkyl group has preferably 1 to 20 carbon atoms, more preferably 1 to 15 carbon atoms, and most preferably 1 to 10 carbon atoms. Specific examples thereof include a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a hexyl group, a heptyl group, an octyl group, a nonyl group, and a decyl group.

**[0067]** The branched alkyl group has preferably 3 to 20 carbon atoms, more preferably 3 to 15 carbon atoms, and most preferably 3 to 10 carbon atoms. Specific examples thereof include an isopropyl group, an isobutyl group, a tert-butyl group, an isopentyl group, a neopentyl group, a 1,1-diethylpropyl group, and a 2,2-dimethylbutyl group.

**[0068]** The chain-like alkenyl group as  $\text{Ra}^{01}$  may be linear or branched, and has preferably 2 to 20 carbon atoms, more preferably 2 to 10 carbon atoms, and still more preferably 2 to 5 carbon atoms. Examples of the linear alkenyl group include a vinyl group, a propenyl group (an allyl group), and a butenyl group. Examples of the branched alkenyl group include a 1-methylvinyl group, a 2-methylvinyl group, a 1-methylpropenyl group, and a 2-methylpropenyl group.

**[0069]** Among these,  $\text{Ra}^{01}$  represents preferably a chain-like alkyl group having 1 to 20 carbon atoms, more preferably a chain-like alkyl group having 1 to 20 carbon atoms, still more preferably a chain-like alkyl group having 1 to 10 carbon atoms, and even still more preferably a chain-like alkyl group having 1 to 5 carbon atoms.

**[0070]** In Formula (a0-pg),  $m_1$  represents preferably 0 to 10, more preferably 0 to 5, still more preferably 0 to 2, and may be 0.

**[0071]** In Formula (a0-pg),  $m_2$  represents preferably 1 to 10, more preferably 1 to 5, still more preferably 1 or 2, and may be 1.

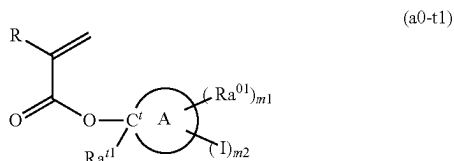
[0072] Here, in Formula (a0-pg),  $m1+m2 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied.

[0073] In Formula (a0-pg), from the viewpoint of achieving high sensitivity, it is preferable that an iodine atom is bonded to at least one of the carbon atoms constituting the carbon-carbon unsaturated bond in the ring skeleton of the cyclic alkene A, more preferable that the carbon-carbon unsaturated bond included in the ring skeleton of the cyclic alkene A is formed between the carbon atom ( $C^\alpha$ ) at a position ( $\alpha$ -position of the carbon atom ( $C^\alpha$ )) adjacent to the carbon atom ( $C^d$ ) forming the ring skeleton of the cyclic alkene A, which is bonded to the carbonyloxy group in Formula (a0-1), and the carbon atom ( $C^\alpha$ ) at a position ( $\beta$ -position of the carbon atom ( $C^d$ ))) adjacent to the carbon atom ( $C^\alpha$ ) and at least one of the carbon atom ( $C^\alpha$ ) or the carbon atom ( $C^\beta$ ) is bonded to an iodine element, and still more preferable that the carbon-carbon unsaturated bond included in the ring skeleton of the cyclic alkene A is formed between the carbon atom ( $C^\alpha$ ) and the carbon atom ( $C^\beta$ ) and an iodine element is bonded to the carbon atom ( $C^\beta$ ) from the viewpoint of achieving high sensitivity and improving the stability of the component (A1) in the resist film. In cases where an iodine element is bonded to the carbon atom ( $C^\beta$ ), since the stability of the carbocation formed by the dissociation of the cyclic alkene A by the action of an acid is increased, the deprotection rate of the constitutional unit (a0) is increased, and high sensitivity is likely to be achieved.

[0074] In Formula (a0-pg), in a case where m1 represents an integer of 2 or greater, a plurality of  $Ra^{01}$ 's may be the same as or different from each other.

[0075] In the present embodiment, the constitutional unit (a0) may be a constitutional unit derived from a compound represented by General Formula (a0-t1) (hereinafter, also referred to as "constitutional unit (a0-t1)").

[Chemical Formula 7]



[0076] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton.  $C'$  represents a tertiary carbon atom.  $Ra^{01}$  represents a linear or branched alkyl group having 1 to 12 carbon atoms.  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms. m1 represents an integer of 0 to 20. m2 represents an integer of 1 to 20. Here,  $m1+m2 \leq$  (a total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied. In a case where m1 represents an integer of 2 or greater, a plurality of  $Ra^{01}$ 's may be the same as or different from each other.]

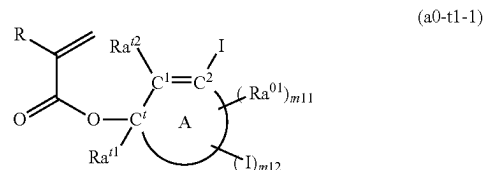
[0077] In Formula (a0-t1), R has the same definition as that for R in Formula (a0-1). In Formula (a0-t1), A,  $Ra^{01}$ , m1, and m2 each have the same definition as that for A,  $Ra^{01}$ , m1, and m2 in Formula (a0-pg).

[0078] In Formula (a0-t1), examples of the linear alkyl group having 1 to 12 carbon atoms as  $Ra^{01}$  include a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a hexyl group, a heptyl group, an octyl group, a nonyl group, and a decyl group. Among these, a linear alkyl group having 1 to 5 carbon atoms is preferable, a methyl group, an ethyl group, or a butyl group is more preferable, and a methyl group or an ethyl group is still more preferable.

[0079] In Formula (a0-t1), examples of the branched alkyl group having 1 to 12 carbon atoms as  $Ra^{01}$  include an isopropyl group, an isobutyl group, a tert-butyl group, an isopentyl group, a neopentyl group, a 1,1-diethylpropyl group, and a 2,2-dimethylbutyl group. Among these, a branched alkyl group having 3 to 5 carbon atoms is preferable, and an isopropyl group is more preferable.

[0080] In the present embodiment, it is preferable that the constitutional unit (a0-t1) is a constitutional unit derived from a compound represented by General Formula (a0-t1-1) (hereinafter, also referred to as "constitutional unit (a0-t1-1)").

[Chemical Formula 8]



[0081] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton.  $C'$  represents a tertiary carbon atom.  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A.  $Ra^{01}$  represents a linear or branched alkyl group having 1 to 12 carbon atoms.  $Ra^{02}$  represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms.  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms. m11 represents an integer of 0 to 20. m12 represents an integer of 0 to 19. Here,  $m11+m12 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied. In a case where m11 represents an integer of 2 or greater, a plurality of  $Ra^{01}$ 's may be the same as or different from each other.]

[0082] In Formula (a0-t1-1). R has the same definition as that for R in Formula (a0-1).

[0083] In Formula (a0-t1-1), A and  $Ra^{01}$  each have the same definition as that for A and  $Ra^{01}$  in Formula (a0-pg).

[0084] In Formula (a0-t1-1),  $Ra^{02}$  has the same definition as that for  $Ra^{01}$  in Formula (a0-t1).

[0085] In Formula (a0-t1-1), the monovalent hydrocarbon group having 1 to 20 carbon atoms as  $Ra^{02}$  has the same definition as the monovalent hydrocarbon group having 1 to 20 carbon atoms as  $Ra^{01}$  in Formula (a0-1). Among these, it is preferable that  $Ra^{02}$  represents a hydrogen atom.

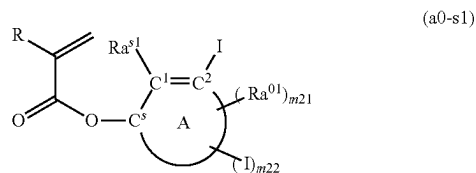
[0086] In Formula (a0-t1-1), m11 represents preferably 0 to 10, more preferably 0 to 5, still more preferably 0 to 2, and may be 0.

[0087] In Formula (a0-t1-1), m12 represents preferably 0 to 10, more preferably 0 to 5, still more preferably 0 or 1, and may be 0.

[0088] Here, in Formula (a0-t1-1),  $m11+m12 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied.

[0089] In the present embodiment, the constitutional unit (a0) may be a constitutional unit derived from a compound represented by General Formula (a0-s1) (hereinafter, also referred to as “constitutional unit (a0-s1)”).

[Chemical Formula 9]



[0090] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton. Cs represents a secondary carbon atom. C1 and C2 each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A. Ra<sup>s1</sup> and Ra<sup>s2</sup> each independently represent a hydrogen atom, an iodine atom, or a monovalent hydrocarbon group having 1 to 20 carbon atoms. Ra<sup>o1</sup> represents a monovalent hydrocarbon group having 1 to 20 carbon atoms. m21 represents an integer of 0 to 20. m22 represents an integer of 0 to 20. Here,  $m21+m22 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied. In addition, in a case where m22 represents 0, at least one of Ra<sup>s1</sup> or Ra<sup>s2</sup> represents an iodine atom. In a case where m21 represents an integer of 2 or greater, a plurality of Ra<sup>o1</sup>'s may be the same as or different from each other.]

[0091] In Formula (a0-s1), R has the same definition as that for R in Formula (a0-1).

[0092] In Formula (a0-s1), A and Ra<sup>o1</sup> each have the same definition as that for A and Ra<sup>o1</sup> in Formula (a0-pg).

[0093] In Formula (a0-s1), the monovalent hydrocarbon group having 1 to 20 carbon atoms as Ra<sup>s1</sup> and Ra<sup>s2</sup> is the same as the monovalent hydrocarbon group having 1 to 20 carbon atoms as Ra<sup>o1</sup> in Formula (a0-1). Among these, it is preferable that one of Ra<sup>s1</sup> and Ra<sup>s2</sup> represents an iodine atom and the other represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms, more preferable that Ra<sup>s1</sup> represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms and Ra<sup>s2</sup> represents an iodine atom, and still more preferable that Ra<sup>s2</sup> represents a monovalent hydrocarbon group having 1 to 20 carbon atoms and Ra<sup>s2</sup> represents an iodine atom.

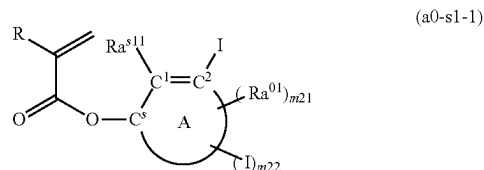
[0094] In Formula (a0-s1), m21 represents preferably 0 to 10, more preferably 0 to 5, still more preferably 0 to 2, and may be 0.

[0095] In Formula (a0-s1), m22 represents preferably 0 to 10, more preferably 0 to 5, still more preferably 0 or 1, and may be 0.

[0096] Here, in Formula (a0-s1),  $m21+m22 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied.

[0097] In the present embodiment, the constitutional unit (a0-s1) may be a constitutional unit derived from a compound represented by General Formula (a0-s1-1) (hereinafter, also referred to as “constitutional unit (a0-s1-1)”).

[Chemical Formula 10]



[0098] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton. Cs represents a secondary carbon atom. C1 and C2 each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A. Ra<sup>s11</sup> represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms. Ra<sup>o1</sup> represents a monovalent hydrocarbon group having 1 to 20 carbon atoms. m21 represents an integer of 0 to 20. m22 represents an integer of 0 to 19. Here,  $m21+m22 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied. In a case where m21 represents an integer of 2 or greater, a plurality of Ra<sup>o1</sup>'s may be the same as or different from each other.]

[0099] In Formula (a0-s1-1), R has the same definition as that for R in Formula (a0-1).

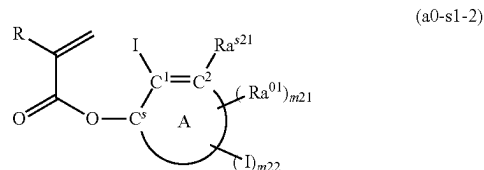
[0100] In Formula (a0-s1-1), A and Ra<sup>o1</sup> each have the same definition as that for A and Ra<sup>o1</sup> in Formula (a0-pg).

[0101] In Formula (a0-s1-1), m21 and m22 each have the same definition as that for m21 and m22 in Formula (a0-s1).

[0102] In Formula (a0-s1-1), the monovalent hydrocarbon group having 1 to 20 carbon atoms as Ra<sup>s11</sup> has the same definition as that for the monovalent hydrocarbon group having 1 to 20 carbon atoms as Ra<sup>o1</sup> in Formula (a0-1). Among these, Ra<sup>s11</sup> represents preferably a chain-like alkyl group having 1 to 10 carbon atoms, more preferably a linear alkyl group having 1 to 5 carbon atoms or a branched alkyl group having 3 to 5 carbon atoms, and still more preferably a linear alkyl group having 1 to 3 carbon atoms.

[0103] In the present embodiment, the constitutional unit (a0-s1) may be a constitutional unit derived from a compound represented by General Formula (a0-s1-2) (hereinafter, also referred to as “constitutional unit (a0-s1-2)”).

[Chemical Formula 11]



[0104] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton. C<sup>s</sup> represents a secondary carbon atom. C<sup>1</sup> and C<sup>2</sup> each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A. Ra<sup>s21</sup> represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms. Ra<sup>01</sup> represents a monovalent hydrocarbon group having 1 to 20 carbon atoms. m21 represents an integer of 0 to 20. m22 represents an integer of 0 to 19. Here, m21+m22≤(the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied. In a case where m21 represents an integer of 2 or greater, a plurality of Ra<sup>01</sup>'s may be the same as or different from each other.]

[0105] In Formula (a0-s1-2), R has the same definition as that for R in Formula (a0-1).

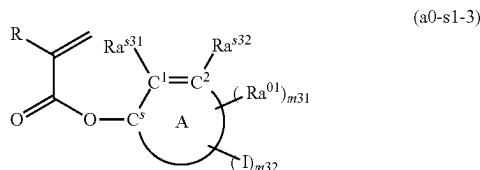
[0106] In Formula (a0-s1-2), A and Ra<sup>01</sup> each have the same definition as that for A and Ra<sup>01</sup> in Formula (a0-pg).

[0107] In Formula (a0-s1-2), m21 and m22 each have the same definition as that for m21 and m22 in Formula (a0-s1).

[0108] In Formula (a0-s1-2), the monovalent hydrocarbon group having 1 to 20 carbon atoms as Ra<sup>s21</sup> has the same definition as that for the monovalent hydrocarbon group having 1 to 20 carbon atoms as Ra<sup>01</sup> in Formula (a0-1). Among these, it is preferable that Ra<sup>s21</sup> represents a hydrogen atom.

[0109] In the present embodiment, the constitutional unit (a0-s1) may be a constitutional unit derived from a compound represented by General Formula (a0-s1-3) (hereinafter, also referred to as "constitutional unit (a0-s1-3)").

[Chemical Formula 12]



[0110] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton. C<sup>s</sup> represents a secondary carbon atom. C<sup>1</sup> and C<sup>2</sup> each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A. Ra<sup>s31</sup> and Ra<sup>s32</sup> each independently represent a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms. Ra<sup>01</sup> represents a monovalent hydrocarbon group having 1 to 20 carbon atoms. m31 represents an integer of 0 to 20. m32 represents an integer of 1 to 20. Here, m31+m32≤(the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied. In a case where m31 represents an integer of 2 or greater, a plurality of Ra<sup>01</sup>'s may be the same as or different from each other.

[0111] In Formula (a0-s1-3), R has the same definition as that for R in Formula (a0-1).

[0112] In Formula (a0-s1-3), A and Ra<sup>01</sup> each have the same definition as that for A and Ra<sup>01</sup> in Formula (a0-pg).

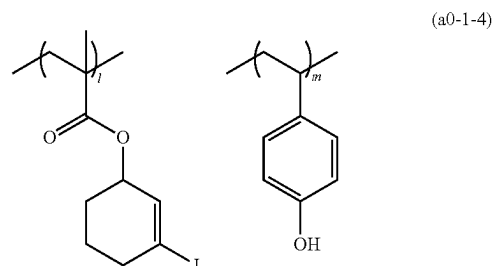
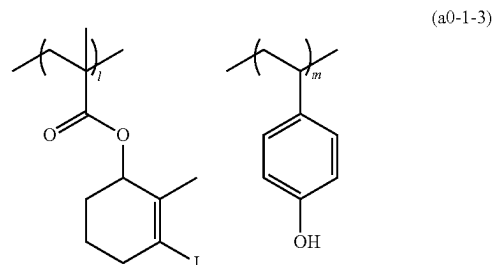
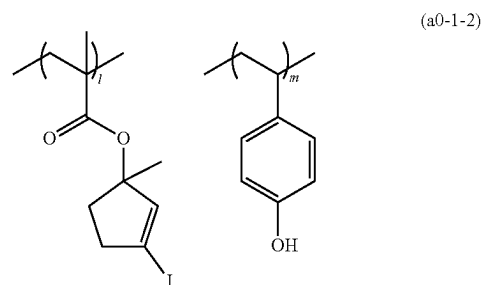
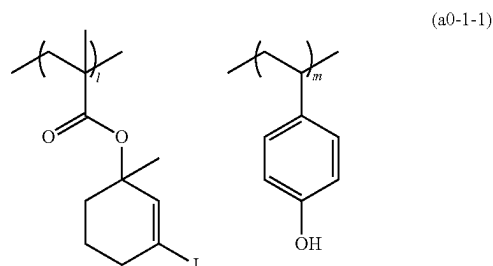
[0113] In Formula (a0-s1-3), m31 represents preferably 0 to 10, more preferably 0 to 5, still more preferably 0 to 2, and may be 0.

[0114] In Formula (a0-s1-3), m32 represents preferably 1 to 10, more preferably 1 to 5, still more preferably 1 or 2, and may be 1.

[0115] Here, in Formula (a0-s1-3), m31+m32≤(the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied.

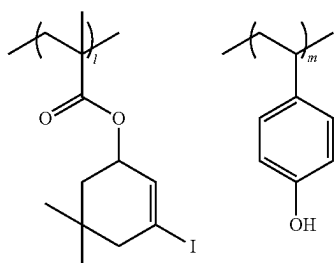
[0116] Specific examples of the constitutional unit (a0) are shown below. In the formulae shown below, R<sup>α</sup> represents a hydrogen atom, a methyl group, or a trifluoromethyl group.

[Chemical Formula 13]

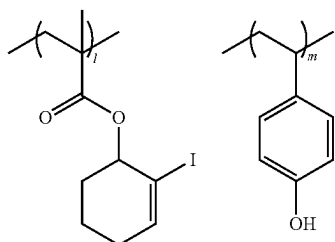




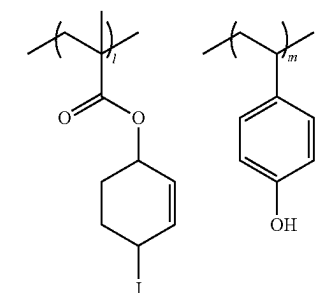
-continued



(a0-1-5)



(a0-1-6)



(a0-1-7)

[0117] The constitutional unit (a0) contained in the component (A1) may be one kind or may be two or more kinds.

[0118] The proportion of the constitutional unit (a0) in the component (A1) is preferably in a range of 40% to 80% by mole, more preferably in a range of 45% to 75% by mole, still more preferably in a range of 50% to 70% by mole, and particularly preferably in a range of 55% to 65% by mole with respect to the total amount (100% by mole) of all constitutional units constituting the component (A1).

[0119] In a case where the content of the constitutional unit (a0) is greater than or equal to the lower limits of the above-described preferable ranges, the sensitivity is likely to be further increased, and reduction of the film thickness is likely to be suppressed. Meanwhile, in a case where the proportion thereof is set to be less than or equal to the above-described upper limits, the constitutional unit (a0) and other constitutional units are likely to be balanced.

<<Other Constitutional Units>>

[0120] The component (A1) may have other constitutional units as necessary in addition to the constitutional unit (a0) described above.

[0121] Examples of the other constitutional units include a constitutional unit (a1) containing an acid decomposable group whose polarity is increased by the action of an acid (here, a constitutional unit corresponding to the constitutional unit (a0) is excluded), a constitutional unit (a10) represented by General Formula (a10-1), a constitutional unit (a2) containing a lactone-containing cyclic group, and

a constitutional unit (a8) derived from a compound represented by General Formula (a8-1).

In Regard to Constitutional Unit (a1):

[0122] The constitutional unit (a1) is a constitutional unit that contains an acid decomposable group whose polarity is increased due to the action of an acid. Here, the constitutional unit corresponding to the constitutional unit (a0) is excluded from the constitutional unit (a1).

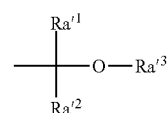
[0123] Examples of the acid dissociable group are the same as those which have been suggested as the acid dissociable groups of the base resin for a chemically amplified resist composition.

[0124] Specific examples of the acid dissociable group of the base resin suggested for a chemically amplified resist composition include "acetal type acid dissociable group", "tertiary alkyl ester type acid dissociable group", and "tertiary alkyloxycarbonyl acid dissociable group" described below.

Acetal Type Acid Dissociable Group:

[0125] Examples of the acid dissociable group that protects a carboxy group or a hydroxyl group in the polar groups include an acid dissociable group represented by General Formula (a1-r-1) (hereinafter, also referred to as "acetal type acid dissociable group").

[Chemical Formula 2]



(a1-r-1)

[0126] [In the formula, Ra<sup>1</sup> and Ra<sup>2</sup> represent a hydrogen atom or an alkyl group. Ra<sup>3</sup> represents a hydrocarbon group, and Ra<sup>3</sup> may be bonded to any of Ra<sup>1</sup> and Ra<sup>2</sup> to form a ring.]

[0127] In Formula (a1-r-1), it is preferable that at least one of Ra<sup>1</sup> or Ra<sup>2</sup> represents a hydrogen atom and more preferable that both Ra<sup>1</sup> and Ra<sup>2</sup> represent a hydrogen atom. In a case where Ra<sup>1</sup> or Ra<sup>2</sup> represents an alkyl group, examples of the alkyl group include the same alkyl groups exemplified as the substituent which may be bonded to the carbon atom at the  $\alpha$ -position in the description of the  $\alpha$ -substituted acrylic acid ester. Among these, an alkyl group having 1 to 5 carbon atoms is preferable. Specific preferred examples thereof include linear or branched alkyl groups. More specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group. Among these, a methyl group or an ethyl group is more preferable, and a methyl group is particularly preferable.

[0128] In Formula (a1-r-1), examples of the hydrocarbon group as Ra<sup>3</sup> include a linear or branched alkyl group and a cyclic hydrocarbon group.

[0129] The linear alkyl group has preferably 1 to 5 carbon atoms, more preferably 1 to 4 carbon atoms, and still more preferably 1 or 2 carbon atoms. Specific examples thereof include a methyl group, an ethyl group, an n-propyl group,

an n-butyl group, and an n-pentyl group. Among these, a methyl group, an ethyl group, or an n-butyl group is preferable, and a methyl group or an ethyl group is more preferable.

[0130] The branched alkyl group has preferably 3 to 10 carbon atoms and more preferably 3 to 5 carbon atoms. Specific examples thereof include an isopropyl group, an isobutyl group, a tert-butyl group, an isopentyl group, a neopentyl group, a 1,1-diethylpropyl group, and a 2,2-dimethylbutyl group. Among these, an isopropyl group is preferable.

[0131] In a case where  $Ra^{13}$  represents a cyclic hydrocarbon group, the hydrocarbon group may be an alicyclic hydrocarbon group or an aromatic hydrocarbon group and may be a polycyclic group or a monocyclic group.

[0132] The alicyclic hydrocarbon group which is a monocyclic group is preferably a group in which one hydrogen atom has been removed from a monocycloalkane. The monocycloalkane has preferably 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane.

[0133] The alicyclic hydrocarbon group which is a polycyclic group is preferably a group in which one hydrogen atom has been removed from a polycycloalkane. The polycycloalkane preferably has 7 to 12 carbon atoms, and specific examples thereof include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclododecane.

[0134] In a case where the cyclic hydrocarbon group as  $Ra^{13}$  is an aromatic hydrocarbon group, the aromatic hydrocarbon group is a hydrocarbon group having at least one aromatic ring.

[0135] The aromatic ring is not particularly limited as long as the aromatic ring is a cyclic conjugated system having  $(4n+2)$   $\pi$  electrons and may be monocyclic or polycyclic. The aromatic ring has preferably 5 to 30 carbon atoms, more preferably 5 to 20 carbon atoms, still more preferably 6 to 15 carbon atoms, and particularly preferably 6 to 12 carbon atoms.

[0136] Specifically, as the aromatic ring, an aromatic hydrocarbon ring such as benzene, naphthalene, anthracene, and phenanthrene; and an aromatic heterocyclic ring in which some carbon atoms constituting the aromatic hydrocarbon ring have been substituted with heteroatoms are exemplary examples. Examples of the heteroatom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom. Specific examples of the aromatic heterocyclic ring include a pyridine ring and a thiophene ring.

[0137] Specific examples of the aromatic hydrocarbon group as  $Ra^{13}$  include a group in which one hydrogen atom has been removed from the above-described aromatic hydrocarbon ring or aromatic heterocyclic ring (such as an aryl group or a heteroaryl group); a group in which one hydrogen atom has been removed from an aromatic compound having two or more aromatic rings (such as biphenyl or fluorene); and a group in which one hydrogen atom of the above-described aromatic hydrocarbon ring or aromatic heterocyclic ring has been substituted with an alkylene group (for example, an arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, a 1-naphthylethyl group, or a 2-naphthylethyl group). The number of carbon atoms in the alkylene group bonded to the aromatic hydrocarbon ring or aromatic het-

erocyclic ring is preferably in a range of 1 to 4, more preferably 1 or 2, and particularly preferably 1.

[0138] The cyclic hydrocarbon group as  $Ra^{13}$  may include a substituent. Examples of the substituent include  $-R^{P1}$ ,  $-R^{P2}-O-R^{P1}$ ,  $-R^{P2}-CO-R^{P1}$ ,  $-R^{P2}-CO-OR^{P1}$ ,  $-R^{P2}-O-CO-R^{P1}$ ,  $-R^{P2}-OH$ ,  $-R^{P2}-CN$ , and  $-R^{P2}-COOH$  (hereinafter, these substituents will also be collectively referred to as " $Ra^{15}$ ").

[0139] Here,  $R^{P1}$  represents a chain-like monovalent saturated hydrocarbon group having 1 to 10 carbon atoms, a monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms, or a monovalent aromatic hydrocarbon group having 6 to 30 carbon atoms. Further,  $R^{P2}$  represents a single bond, a chain-like divalent saturated hydrocarbon group having 1 to 10 carbon atoms, a divalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms, or a divalent aromatic hydrocarbon group having 6 to 30 carbon atoms. Here, some or all hydrogen atoms in the chain-like saturated hydrocarbon group, the aliphatic cyclic saturated hydrocarbon group, and the aromatic hydrocarbon group as  $R^{P1}$  and  $R^{P2}$  may be substituted with fluorine atoms. The aliphatic cyclic hydrocarbon group may have one or more of a single kind of substituents or one or more of each of plural kinds of the substituents.

[0140] Examples of the chain-like monovalent saturated hydrocarbon group having 1 to 10 carbon atoms include a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a hexyl group, a heptyl group, an octyl group, and a decyl group.

[0141] Examples of the monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms include a monocyclic aliphatic saturated hydrocarbon group such as a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, a cyclodecyl group, or a cyclododecyl group; and a polycyclic aliphatic saturated hydrocarbon group such as a bicyclo[2.2.2]octanyl group, a tricyclo[5.2.1.0<sup>2,5</sup>]decanyl group, a tricyclo[3.3.1.1<sup>3,7</sup>]decanyl group, a tetracyclo[6.2.1.1.3.6.0<sup>2,7</sup>]dodecanyl group, or an adamantyl group.

[0142] Examples of the monovalent aromatic hydrocarbon group having 6 to 30 carbon atoms include a group formed by removing one hydrogen atom from an aromatic hydrocarbon ring such as benzene, biphenyl, fluorene, naphthalene, anthracene, or phenanthrene.

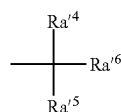
[0143] In a case where  $Ra^{13}$  is bonded to any of  $Ra^1$  and  $Ra^2$  to form a ring, the cyclic group is preferably a 4- to 7-membered ring and more preferably a 4- to 6-membered ring. Specific examples of the cyclic group include a tetrahydropyranyl group and a tetrahydrofuranyl group.

#### Tertiary Alkyl Ester Type Acid Dissociable Group:

[0144] Examples of the acid dissociable group that protects a carboxy group among the polar groups include an acid dissociable group represented by General Formula (a1-r-2).

[0145] Among examples of the acid dissociable group represented by Formula (a1-r-2), a group formed of an alkyl group is referred to as "tertiary alkyl ester type acid dissociable group" for convenience.

[Chemical Formula 15]



(a1-r-2)

[0146] [In the formula,  $Ra'^4$  to  $Ra'^6$  each represent a hydrocarbon group, and  $Ra'^5$  and  $Ra'^6$  may be bonded to each other to form a ring.]

[0147] Examples of the hydrocarbon group as  $Ra'^4$  include a linear or branched alkyl group, a chain-like or cyclic alkenyl group, a chain-like alkynyl group, and a cyclic hydrocarbon group.

[0148] Examples of the linear or branched alkyl group and the cyclic hydrocarbon group (an alicyclic hydrocarbon group which is a monocyclic group, an alicyclic hydrocarbon group which is a polycyclic group, or an aromatic hydrocarbon group) as  $Ra'^4$  include the same groups as those for  $Ra'^3$ .

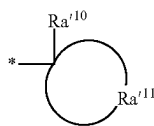
[0149] As the chain-like or cyclic alkenyl group as  $Ra'^4$ , an alkenyl group having 2 to 10 carbon atoms is preferable.

[0150] Examples of the hydrocarbon group as  $Ra'^5$  or  $Ra'^6$  include the same groups as those for  $Ra'^3$ .

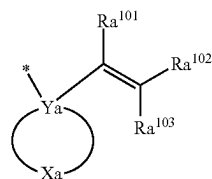
[0151] In a case where  $Ra'^5$  and  $Ra'^6$  are bonded to each other to form a ring, suitable examples thereof include a group represented by General Formula (a1-r2-1), a group represented by General Formula (a1-r2-2), and a group represented by General Formula (a1-r2-3).

[0152] Meanwhile, in a case where  $Ra'^4$  to  $Ra'^6$  independently represent a hydrocarbon group without being bonded to one another, suitable examples thereof include a group represented by General Formula (a1-r2-4).

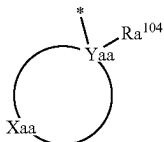
[Chemical Formula 16]



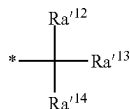
(a1-r2-1)



(a1-r2-2)



a1-r2-3)



(a1-r2-4)

[0153] [In Formula (a1-r2-1),  $Ra'^{10}$  represents a linear or branched alkyl group having 1 to 12 carbon atoms, in which a part thereof may be substituted with a halogen atom or a heteroatom-containing group.  $Ra'^{11}$  represents a group that forms an aliphatic cyclic group with the carbon atom to which  $Ra'^{10}$  has been bonded. In Formula (a1-r2-2), Ya represents a carbon atom. Xa represents a group that forms a cyclic hydrocarbon group with Ya. Some or all hydrogen atoms in this cyclic hydrocarbon group may be substituted.  $Ra'^{101}$  to  $Ra'^{103}$  each independently represent a hydrogen atom, a chain-like monovalent saturated hydrocarbon group having 1 to 10 carbon atoms, or a monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms. Some or all hydrogen atoms in the chain-like saturated hydrocarbon group and the aliphatic cyclic saturated hydrocarbon group may be substituted. Two or more of  $Ra'^{101}$  to  $Ra'^{103}$  may be bonded to each other to form a cyclic structure. In Formula (a1-r2-3), Yaa represents a carbon atom. Xaa represents a group that forms an aliphatic cyclic group with Yaa.  $Ra'^{104}$  represents an aromatic hydrocarbon group which may have a substituent. In Formula (a1-r2-4),  $Ra'^{12}$  and  $Ra'^{13}$  each independently represent a chain-like monovalent saturated hydrocarbon group having 1 to 10 carbon atoms. Some or all hydrogen atoms in this chain-like saturated hydrocarbon group may be substituted.  $Ra'^{14}$  represents a hydrocarbon group which may have a substituent. \* represents a bonding site (the same applies hereinafter).]

[0154] In Formula (a1-r2-1),  $Ra'^{10}$  represents a linear or branched alkyl group having 1 to 12 carbon atoms, in which a part thereof may be substituted with a halogen atom or a heteroatom-containing group.

[0155] The linear alkyl group as  $Ra'^{10}$  has 1 to 12 carbon atoms, preferably 1 to 10 carbon atoms, and particularly preferably 1 to 5 carbon atoms.

[0156] Examples of the branched alkyl group as  $Ra'^{10}$  include those for  $Ra'^3$  described above.

[0157] The alkyl group in  $Ra'^{10}$  may be partially substituted with a halogen atom or a heteroatom-containing group. For example, some hydrogen atoms constituting the alkyl group may be substituted with a halogen atom or a heteroatom-containing group. Further, some carbon atoms (methylene group or the like) constituting the alkyl group may be substituted with a heteroatom-containing group.

[0158] Examples of the heteroatoms here include an oxygen atom, a nitrogen atom, and a sulfur atom. Examples of the heteroatom-containing group include  $(-O-)$ ,  $-C(=O)-O-$ ,  $-O-C(=O)-$ ,  $-C(=O)-$ ,  $-O-C(=O)-O-$ ,  $-C(=O)-NH-$ ,  $-NH-$ ,  $-S-$ ,  $-S(=O)_2-$ , and  $-S(=O)_2-O-$ .

[0159] In Formula (a1-r2-1), preferred examples of  $Ra'^{11}$  (an aliphatic cyclic group that is formed together with a carbon atom to which  $Ra'^{10}$  is bonded) include the groups described as the alicyclic hydrocarbon group (alicyclic hydrocarbon group) which is a monocyclic group or a polycyclic group as  $Ra'^3$  in Formula (a1-r-1). Among these, it is preferably a monocyclic alicyclic hydrocarbon group, and specifically, it is more preferably a cyclopentyl group or a cyclohexyl group.

[0160] In Formula (a1-r2-2), examples of the cyclic hydrocarbon group that is formed by Xa together with Ya include a group in which one or more hydrogen atoms have been further removed from the cyclic monovalent hydrocarbon group (alicyclic hydrocarbon group) as  $Ra'^3$  in Formula (a1-r-1).

[0161] The cyclic hydrocarbon group that is formed by Xa together with Ya may have a substituent. Examples of the substituent include those exemplified as the substituents that the cyclic hydrocarbon group as Ra<sup>13</sup> may have.

[0162] In Formula (a1-r2-2), examples of the chain-like monovalent saturated hydrocarbon group having 1 to 10 carbon atoms as Ra<sup>101</sup> to Ra<sup>103</sup> include a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a hexyl group, a heptyl group, an octyl group, and a decyl group.

[0163] Examples of the monovalent aliphatic cyclic saturated hydrocarbon group having 3 to 20 carbon atoms as Ra<sup>101</sup> to Ra<sup>103</sup> include a monocyclic aliphatic saturated hydrocarbon group such as a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, a cyclodecyl group, or a cyclododecyl group; and a polycyclic aliphatic saturated hydrocarbon group such as a bicyclo[2.2.2]octanyl group, a tricyclo[5.2.1.02,6]decanyl group, a tricyclo[3.3.1.13,7]decanyl group, a tetracyclo[6.2.1.13,6.02,7]dodecanyl group, or an adamantyl group.

[0164] From the viewpoint of ease of synthesis, Ra<sup>101</sup> to Ra<sup>103</sup> represent preferably a hydrogen atom or a chain-like monovalent saturated hydrocarbon group having 1 to 10 carbon atoms, more preferably a hydrogen atom, a methyl group, or an ethyl group, and particularly preferably a hydrogen atom.

[0165] Examples of the substituent included in the chain-like saturated hydrocarbon group or the aliphatic cyclic saturated hydrocarbon group represented by Ra<sup>101</sup> to Ra<sup>103</sup> include the same substituents as those for Ra<sup>x5</sup>.

[0166] Examples of the group having a carbon-carbon double bond generated by two or more of Ra<sup>101</sup> to Ra<sup>103</sup> being bonded to each other to form a cyclic structure include a cyclopentenyl group, a cyclohexenyl group, a methylcyclopentenyl group, a methylcyclohexenyl group, a cyclopentylidenethenyl group, and a cyclohexylidenethenyl group. Among these, from the viewpoint of ease of synthesis, a cyclopentenyl group, a cyclohexenyl group, or a cyclopentylidenethenyl group is preferable.

[0167] In Formula (a1-r2-3), as the aliphatic cyclic group that is formed by Xaa together with Yaa, the group described as the alicyclic hydrocarbon group which is a monocyclic group or a polycyclic group as Ra<sup>13</sup> in Formula (a1-r-1) is preferable.

[0168] In Formula (a1-r2-3), examples of the aromatic hydrocarbon group as Ra<sup>104</sup> include a group in which one or more hydrogen atoms have been removed from an aromatic hydrocarbon ring having 5 to 30 carbon atoms. Among the examples, Ra<sup>104</sup> represents preferably a group in which one or more hydrogen atoms have been removed from an aromatic hydrocarbon ring having 6 to 15 carbon atoms, more preferably a group in which one or more hydrogen atoms have been removed from benzene, naphthalene, anthracene, or phenanthrene, still more preferably a group in which one or more hydrogen atoms have been removed from benzene, naphthalene, or anthracene, particularly preferably a group in which one or more hydrogen atoms have been removed from benzene or naphthalene, and most preferably a group in which one or more hydrogen atoms have been removed from benzene.

[0169] Examples of the substituent that Ra<sup>104</sup> in Formula (a1-r2-3) may have include a methyl group, an ethyl group, a propyl group, a hydroxyl group, a carboxyl group, a

halogen atom, an alkoxy group (such as a methoxy group, an ethoxy group, a propoxy group, or a butoxy group), and an alkyloxycarbonyl group.

[0170] In Formula (a1-r2-4), Ra<sup>112</sup> and Ra<sup>113</sup> each independently represent a chain-like monovalent saturated hydrocarbon group having 1 to 10 carbon atoms. Examples of the chain-like monovalent saturated hydrocarbon group having 1 to 10 carbon atoms as Ra<sup>112</sup> and Ra<sup>113</sup> include those exemplified as the chain-like monovalent saturated hydrocarbon group having 1 to 10 carbon atoms as Ra<sup>101</sup> to Ra<sup>103</sup>. Some or all hydrogen atoms in this chain-like saturated hydrocarbon group may be substituted.

[0171] Ra<sup>112</sup> and Ra<sup>113</sup> represent preferably a hydrogen atom or an alkyl group having 1 to 5 carbon atoms, more preferably an alkyl group having 1 to 5 carbon atoms, still more preferably a methyl group or an ethyl group, and particularly preferably a methyl group.

[0172] In a case where the chain-like saturated hydrocarbon group represented by Ra<sup>112</sup> and Ra<sup>113</sup> is substituted, examples of the substituent thereof include the same substituents as those for Ra<sup>x5</sup>.

[0173] In Formula (a1-r2-4), Ra<sup>114</sup> represents a hydrocarbon group which may have a substituent. Examples of the hydrocarbon group as Ra<sup>114</sup> include a linear or branched alkyl group and a cyclic hydrocarbon group.

[0174] The linear alkyl group as Ra<sup>114</sup> has preferably 1 to 5 carbon atoms, more preferably 1 to 4 carbon atoms, and still more preferably 1 or 2 carbon atoms. Specific examples thereof include a methyl group, an ethyl group, an n-propyl group, an n-butyl group, and an n-pentyl group. Among these, a methyl group, an ethyl group, or an n-butyl group is preferable, and a methyl group or an ethyl group is more preferable.

[0175] The branched alkyl group as Ra<sup>114</sup> has preferably 3 to 10 carbon atoms and more preferably 3 to 5 carbon atoms. Specific examples thereof include an isopropyl group, an isobutyl group, a tert-butyl group, an isopentyl group, a neopentyl group, a 1,1-diethylpropyl group, and a 2,2-dimethylbutyl group. Among these, an isopropyl group is preferable.

[0176] In a case where Ra<sup>114</sup> represents a cyclic hydrocarbon group, the hydrocarbon group may be an alicyclic hydrocarbon group or an aromatic hydrocarbon group and may be a polycyclic group or a monocyclic group.

[0177] The alicyclic hydrocarbon group which is a monocyclic group is preferably a group in which one hydrogen atom has been removed from a monocycloalkane. The monocycloalkane has preferably 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane.

[0178] The alicyclic hydrocarbon group which is a polycyclic group is preferably a group in which one hydrogen atom has been removed from a polycycloalkane. The polycycloalkane preferably has 7 to 12 carbon atoms, and specific examples thereof include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclododecane.

[0179] Examples of the aromatic hydrocarbon group as Ra<sup>114</sup> include the same groups as those for the aromatic hydrocarbon group as Ra<sup>104</sup>. Among these, Ra<sup>114</sup> represents preferably a group in which one or more hydrogen atoms have been removed from an aromatic hydrocarbon ring having 6 to 15 carbon atoms, more preferably a group in which one or more hydrogen atoms have been removed from benzene, naphthalene, anthracene, or phenanthrene, still

more preferably a group in which one or more hydrogen atoms have been removed from benzene, naphthalene, or anthracene, particularly preferably a group in which one or more hydrogen atoms have been removed from naphthalene or anthracene, and most preferably a group in which one or more hydrogen atoms have been removed from naphthalene.

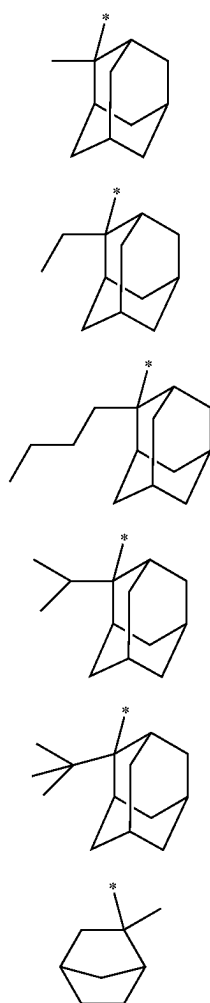
**[0180]** Examples of the substituent that  $Ra^{14}$  may have include the same groups as those for the substituent that  $Ra^{104}$  may have.

**[0181]** In a case where  $Ra^{14}$  in Formula (a1-r2-4) represents a naphthyl group, the position bonded to the tertiary carbon atom in Formula (a1-r2-4) may be the 1-position or the 2-position of the naphthyl group.

**[0182]** In a case where  $Ra^{14}$  in Formula (a1-r2-4) represents an anthryl group, the position bonded to the tertiary carbon atom in Formula (a1-r2-4) may be the 1-position, the 2-position, or the 9-position of the anthryl group.

**[0183]** Specific examples of the group represented by Formula (a1-r2-1) are shown below.

[Chemical Formula 17]



(r-pr-m1)

(r-pr-m2)

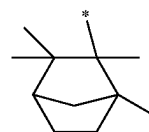
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(r-pr-m4)

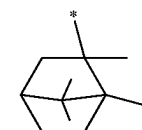
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(r-pr-m6)

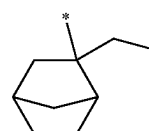
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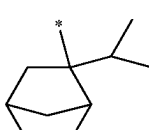
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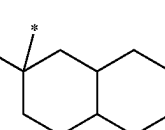
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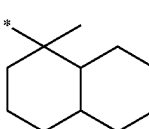
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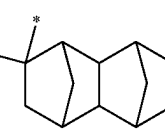
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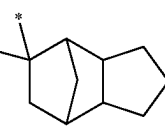
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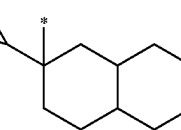
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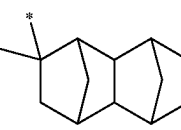
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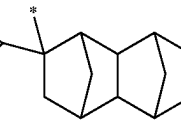
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(r-pr-m15)



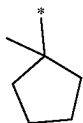
(r-pr-m16)



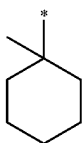
(r-pr-m17)

[Chemical Formula 18]

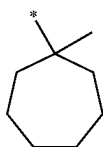
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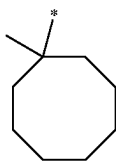
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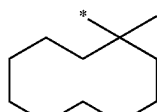
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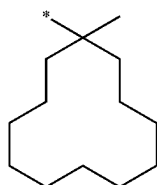
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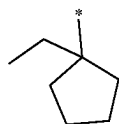
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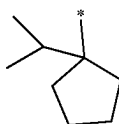
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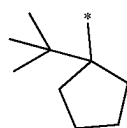
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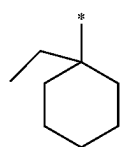
(r-pr-s7)



(r-pr-s8)

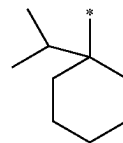


(r-pr-s9)

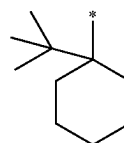


(r-pr-s10)

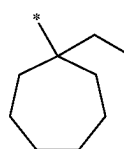
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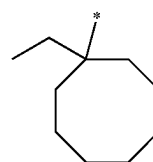
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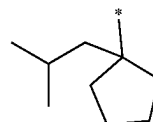
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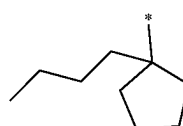
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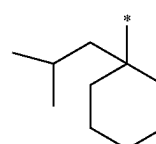
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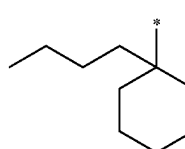
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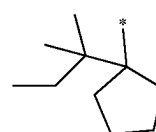
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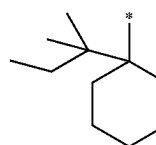
(r-pr-s17)



(r-pr-s18)



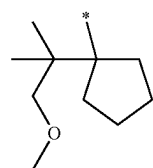
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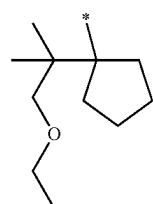
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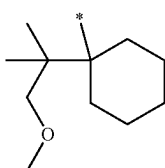
[Chemical Formula 19]



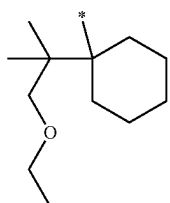
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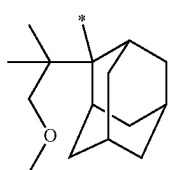
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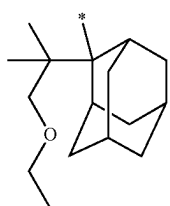
(r-pr-sp3)



(r-pr-sp4)



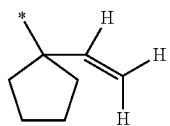
(r-pr-sp5)



(r-pr-sp6)

[0184] Specific examples of the group represented by Formula (a1-r2-2) are shown below.

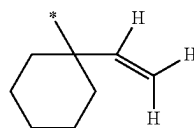
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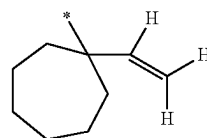
(r-pr-sv1)

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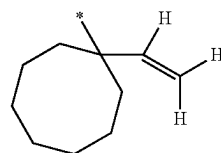
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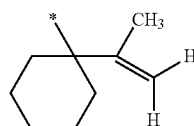
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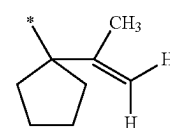
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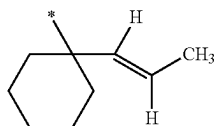
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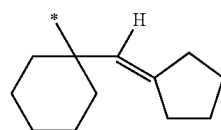
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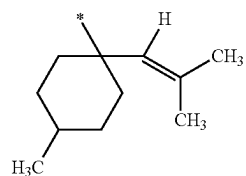
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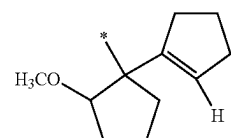
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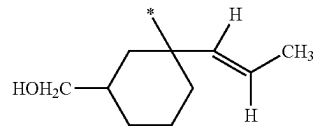
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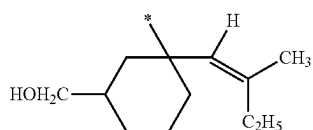
(r-pr-sv10)



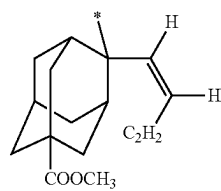
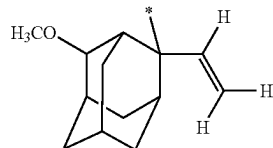
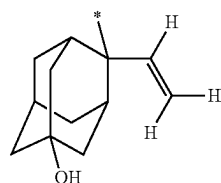
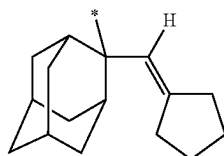
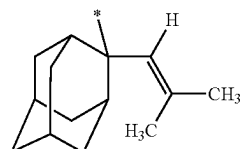
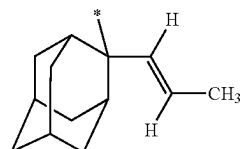
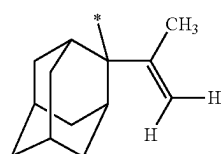
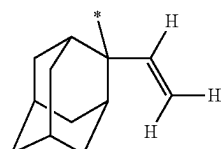
(r-pr-sv11)



-continued



[Chemical Formula 21]



(r-pr-sv12)

(r-pr-mv1)

(r-pr-mv2)

(r-pr-mv3)

(r-pr-mv4)

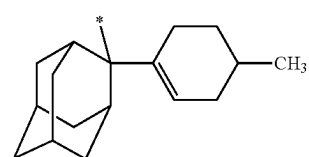
(p-pr-mv5)

(r-pr-mv6)

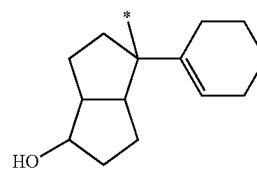
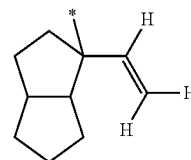
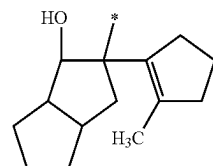
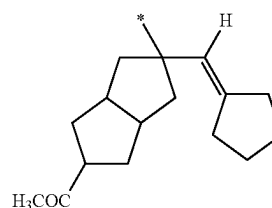
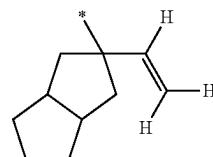
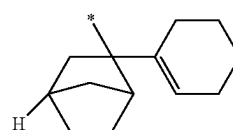
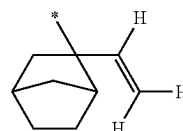
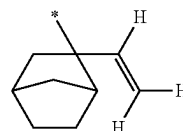
(r-pr-mv7)

(r-pr-mv8)

-continued



[Chemical Formula 22]



(r-pr-mv9)

(r-pr-mv10)

(r-pr-mv11)

(r-pr-mv12)

(r-pr-mv13)

(r-pr-mv14)

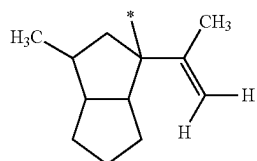
(r-pr-mv15)

(r-pr-mv-16)

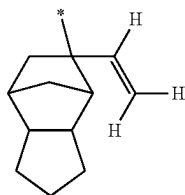
(r-pr-mv17)



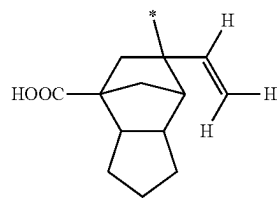
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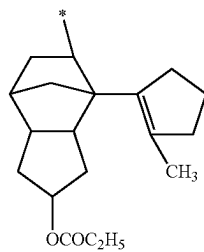
(r-pr-mv18)



(r-pr-mv19)



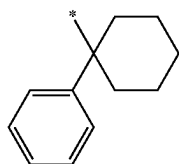
(r-pr-mv20)



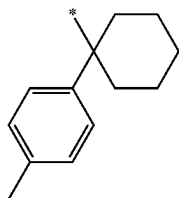
(r-pr-mv21)

[0185] Specific examples of the group represented by Formula (a1-r2-3) are shown below.

[Chemical Formula 23]

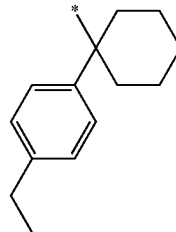


(r-pr-sa1)

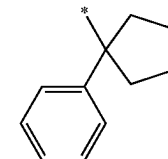


(r-pr-sa2)

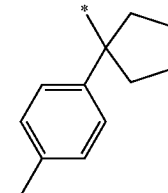
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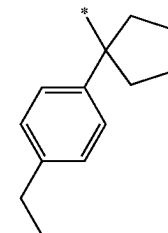
(r-pr-sa3)



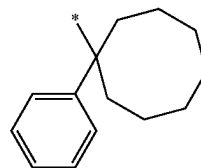
(r-pr-sa4)



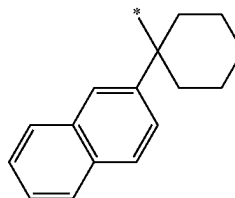
(r-pr-sa5)



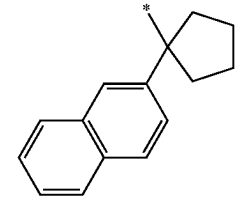
(r-pr-sa6)



(r-pr-sa7)

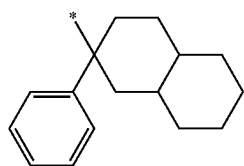


(r-pr-sa8)

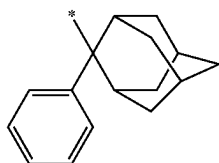


(r-pr-sa9)

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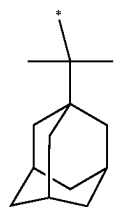
(r-pr-ma1)



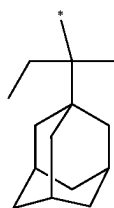
(r-pr-ma2)

[0186] Specific examples of the group represented by Formula (a1-r2-4) are shown below.

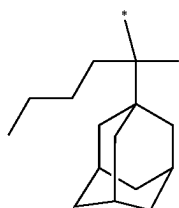
[Chemical Formula 24]



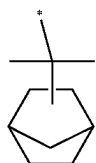
(r-pr-cm1)



(r-pr-cm2)

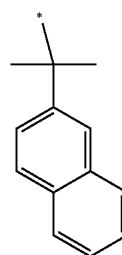


(r-pr-cm3)

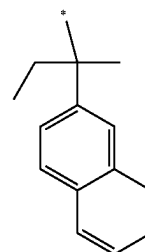


(r-pr-cm4)

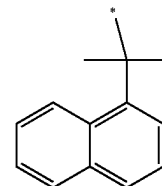
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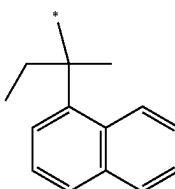
(r-pr-cm5)



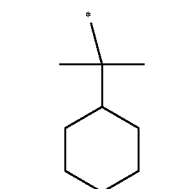
(r-pr-cm6)



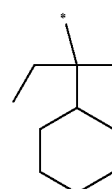
(r-pr-cm7)



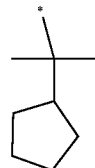
(r-pr-cm8)



(r-pr-cs1)



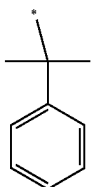
(r-pr-cs2)



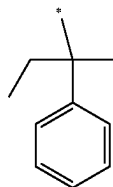
(r-pr-cs3)

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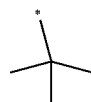
(r-pr-cs4)



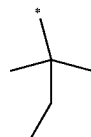
(r-pr-cs5)



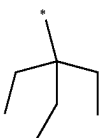
(r-pr-c1)



(r-pr-c2)



(r-pr-c3)

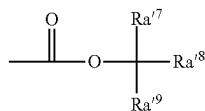


## Tertiary Alkyloxycarbonyl Acid Dissociable Group:

[0187] Examples of the acid dissociable group that protects a hydroxyl group among the polar groups include an acid dissociable group (hereinafter, also referred to as “tertiary alkyloxycarbonyl acid dissociable group” for convenience) represented by General Formula (a1-r-3).

[Chemical Formula 25]

(a1-r-3)



[0188] [In the formula,  $Ra^{7}$  to  $Ra^{9}$  each represent an alkyl group.]

[0189] In Formula (a1-r-3),  $Ra^{7}$  to  $Ra^{9}$  each represent preferably an alkyl group having 1 to 5 carbon atoms and more preferably an alkyl group having 1 to 3 carbon atoms.

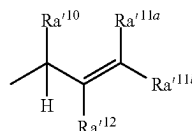
[0190] Further, the total number of carbon atoms in each alkyl group is preferably in a range of 3 to 7, more preferably in a range of 3 to 5, and most preferably 3 or 4.

## Secondary Alkyl Ester Type Acid Dissociable Group:

[0191] Examples of the acid dissociable group that protects a carboxy group among the polar groups include an acid dissociable group represented by General Formula (a1-r-4).

[Chemical Formula 26]

(a1-r-4)



[0192] [In the formula,  $Ra^{10}$  represents a hydrocarbon group.  $Ra^{11a}$  and  $Ra^{11b}$  each independently represent a hydrogen atom, a halogen atom, or an alkyl group.  $Ra^{112}$  represents a hydrogen atom or a hydrocarbon group.  $Ra^{10}$  and  $Ra^{11a}$  or  $Ra^{11b}$  may be bonded to each other to form a ring.  $Ra^{11a}$  or  $Ra^{11b}$  and  $Ra^{112}$  may be bonded to each other to form a ring.]

[0193] Examples of the hydrocarbon group as  $Ra^{10}$  or  $Ra^{112}$  in the formula include the same groups as those for  $Ra^{*3}$ .

[0194] Examples of the alkyl group as  $Ra^{11a}$  and  $Ra^{11b}$  in the formula include the same groups as those for the alkyl group as  $Ra^{*1}$ .

[0195] In the formula, the hydrocarbon group as  $Ra^{10}$  or  $Ra^{112}$  and the alkyl group as  $Ra^{11a}$  and  $Ra^{11b}$  may have a substituent. Examples of the substituent include  $Ra^{*5}$  described above.

[0196]  $Ra^{10}$  and  $Ra^{11a}$  or  $Ra^{11b}$  may be bonded to each other to form a ring. The ring may be a polycyclic ring or a monocyclic ring, and may be an alicyclic ring or an aromatic ring.

[0197] The alicyclic ring and the aromatic ring may have a heteroatom.

[0198] Among the examples described above, as the ring formed by  $Ra^{10}$  and  $Ra^{11a}$  or  $Ra^{11b}$  being bonded to each other, monocycloalkene, a ring in which some carbon atoms of monocycloalkene are substituted with heteroatoms (such as an oxygen atom and a sulfur atom), or monocycloalkadiene is preferable, cycloalkene having 3 to 6 carbon atoms is preferable, and cyclopentene or cyclohexene is preferable.

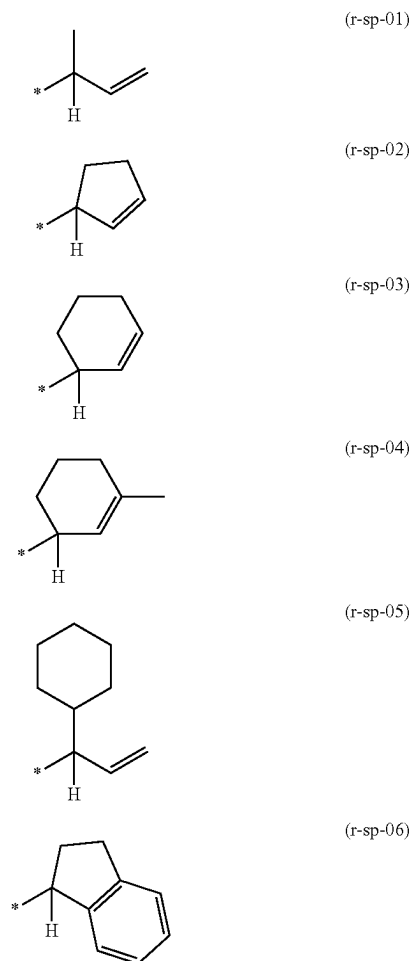
[0199] The ring formed by  $Ra^{10}$  and  $Ra^{11a}$  or  $Ra^{11b}$  being bonded to each other may be a condensed ring. Specific examples of the condensed ring include indane.

[0200] The ring formed by  $Ra^{10}$  and  $Ra^{11a}$  or  $Ra^{11b}$  being bonded to each other may have a substituent. Examples of the substituent include  $Ra^{*5}$  described above.

[0201]  $Ra^{11a}$  or  $Ra^{11b}$  and  $Ra^{112}$  may be bonded to each other to form a ring, and examples of the ring include the rings formed by  $Ra^{10}$  and  $Ra^{11a}$  or  $Ra^{11b}$  being bonded to each other.

[0202] Specific examples of the group represented by Formula (a1-r-4) are shown below.

[Chemical Formula 27]

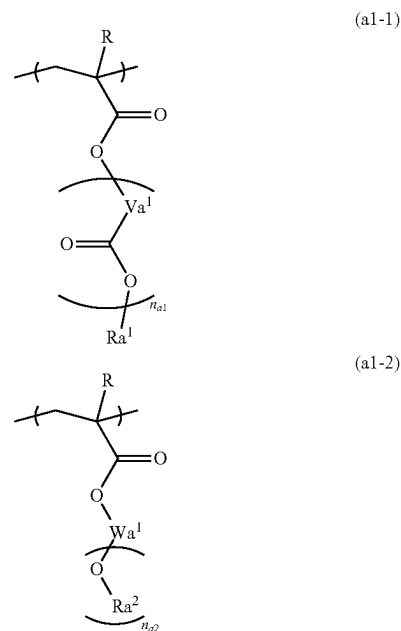


[0203] Examples of the constitutional unit (a1) include a constitutional unit derived from acrylic acid ester in which the hydrogen atom bonded to the carbon atom at the  $\alpha$ -position may be substituted with a substituent; a constitutional unit derived from acrylamide; a constitutional unit in which at least some hydrogen atoms in a hydroxyl group of a constitutional unit derived from hydroxystyrene or a hydroxystyrene derivative are protected by a substituent containing the acid decomposable group; and a constitutional unit in which at least some hydrogen atoms in  $\text{—C(=O)—OH}$  of a constitutional unit derived from vinylbenzoic acid or a vinylbenzoic acid derivative are protected by a substituent containing the acid decomposable group.

[0204] Among the examples, as the constitutional unit (a1), a constitutional unit derived from acrylic acid ester in which the hydrogen atom bonded to the carbon atom at the  $\alpha$ -position may be substituted with a substituent is preferable.

[0205] Specific preferred examples of such a constitutional unit (a1) include constitutional units represented by General Formula (a1-1) or (a1-2) shown below.

[Chemical Formula 28]



[0206] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms.  $\text{Va}^1$  represents a divalent hydrocarbon group which may have an ether bond.  $n_{a1}$  represents an integer of 0 to 2.  $\text{Ra}^1$  represents an acid dissociable group represented by General Formula (a1-r-1) or (a1-r-2).  $\text{Wa}^1$  represents a  $(n_{a2}+1)$ -valent hydrocarbon group,  $n_{a2}$  represents an integer of 1 to 3, and  $\text{Ra}^2$  represents an acid dissociable group represented by General Formula (a1-r-1) or (a1-r-3).]

[0207] In Formula (a1-1), as the alkyl group having 1 to 5 carbon atoms as R, a linear or branched alkyl group having 1 to 5 carbon atoms is preferable, and specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group. The halogenated alkyl group having 1 to 5 carbon atoms is a group in which some or all hydrogen atoms in the alkyl group having 1 to 5 carbon atoms have been substituted with halogen atoms. As the halogen atom, a fluorine atom is particularly preferable.

[0208] R represents preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms, and most preferably a hydrogen atom or a methyl group from the viewpoint of the industrial availability.

[0209] In Formula (a1-1), the divalent hydrocarbon group as  $\text{Va}^1$  may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group.

[0210] The aliphatic hydrocarbon group as the divalent hydrocarbon group represented by  $\text{Va}^1$  may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated.

[0211] More specific examples of the aliphatic hydrocarbon group include a linear or branched aliphatic hydrocarbon group and an aliphatic hydrocarbon group having a ring in the structure thereof.

[02112] The linear aliphatic hydrocarbon group has preferably 1 to 10 carbon atoms, more preferably 1 to 6 carbon atoms, still more preferably 1 to 4 carbon atoms, and most preferably 1 to 3 carbon atoms.

[02113] As the linear aliphatic hydrocarbon group, a linear alkylene group is preferable, and specific examples thereof include a methylene group  $[-CH_2-]$ , an ethylene group  $[-(CH_2)_2-]$ , a trimethylene group  $[-(CH_2)_3-]$ , a tetramethylene group  $[-(CH_2)_4-]$ , and a pentamethylene group  $[-(CH_2)_5-]$ .

[02114] The branched aliphatic hydrocarbon group has preferably 2 to 10 carbon atoms, more preferably 3 to 6 carbon atoms, still more preferably 3 or 4 carbon atoms, and most preferably 3 carbon atoms.

[02115] As the branched aliphatic hydrocarbon group, a branched alkylene group is preferable, and specific examples thereof include alkylalkylene groups, for example, alkylmethylenes groups such as  $-CH(CH_3)-$ ,  $-CH(CH_2CH_3)-$ ,  $-C(CH_3)_2-$ ,  $-C(CH_3)(CH_2CH_3)-$ ,  $-C(CH_3)(CH_2CH_2CH_3)-$ , and  $-C(CH_2CH_3)_2-$ ; alkyl-ethylene groups such as  $-CH(CH_3)CH_2-$ ,  $-CH(CH_3)CH(CH_3)-$ ,  $-C(CH_3)_2CH_2-$ ,  $-CH(CH_2CH_3)CH_2-$ , and  $-C(CH_2CH_3)_2CH_2-$ ; alkyltrimethylene groups such as  $-CH(CH_3)CH_2CH_2-$  and  $-CH_2CH(CH_3)CH_2-$ ; and alkyltetramethylene groups such as  $-CH(CH_3)CH_2CH_2CH_2-$  and  $-CH_2CH(CH_3)CH_2CH_2-$ . As the alkyl group in the alkylalkylene group, a linear alkyl group having 1 to 5 carbon atoms is preferable.

[02116] Examples of the aliphatic hydrocarbon group having a ring in the structure thereof include an alicyclic hydrocarbon group (a group in which two hydrogen atoms have been removed from an aliphatic hydrocarbon ring), a group in which the alicyclic hydrocarbon group is bonded to the terminal of the linear or branched aliphatic hydrocarbon group, and a group in which the alicyclic hydrocarbon group is interposed in the middle of the linear or branched aliphatic hydrocarbon group. Examples of the linear or branched aliphatic hydrocarbon group include the same groups as those for the linear aliphatic hydrocarbon group or the branched aliphatic hydrocarbon group.

[02117] The alicyclic hydrocarbon group has preferably 3 to 20 carbon atoms and more preferably 3 to 12 carbon atoms.

[02118] The alicyclic hydrocarbon group may be monocyclic or polycyclic. The monocyclic alicyclic hydrocarbon group is preferably a group obtained by removing two hydrogen atoms from a monocycloalkane. The monocycloalkane has preferably 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane. As the polycyclic alicyclic hydrocarbon group, a group in which two hydrogen atoms have been removed from a polycycloalkane is preferable. As the polycycloalkane, a group having 7 to 12 carbon atoms is preferable. Specific examples thereof include adamantane, norbornane, isobornane, tricyclodecane, and tetracyclodecane.

[02119] The aromatic hydrocarbon group as the divalent hydrocarbon group represented by  $Va^1$  is a hydrocarbon group having an aromatic ring.

[0220] The aromatic hydrocarbon group has preferably 3 to 30 carbon atoms, more preferably 5 to 30 carbon atoms, still more preferably 5 to 20 carbon atoms, particularly preferably 6 to 15 carbon atoms, and most preferably 6 to 12 carbon atoms. Here, the number of carbon atoms in a substituent is not included in the number of carbon atoms.

[0221] Specific examples of the aromatic ring contained in the aromatic hydrocarbon group include aromatic hydrocarbon rings such as benzene, biphenyl, fluorene, naphthalene, anthracene, and phenanthrene; and aromatic heterocyclic rings in which some carbon atoms constituting the above-described aromatic hydrocarbon rings have been substituted with heteroatoms. Examples of the heteroatom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom.

[0222] Specific examples of the aromatic hydrocarbon group include a group in which two hydrogen atoms have been removed from the above-described aromatic hydrocarbon ring (an arylene group); and a group in which one hydrogen atom of a group (an aryl group) formed by removing one hydrogen atom from the aromatic hydrocarbon ring has been substituted with an alkylene group (for example, a group formed by further removing one more hydrogen atom from an aryl group in an arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, a 1-naphthylethyl group, or a 2-naphthylethyl group). The alkylene group (alkyl chain in the arylalkyl group) has preferably 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and particularly preferably 1 carbon atom.

[0223] In Formula (a1-1),  $Ra^1$  represents an acid dissociable group represented by Formula (a1-r-1) or (a1-r-2).

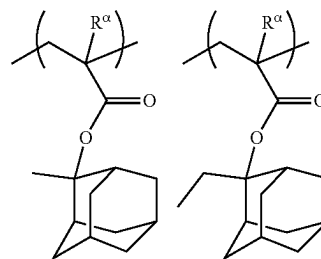
[0224] In Formula (a1-2), the  $(n_{a2}+1)$ -valent hydrocarbon group as  $Wa^1$  may be an aliphatic hydrocarbon group or an aromatic hydrocarbon group. The aliphatic hydrocarbon group indicates a hydrocarbon group that has no aromaticity and may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated. Examples of the aliphatic hydrocarbon group include a linear or branched aliphatic hydrocarbon group, an aliphatic hydrocarbon group having a ring in the structure thereof, and a group obtained by combining the linear or branched aliphatic hydrocarbon group and the aliphatic hydrocarbon group having a ring in the structure thereof.

[0225] The valency of  $n_{a2}+1$  is preferably divalent to tetravalent and more preferably divalent or trivalent.

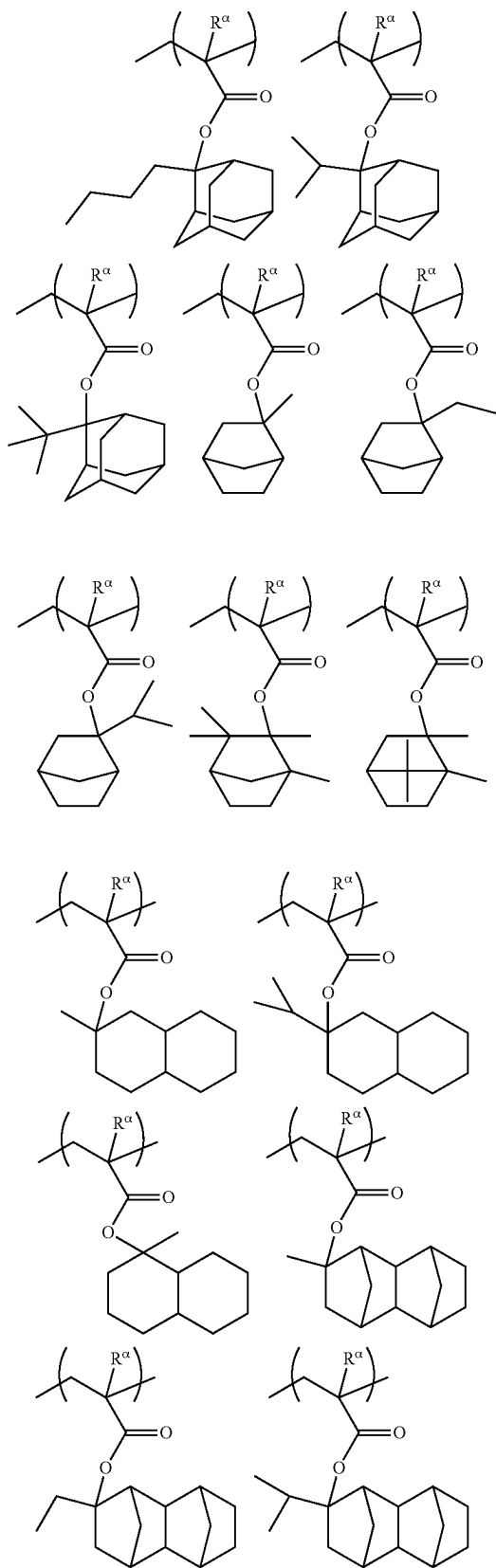
[0226] In the Formula (a1-2),  $Ra^2$  represents an acid dissociable group represented by General Formula (a1-r-1) or (a1-r-3).

[0227] Specific examples of the constitutional unit (a1) are shown below. In the formulae shown below,  $R^\alpha$  represents a hydrogen atom, a methyl group, or a trifluoromethyl group.

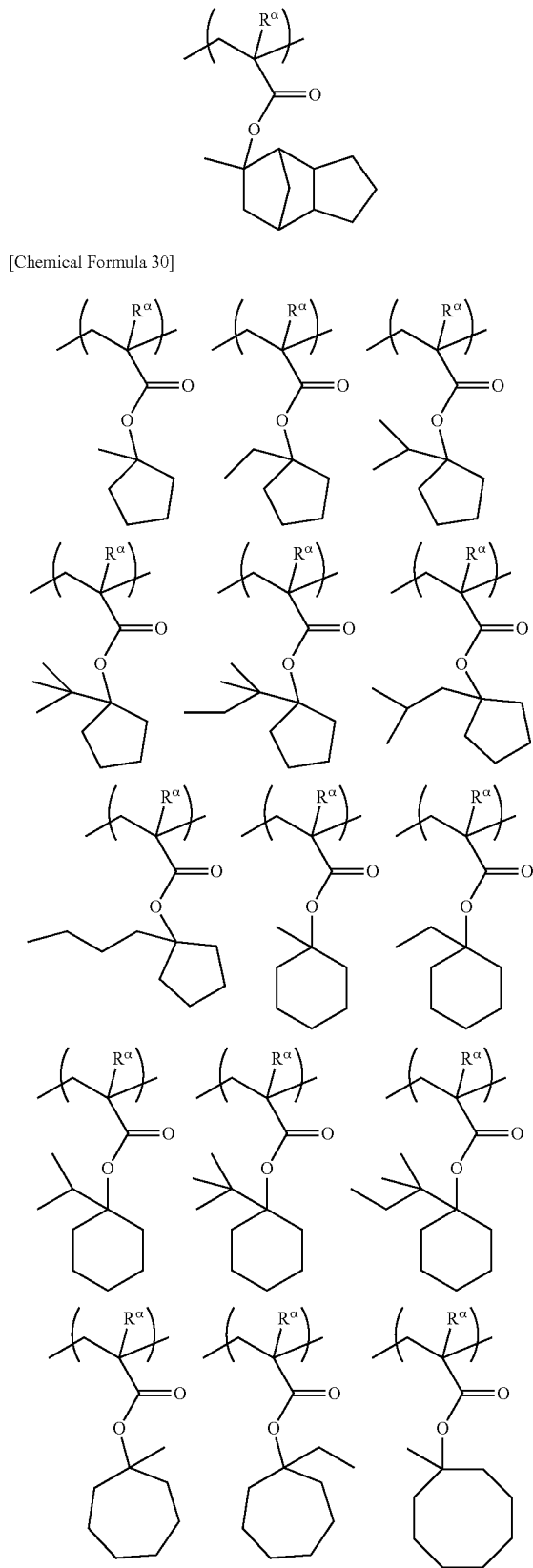
[Chemical Formula 29]



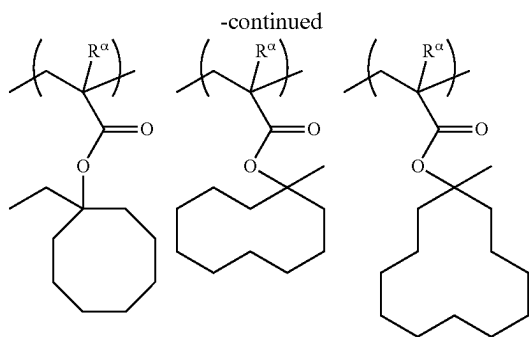
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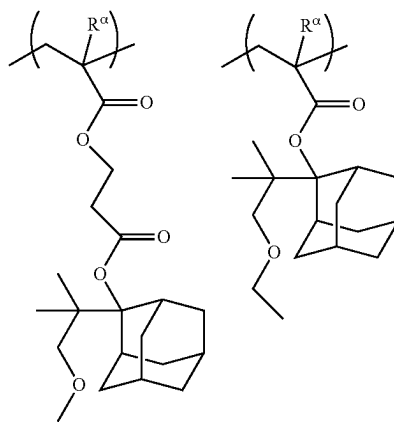
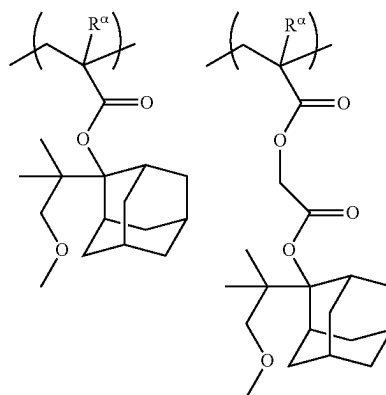
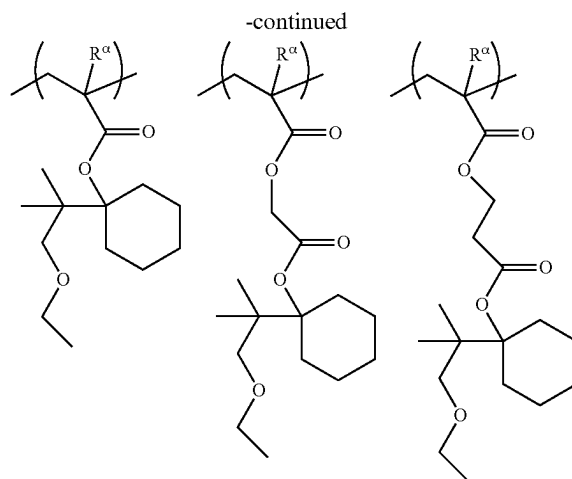
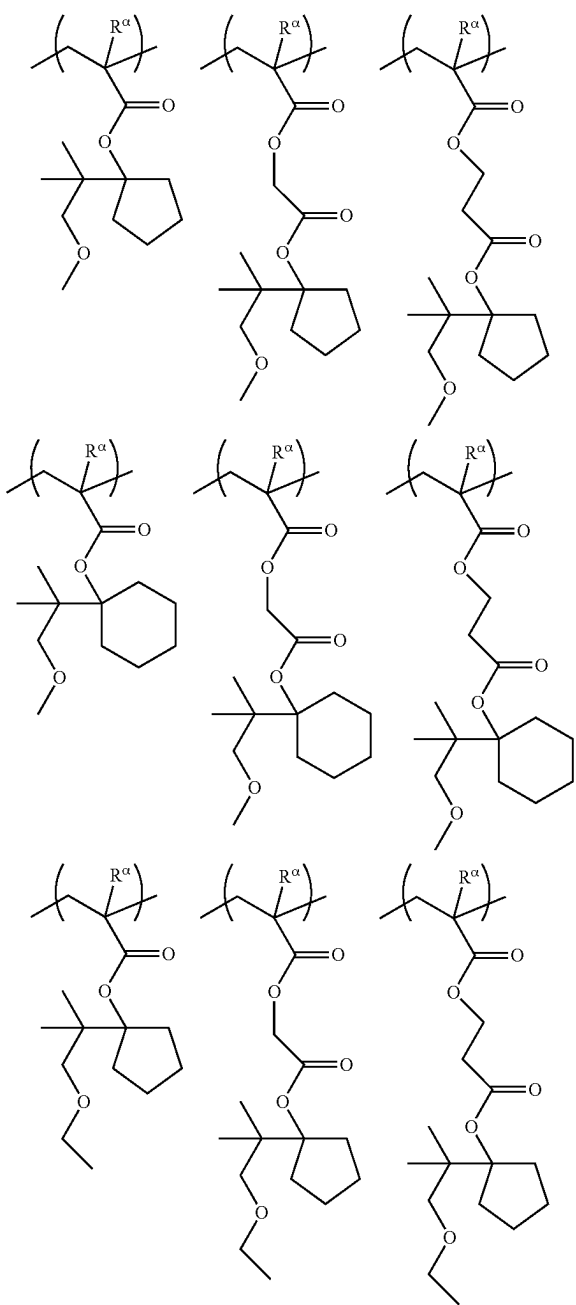
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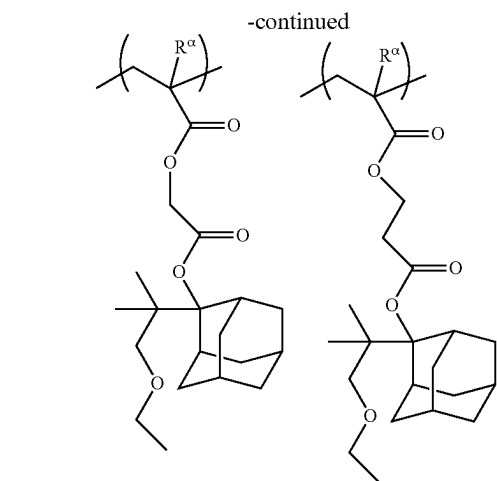


[Chemical Formula 30]

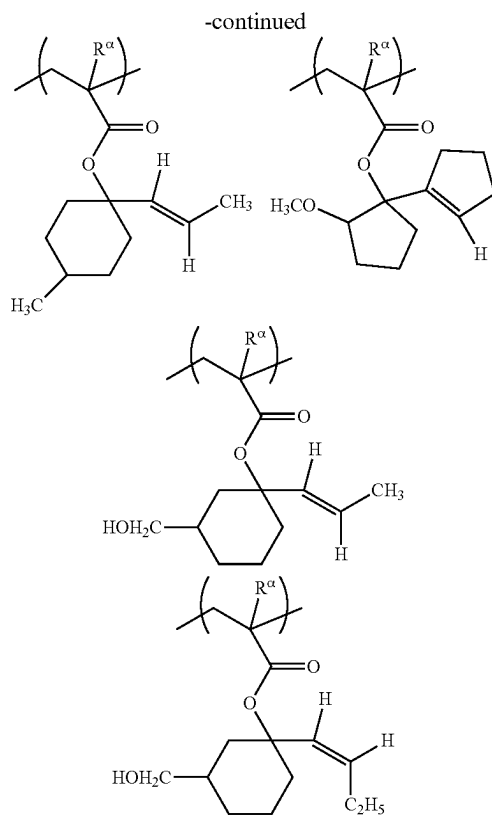
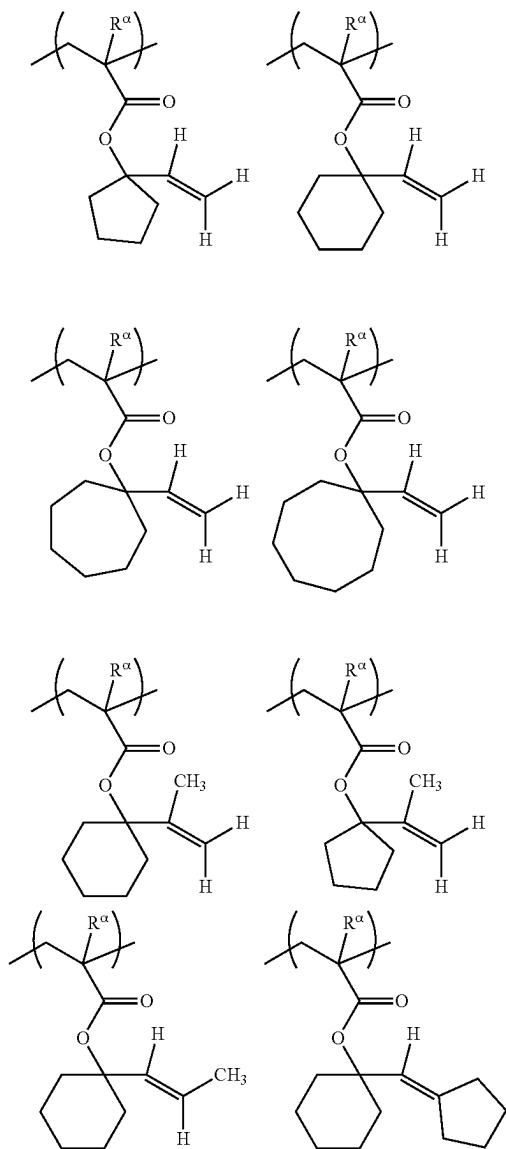


[Chemical Formula 31]

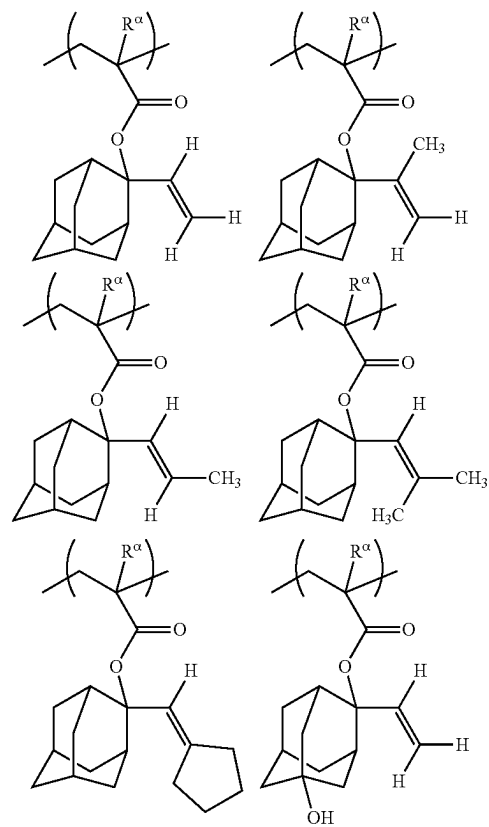




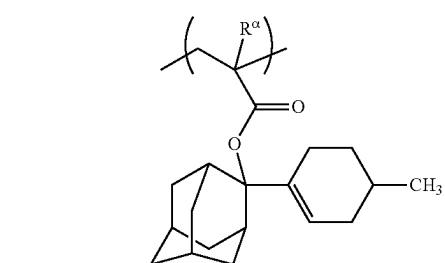
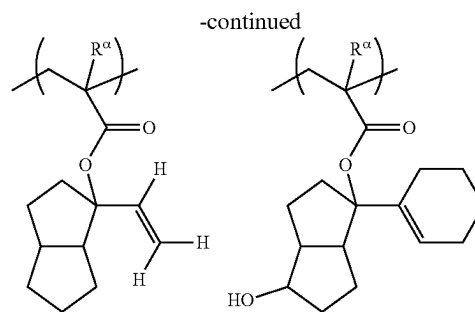
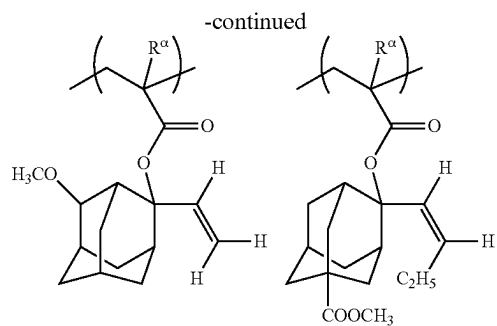
[Chemical Formula 32]



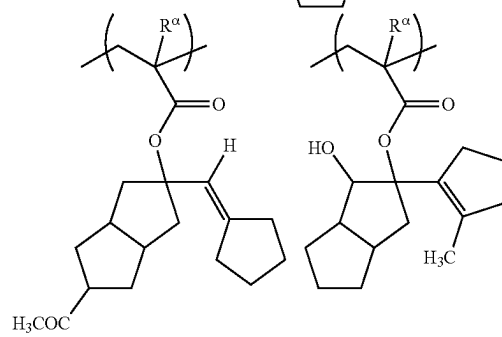
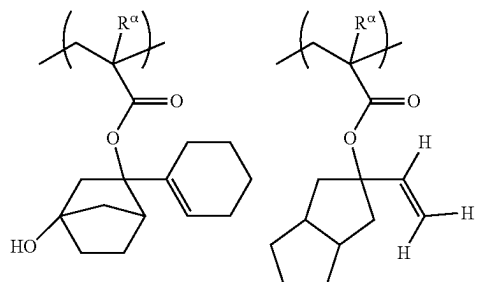
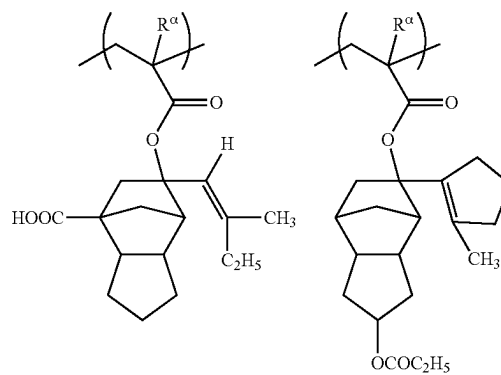
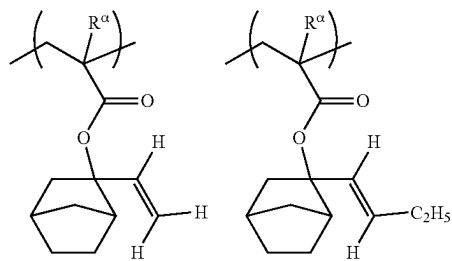
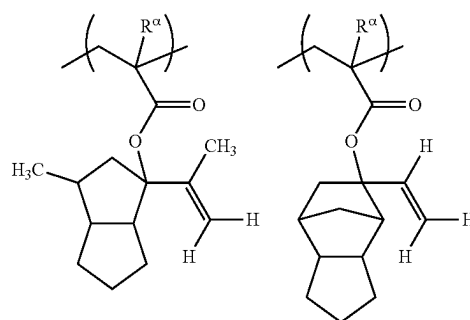
[Chemical Formula 33]



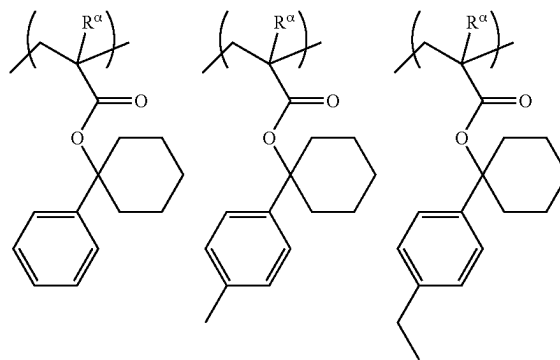


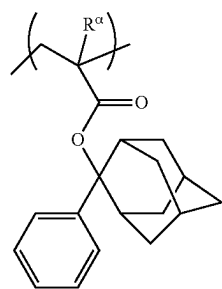
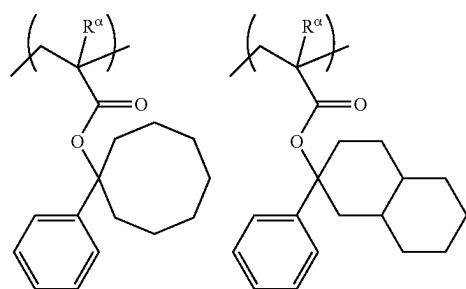
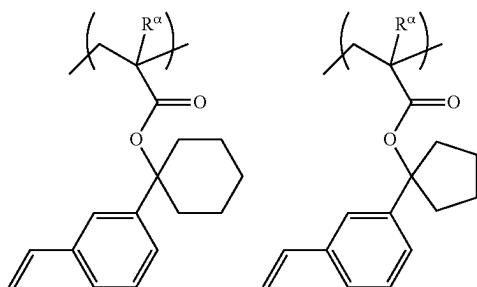
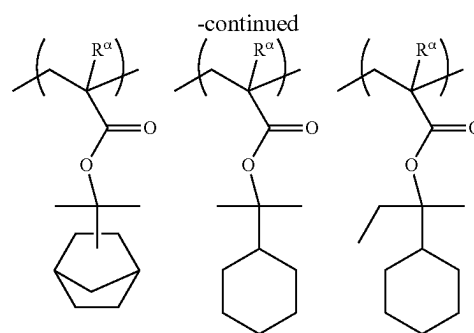
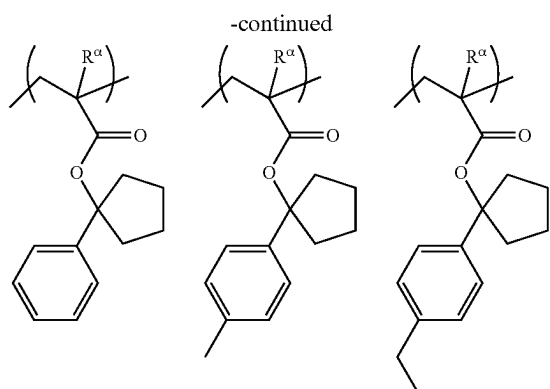


[Chemical Formula 34]

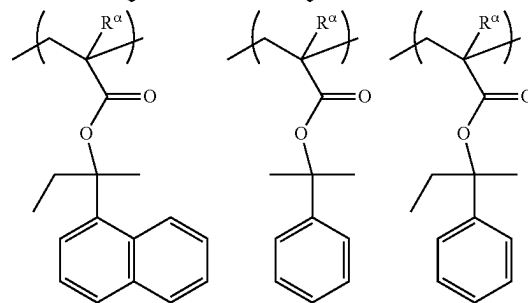
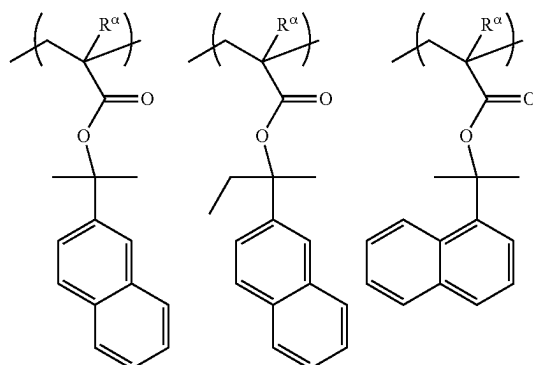
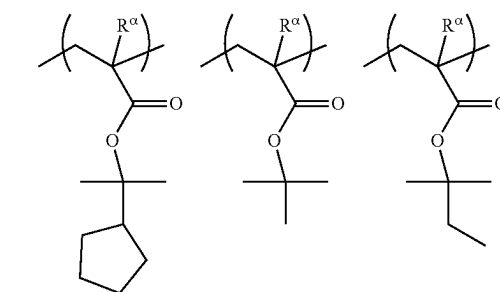
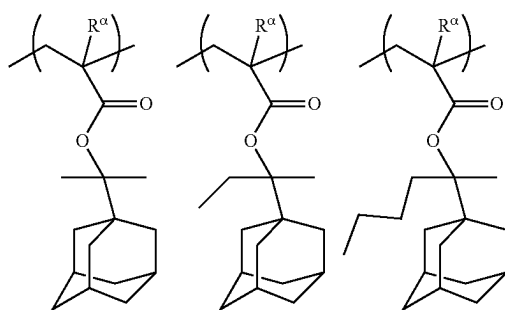


[Chemical Formula 35]





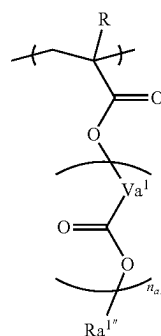
[Chemical Formula 36]



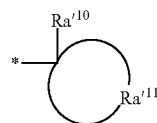
**[0228]** The constitutional unit (a1) included in the component (A1) may be used alone or two or more kinds thereof.

**[0229]** Since the lithography characteristics (CDU and the like) are easily improved using electron beams or EUV, a constitutional unit represented by Formula (a1-1) is more preferable as the constitutional unit (a1).

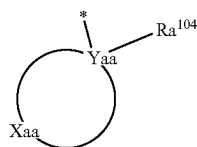
[Chemical Formula 37]



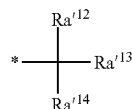
(a1-1-1)



(a1-r2-1)



(a1-r2-3)



(a1-r2-4)

[0230] [In the formulae,  $Ra^{1m}$  represents an acid dissociable group represented by General Formula (a1-r2-1), (a1-r2-3), or (a1-r2-4). \* represents a bonding site.]

[0231]  $R$ ,  $Va^1$ , and  $n_{a1}$  in Formula (a1-1-1) each have the same definition as that for  $R$ ,  $Va^1$ , and  $n_{a1}$  in Formula (a1-1).

[0232] The description of the acid dissociable group represented by General Formula (a1-r2-1), (a1-r2-3), or (a1-r2-4) is the same as described above. Among these, from the viewpoint of suitably increasing the reactivity in the application for EB or EUV, an acid dissociable group which is a cyclic group is preferably selected, and an acid dissociable group represented by General Formula (a1-r2-1) is more preferable.

[0233] In a case where the component (A1) has the constitutional unit (a1), the proportion of the constitutional unit (a1) in the component (A1) is preferably in a range of 5% to 95% by mole, more preferably in a range of 10% to 90% by mole, still more preferably in a range of 30% to 70% by mole, and particularly preferably in a range of 40% to 60% by mole with respect to the total amount (100% by mole) of all constitutional units constituting the component (A1).

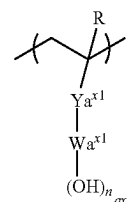
[0234] In a case where the proportion of the constitutional unit (a1) is set to be greater than or equal to the lower limits of the above-described preferable ranges, lithography characteristics such as the sensitivity, CDU, the resolution, and reduction of the roughness are improved. Further, in a case where the proportion of the constitutional unit (a1) is less than or equal to the upper limits of the above-described

preferable ranges, the constitutional unit (a1) and other constitutional units can be balanced, and the lithography characteristics are improved.

In Regard to Constitutional Unit (a10):

[0235] The constitutional unit (a10) is a constitutional unit represented by General Formula (a10-1) (here, a constitutional unit corresponding to the constitutional unit (a1) is excluded).

[Chemical Formula 38]



(a10-1)

[0236] [In the formula,  $R$  represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms.  $Ya^{x1}$  represents a single bond or a divalent linking group.  $Wa^{x1}$  represents an aromatic hydrocarbon group which may have a substituent.  $n_{ax1}$  represents an integer of 1 or greater.]

[0237] In the Formula (a10-1),  $R$  represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms.

[0238]  $R$  represents preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms, and from the viewpoint of industrial availability, more preferably a hydrogen atom, a methyl group, or trifluoromethyl group, still more preferably a hydrogen atom or a methyl group, and particularly preferably a hydrogen atom.

[0239] In Formula (a10-1),  $Ya^{x1}$  represents a single bond or a divalent linking group.

[0240] In the chemical formula, the divalent linking group as  $Ya^{x1}$  is not particularly limited, and suitable examples thereof include a divalent hydrocarbon group which may have a substituent and a divalent linking group having a heteroatom.

[0241]  $Ya^{x1}$  represents preferably a single bond, an ester bond [ $-C(=O)-O-$  or  $-O-C(=O)-$ ], an ether bond ( $-O-$ ), a linear or branched alkylene group, or a combination thereof and more preferably a single bond or an ester bond [ $-C(=O)-O-$  or  $-O-C(=O)-$ ].

[0242] In Formula (a10-1),  $Wa^{x1}$  represents an aromatic hydrocarbon group which may have a substituent.

[0243] Examples of the aromatic hydrocarbon group as  $Wa^{x1}$  include a group in which  $(n_{ax1}+1)$  hydrogen atoms have been removed from an aromatic ring which may have a substituent. The aromatic ring is not particularly limited as long as the aromatic ring is a cyclic conjugated system having  $(4n+2)$   $\pi$  electrons. The aromatic ring has preferably 5 to 30 carbon atoms, more preferably 5 to 20 carbon atoms, still more preferably 6 to 15 carbon atoms, and particularly preferably 6 to 12 carbon atoms. Specific examples of the aromatic ring include aromatic hydrocarbon rings such as benzene, naphthalene, anthracene, and phenanthrene; and an aromatic heterocyclic ring obtained by substituting some carbon atoms constituting the above-described aromatic

hydrocarbon ring with a heteroatom. Examples of the heteroatom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom. Specific examples of the aromatic heterocyclic ring include a pyridine ring and a thiophene ring.

**[0244]** Further, examples of the aromatic hydrocarbon group as  $Wa^{x1}$  also include a group in which  $(n_{ax1}+1)$  hydrogen atoms have been removed from an aromatic compound having an aromatic ring (for example, biphenyl or fluorene) which may have two or more substituents.

**[0245]** Among the examples,  $Wa^{x1}$  represents preferably a group in which  $(n_{ax1}+1)$  hydrogen atoms have been removed from benzene, naphthalene, anthracene, or biphenyl, more preferably a group in which  $(n_{ax1}+1)$  hydrogen atoms have been removed from benzene or naphthalene, and still more preferably a group in which  $(n_{ax1}+1)$  hydrogen atoms have been removed from benzene.

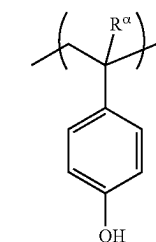
**[0246]** The aromatic hydrocarbon group as  $Wa^{x1}$  may or may not have a substituent. Examples of substituents include an alkyl group, an alkoxy group, a halogen atom, and a halogenated alkyl group. Examples of the alkyl group, the alkoxy group, the halogen atom, and the halogenated alkyl group as the substituent include the same groups as those for the substituent of the cyclic alicyclic hydrocarbon group as  $Ya^{x1}$ . The substituent is preferably a linear or branched alkyl group having 1 to 5 carbon atoms, more preferably a linear or branched alkyl group having 1 to 3 carbon atoms, still more preferably an ethyl group or a methyl group, and particularly preferably a methyl group. It is preferable that the aromatic hydrocarbon group as  $Wa^{x1}$  has no substituent.

**[0247]** In Formula (a10-1),  $n_{ax1}$  represents an integer of 1 or greater, preferably an integer of 1 to 10, more preferably an integer of 1 to 5, still more preferably 1, 2, or 3, and particularly preferably 1 or 2.

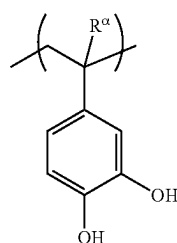
**[0248]** Specific examples of the constitutional unit (a10) represented by Formula (a10-1) are described below.

**[0249]** In the formulae shown below,  $R^{\alpha}$  represents a hydrogen atom, a methyl group, or a trifluoromethyl group.

[Chemical Formula 39]

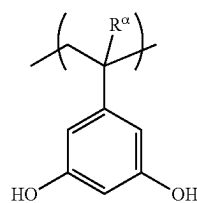


(a10-1-1)

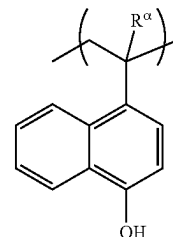


(a10-1-2)

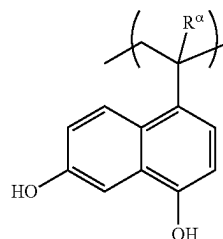
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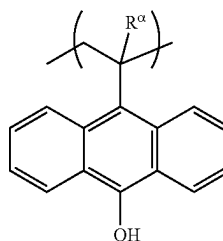
(a10-1-3)



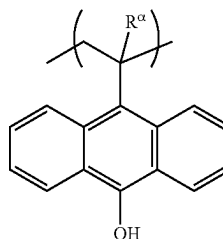
(a10-1-4)



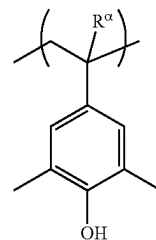
(a10-1-5)



(a10-1-6)

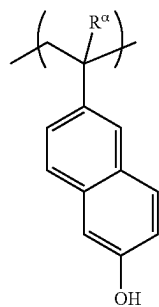


(a10-1-7)

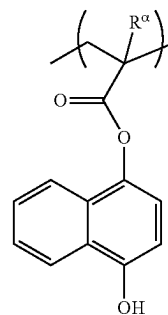
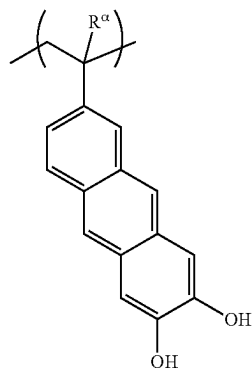
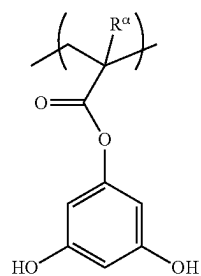
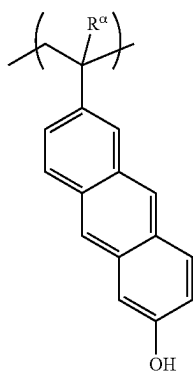
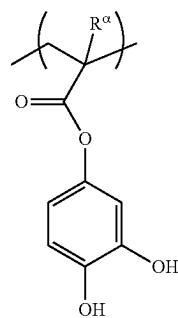
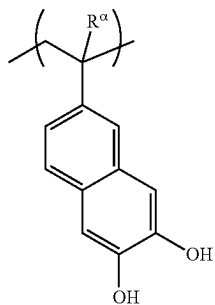
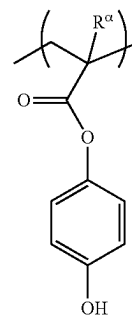


(a10-1-8)

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[Chemical Formula 40]

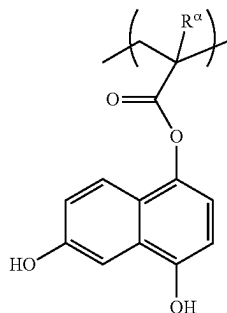


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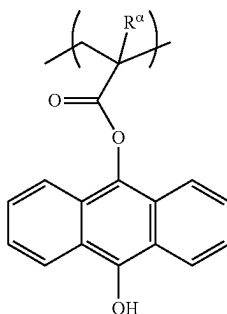


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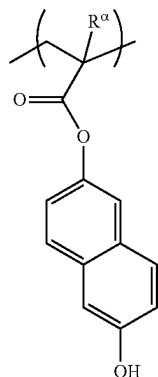
[Chemical Formula 41]



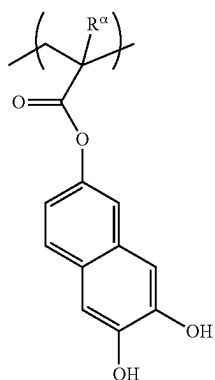
(a10-1-17)



(a10-1-18)



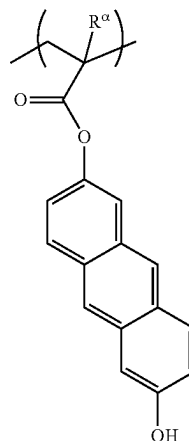
(a10-1-19)



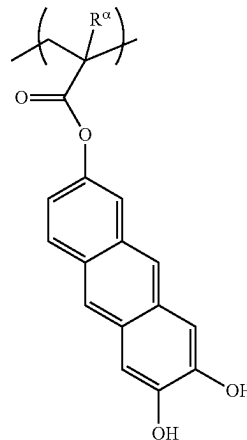
(a10-1-20)

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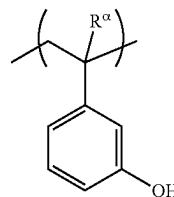
(a10-1-21)



(a10-1-22)



(a10-1-23)



**[0250]** The constitutional unit (a10) included in the component (A1) may be used alone or two or more kinds thereof.

**[0251]** In a case where the component (A1) has the constitutional unit (a10), the proportion of the constitutional unit (a10) in the component (A1) is preferably in a range of 20% to 60% by mole, more preferably in a range of 25% to 55% by mole, still more preferably in a range of 30% to 50% by mole, and particularly preferably in a range of 35% to 45% by mole with respect to the total amount (100% by mole) of all constitutional units constituting the component (A1).

**[0252]** In a case where the proportion of the constitutional unit (a10) is set to be greater than or equal to the above-described lower limits, the sensitivity is likely to be enhanced. Meanwhile, in a case where the proportion thereof is set to be less than or equal to the above-described upper limits, the constitutional unit (a10) and other constitutional units are likely to be balanced.

In Regard to Constitutional Unit (a2):

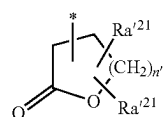
**[0253]** The component (A1) may further have a constitutional unit (a2) (here, a constitutional unit corresponding to the constitutional unit (a1) is excluded) containing a lactone-containing cyclic group.

**[0254]** In a case where the component (A1) is used to form a resist film, the lactone-containing cyclic group of the constitutional unit (a2) is effective for increasing the adhesiveness of the resist film to the substrate. Further, in a case where the component (A1) contains the constitutional unit (a2), the lithography characteristics and the like are improved due to the effects of appropriately adjusting the acid diffusion length, increasing the adhesiveness of the resist film to the substrate, and appropriately adjusting the solubility during the development.

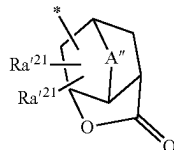
**[0255]** The term “lactone-containing cyclic group” indicates a cyclic group that has a ring (lactone ring) containing  $\text{—O—C(=O)—}$  in the ring skeleton. In a case where the lactone ring is counted as the first ring and the group contains only the lactone ring, the group is referred to as a monocyclic group. Further, in a case where the group has other ring structures, the group is referred to as a polycyclic group regardless of the structures. The lactone-containing cyclic group may be a monocyclic group or a polycyclic group.

**[0256]** The lactone-containing cyclic group in the constitutional unit (a2) is not particularly limited, and an optional constitutional unit can be used. Specific examples thereof include groups each represented by General Formulae (a2-r-1) to (a2-r-7).

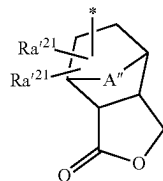
[Chemical Formula 42]



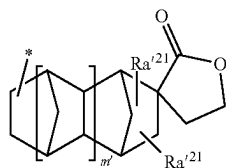
(a2-r-1)



(a2-r-2)

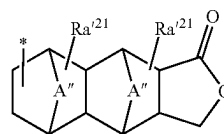


(a2-r-3)

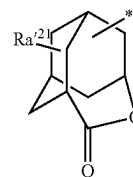


(a2-r-4)

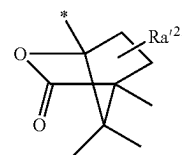
-continued



(a2-r-5)



(a2-r-6)



(a2-r-7)

**[0257]** [In the formulae,  $\text{Ra}'^{21}$ 's each independently represent a hydrogen atom, an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group,  $\text{—COOR''}$ ,  $\text{—OC(=O)R''}$ , a hydroxyalkyl group, or a cyano group,  $\text{R''}$  represents a hydrogen atom, an alkyl group, or a lactone-containing cyclic group,  $\text{A''}$  represents an alkylene group having 1 to 5 carbon atoms which may have an oxygen atom ( $\text{—O—}$ ) or a sulfur atom ( $\text{—S—}$ ), an oxygen atom, or a sulfur atom,  $n'$  represents an integer of 0 to 2, and  $m'$  is 0 or 1. \* represents a bonding site (the same applies hereinafter).]

**[0258]** In General Formulae (a2-r-1) to (a2-r-7), it is preferable that the alkyl group as  $\text{Ra}'^{21}$  is an alkyl group having 1 to 6 carbon atoms. It is preferable that the alkyl group is linear or branched. Specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, a neopentyl group, and a hexyl group. Among these, a methyl group or ethyl group is preferable, and a methyl group is particularly preferable.

**[0259]** It is preferable that the alkoxy group as  $\text{Ra}'^{21}$  is an alkoxy group having 1 to 6 carbon atoms. Further, it is preferable that the alkoxy group is linear or branched. Specific examples of the alkoxy groups include a group formed by linking the above-described alkyl group exemplified as the alkyl group represented by  $\text{Ra}'^{21}$  to an oxygen atom ( $\text{—O—}$ ).

**[0260]** As the halogen atom as  $\text{Ra}'^{21}$ , a fluorine atom is preferable.

**[0261]** Examples of the halogenated alkyl group as  $\text{Ra}'^{21}$  include groups in which some or all hydrogen atoms in the alkyl group as  $\text{Ra}'^{21}$  are substituted with the halogen atoms. As the halogenated alkyl group, a fluorinated alkyl group is preferable, and a perfluoroalkyl group is particularly preferable.

**[0262]** In  $\text{—COOR''}$  and  $\text{—OC(=O)R''}$  as  $\text{Ra}'^{21}$ , any  $\text{R''}$  represents a hydrogen atom, an alkyl group, or a lactone-containing cyclic group.

**[0263]** The alkyl group as  $\text{R''}$  may be linear, branched, or cyclic, and it preferably has 1 to 15 carbon atoms.

[0264] In a case where R" represents a linear or branched alkyl group, an alkyl group having 1 to 10 carbon atoms is preferable, an alkyl group having 1 to 5 carbon atoms is more preferable, and a methyl group or an ethyl group is particularly preferable.

[0265] In a case where R" represents a cyclic alkyl group, the number of carbon atoms thereof is preferably in a range of 3 to 15, more preferably in a range of 4 to 12, and most preferably in a range of 5 to 10. Specific examples thereof include groups in which one or more hydrogen atoms have been removed from a monocycloalkane, which may or may not be substituted with a fluorine atom or a fluorinated alkyl group; and groups in which one or more hydrogen atoms have been removed from a polycycloalkane such as bicycloalkane, tricycloalkane, or tetracycloalkane. More specific examples thereof include groups in which one or more hydrogen atoms have been removed from a monocycloalkane such as cyclopentane or cyclohexane; and groups in which one or more hydrogen atoms have been removed from a polycycloalkane such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane.

[0266] Examples of the lactone-containing cyclic group as R" include the same groups as those for the groups each represented by General Formulae (a2-r-1) to (a2-r-7).

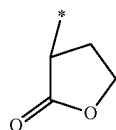
[0267] As the hydroxyalkyl group as Ra<sup>21</sup>, a hydroxyalkyl group having 1 to 6 carbon atoms is preferable, and specific examples thereof include a group in which at least one hydrogen atom in the alkyl group as Ra<sup>21</sup> has been substituted with a hydroxyl group.

[0268] Among the examples, it is preferable that Ra<sup>21</sup>'s each independently represent a hydrogen atom or a cyano group.

[0269] In General Formulae (a2-r-2), (a2-r-3) and (a2-r-5), as the alkylene group having 1 to 5 carbon atoms as A", a linear or branched alkylene group is preferable, and examples thereof include a methylene group, an ethylene group, an n-propylene group, and an isopropylene group. In a case where the alkylene group has an oxygen atom or a sulfur atom, specific examples thereof include groups in which —O— or —S— is interposed in the terminal of the alkylene group or between the carbon atoms of the alkylene group. Further, examples thereof include —O—CH<sub>2</sub>—, —CH<sub>2</sub>—O—CH<sub>2</sub>—, —S—CH<sub>2</sub>—, and —CH<sub>2</sub>—S—CH<sub>2</sub>—. A" represents preferably an alkylene group having 1 to 5 carbon atoms or —O—, more preferably an alkylene group having 1 to 5 carbon atoms, and most preferably a methylene group.

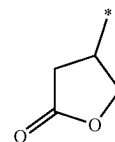
[0270] Specific examples of the groups each represented by General Formulae (a2-r-1) to (a2-r-7) are shown below.

[Chemical Formula 43]

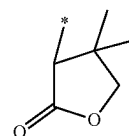


(r-lc-1-1)

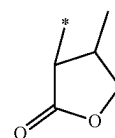
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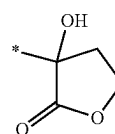
(r-lc-1-2)



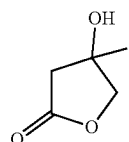
(r-lc-1-3)



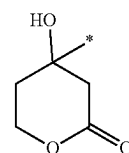
(r-lc-1-4)



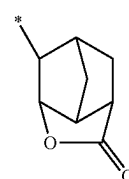
(r-lc-1-5)



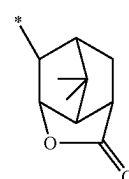
(r-lc-1-6)



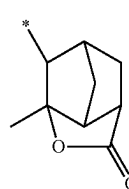
(r-lc-1-7)



(r-lc-2-1)



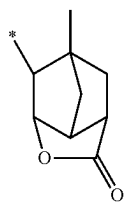
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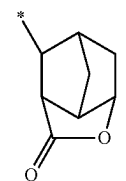
(r-lc-2-3)



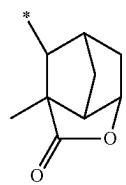
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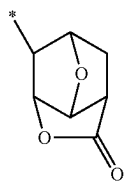
(r-lc-2-4)



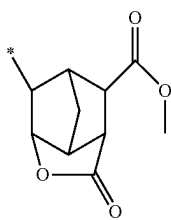
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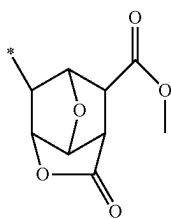
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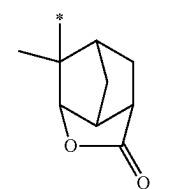
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(r-lc-2-8)

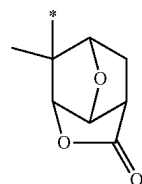


(r-lc-2-9)

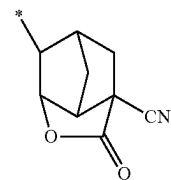


(r-lc-2-10)

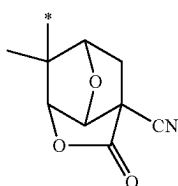
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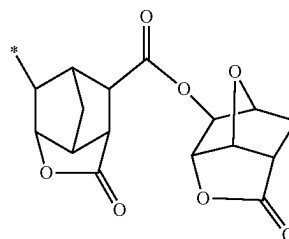
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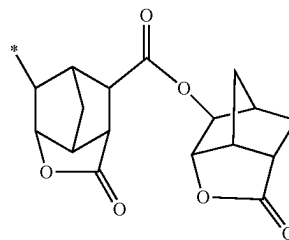
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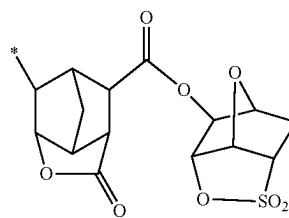
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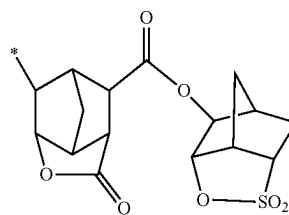
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(r-lc-2-15)

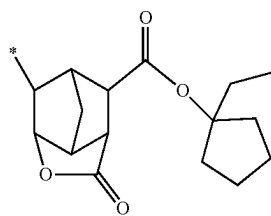


(r-lc-2-16)

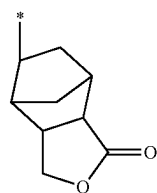


(r-lc-2-17)

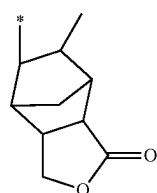
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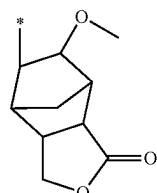
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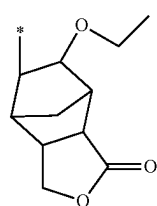
(r-lc-3-1)



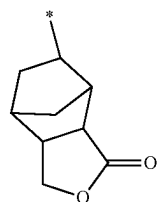
(r-lc-3-2)



(r-lc-3-3)

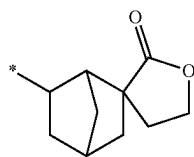


(r-lc-3-4)



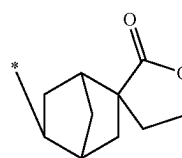
(r-lc-3-5)

[Chemical Formula 44]

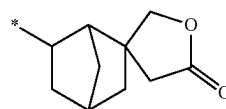


(r-lc-4-1)

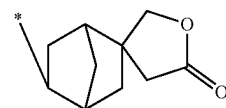
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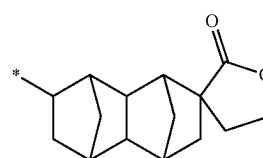
(r-lc-4-2)



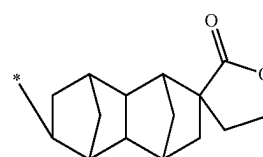
(r-lc-4-3)



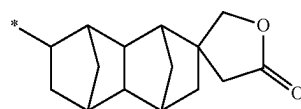
(r-lc-4-4)



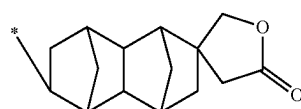
(r-lc-4-5)



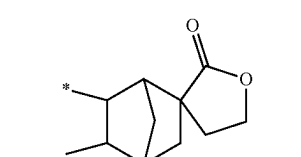
(r-lc-4-6)



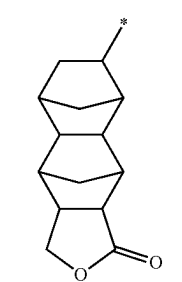
(r-lc-4-7)



(r-lc-4-8)

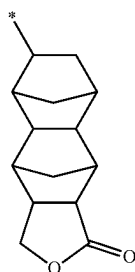


(r-lc-4-9)

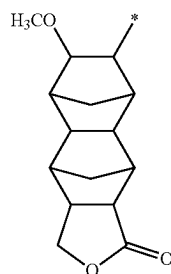


(r-lc-5-1)

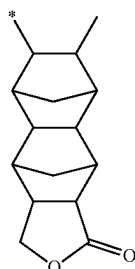
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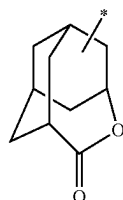
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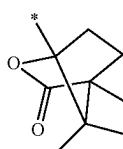
(r-lc-5-3)



(r-lc-5-4)



(r-lc-6-1)



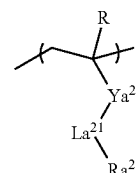
(r-lc-7-1)

[0271] As the constitutional unit (a2), a constitutional unit derived from acrylic acid ester in which the hydrogen atom

bonded to the carbon atom at the  $\alpha$ -position may be substituted with a substituent is preferable.

[0272] It is preferable that such a constitutional unit (a2) is a constitutional unit represented by General Formula (a2-1).

[Chemical Formula 45]



(a2-1)

[0273] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms.  $Ya^{21}$  represents a single bond or a divalent linking group.  $La^{21}$  represents  $-O-$ ,  $-COO-$ ,  $-CON(R')-$ ,  $-OCO-$ ,  $-CONHCO-$ , or  $-CONHCS-$ , and  $R'$  represents a hydrogen atom or a methyl group. In a case where  $La^{21}$  represents  $-O-$ ,  $Ya^{21}$  does not represent  $-CO-$ .  $Ra^{21}$  represents a lactone-containing cyclic group.]

[0274] In Formula (a2-1), R has the same definition as described above. R represents preferably a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a fluorinated alkyl group having 1 to 5 carbon atoms and particularly preferably a hydrogen atom or a methyl group from the viewpoint of the industrial availability.

[0275] In Formula (a2-1), the divalent linking group as  $Ya^{21}$  is not particularly limited, and suitable examples thereof include a divalent hydrocarbon group which may have a substituent and a divalent linking group having a heteroatom.

[0276] It is preferable that  $Ya^{21}$  represents a single bond, an ester bond [ $-C(=O)-O-$ ], an ether bond ( $-O-$ ), a linear or branched alkylene group, or a combination thereof.

[0277] In Formula (a2-1), it is preferable that  $Ya^{21}$  represents a single bond and  $La^{21}$  represents  $-COO-$  or  $-OCO-$ .

[0278] In Formula (a2-1),  $Ra^{21}$  represents a lactone-containing cyclic group.

[0279] Suitable examples of the lactone-containing cyclic group as  $Ra^{21}$  include groups each represented by General Formulae (a2-r-1) to (a2-r-7).

[0280] The constitutional unit (a2) included in the component (A1) may be used alone or two or more kinds thereof.

[0281] In a case where the component (A1) has the constitutional unit (a2), the proportion of the constitutional unit (a2) is preferably in a range of 1% to 20% by mole, more preferably in a range of 1% to 15% by mole, and still more preferably in a range of 1% to 10% by mole with respect to the total amount (100% by mole) of all constitutional units constituting the component (A1).

[0282] In a case where the proportion of the constitutional unit (a2) is set to be greater than or equal to the lower limits of the above-described preferable ranges, the effect to be obtained by allowing the component (A1) to have the constitutional unit (a2) is sufficiently obtained by the above-described effects. Further, in a case where the proportion thereof is set to be less than or equal to the upper limits of

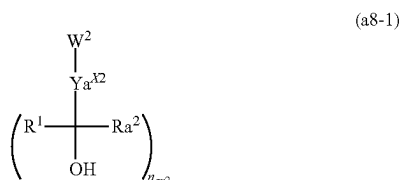
the above-described preferable ranges, the constitutional unit (a2) and other constitutional units can be balanced, and the lithography characteristics are improved.

In Regard to Constitutional Unit (a8):

**[0283]** The constitutional unit (a8) is a constitutional unit derived from a compound represented by General Formula (a8-1).

**[0284]** Here, constitutional units corresponding to the constitutional unit (a0) are excluded.

[Chemical Formula 46]



**[0285]** [In the formula, W² represents a polymerizable group-containing group. Yₐˣ² represents a single bond or an (nₐₓ²+1)-valent linking group. Yₐˣ² and W² may form a condensed ring. R¹ represents a fluorinated alkyl group having 1 to 12 carbon atoms. R² represents an organic group having 1 to 12 carbon atoms, which may have a fluorine atom, or a hydrogen atom. R² and Yₐˣ² may be bonded to each other to form a ring structure. nₐₓ² represents an integer of 1 to 3.]

**[0286]** The term “polymerizable group” in the polymerizable group-containing group as W² denotes a group that enables a compound containing a polymerizable group to be polymerized by radical polymerization or the like, which is, for example, a group having a multiple bond between carbon atoms, such as an ethylenic double bond.

**[0287]** The polymerizable group-containing group may be a group formed of only a polymerizable group or a group formed of a polymerizable group and a group other than the polymerizable group. Examples of the group other than the polymerizable group include a divalent hydrocarbon group which may have a substituent and a divalent linking group having a heteroatom.

**[0288]** Suitable examples of the polymerizable group-containing group include a group represented by Chemical Formula: C(RX¹¹)(Rˣ¹²)=C(Rˣ¹³)—Yₐˣ⁰.

**[0289]** In the chemical formula, Rˣ¹¹, Rˣ¹², and Rˣ¹³ each represent a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, and Yₐˣ⁰ represents a single bond or a divalent linking group.

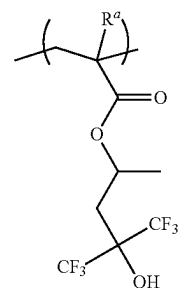
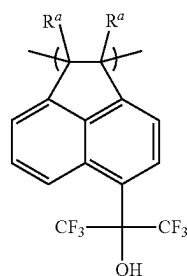
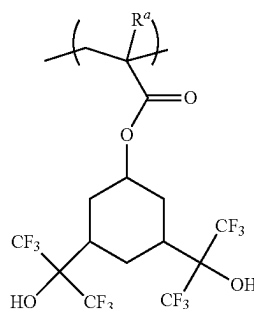
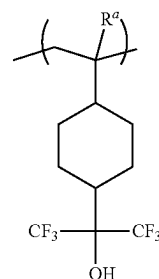
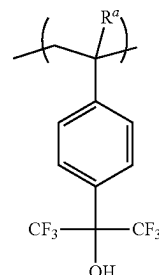
**[0290]** Examples of the condensed ring formed by Yₐˣ² and W² include a condensed ring formed by a polymerizable group of the W² moiety and by Yₐˣ² and a condensed ring formed by a group other than the polymerizable group of the W² moiety and by Yₐˣ².

**[0291]** The condensed ring formed by Yₐˣ² and W² may have a substituent.

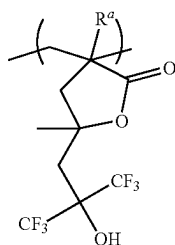
**[0292]** Specific examples of the constitutional unit (a8) are shown below.

**[0293]** In the following formulae, Ra represents a hydrogen atom, a methyl group, or a trifluoromethyl group.

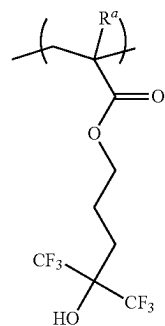
[Chemical Formula 47]



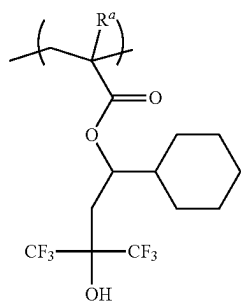
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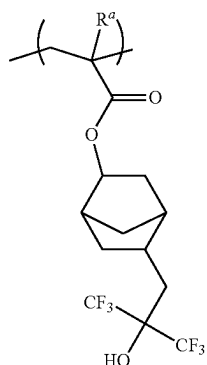
(a8-1-06)



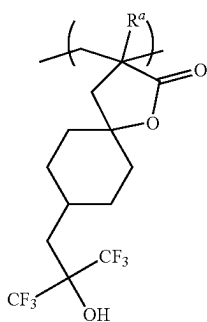
(a8-1-07)



(a8-1-08)

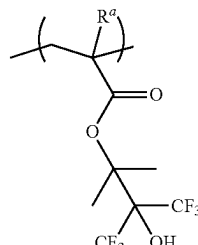


(a8-1-09)

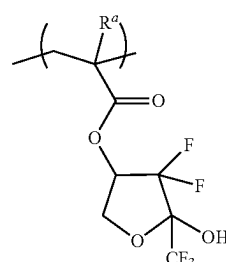


(a8-1-10)

-continued



(a8-1-11)



(a8-1-12)

**[0294]** Among the examples, the constitutional unit (a8) is preferably at least one selected from the group consisting of constitutional units each represented by Chemical Formulae (a8-1-01) to (a8-1-04), (a8-1-06), (a8-1-08), (a8-1-09), and (a8-1-10) and more preferably at least one selected from the group consisting of constitutional units each represented by Chemical Formulae (a8-1-01) to (a8-1-04) and (a8-1-09).

**[0295]** The constitutional unit (a8) contained in the component (A1) may be used alone or two or more kinds thereof.

**[0296]** The proportion of the constitutional unit (a8) in the component (A1) is preferably 50% by mole or less and more preferably in a range of 0% to 30% by mole with respect to the total amount (100% by mole) of all constitutional units constituting the component (A1).

**[0297]** The component (A1) contained in the resist composition may be used alone or in combination of two or more kinds thereof.

**[0298]** In the resist composition of the present embodiment, examples of the component (A1) include a polymer compound having a repeating structure of the constitutional unit (a0).

**[0299]** Among the examples, suitable examples of the component (A1) include a polymer compound having a repeating structure of the constitutional unit (a0) and the constitutional unit (a10).

**[0300]** In the polymer compound having a repeating structure of the constitutional unit (a0) and the constitutional unit (a10), the proportion of the constitutional unit (a0) is preferably in a range of 40% to 80% by mole, more preferably in a range of 45% to 75% by mole, still more preferably in a range of 50% to 70% by mole, and particularly preferably in a range of 55% to 65% by mole with respect to the total amount (100% by mole) of all constitutional units constituting the polymer compound.

**[0301]** In addition, the proportion of the constitutional unit (a10) in the polymer compound is preferably in a range of 20% to 60% by mole, more preferably in a range of 25% to 55% by mole, still more preferably in a range of 30% to 50% by mole, and particularly preferably in a range of 35% to

45% by mole with respect to the total amount (100% by mole) of all constitutional units constituting the polymer compound.

[0302] Such a component (A1) can be produced by dissolving a monomer, from which each constitutional unit is derived, in a polymerization solvent and adding a radical polymerization initiator such as azobisisobutyronitrile (AIBN) or dimethyl azobisisobutyrate (for example, V-601) to the solution so that the polymerization is carried out.

[0303] Alternatively, the component (A1) can be produced by dissolving, in a polymerization solvent, a monomer from which the constitutional unit (a0) is derived and, as necessary, a monomer from which a constitutional unit (for example, the constitutional unit (a10)) other than the constitutional unit (a0) is derived, adding such a radical polymerization initiator thereto to carry out polymerization, and carrying out a deprotection reaction.

[0304] Further, a  $-\text{C}(\text{CF}_3)_2-\text{OH}$  group may be introduced to the terminal during the polymerization using a combination of chain transfer agents such as  $\text{HS}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{C}(\text{CF}_3)_2-\text{OH}$ . As described above, a copolymer into which a hydroxyalkyl group, formed by substitution of some hydrogen atoms in the alkyl group with fluorine atoms, has been introduced is effective for reducing development defects and reducing line edge roughness (LER: uneven irregularities of a line side wall).

[0305] The weight-average molecular weight (Mw) (in terms of polystyrene according to gel permeation chromatography (GPC)) of the component (A1) is not particularly limited, but is preferably in a range of 1000 to 50000, more preferably in a range of 2000 to 30000, and still more preferably in a range of 3000 to 20000.

[0306] In a case where the Mw of the component (A1) is less than or equal to the upper limits of the above-described preferable ranges, the resist composition exhibits a satisfactory solubility in a resist solvent for a resist enough to be used as a resist. On the contrary, in a case where the Mw of the component (A1) is greater than or equal to the lower limits of the above-described preferable ranges, the dry etching resistance and the cross-sectional shape of the resist pattern are excellent.

[0307] Further, the dispersity (Mw/Mn) of the component (A1) is not particularly limited, but is preferably in a range of 1.0 to 4.0, more preferably in a range of 1.0 to 3.0, and particularly preferably in a range of 1.0 to 2.0. Further, Mn represents the number average molecular weight.

#### In Regard to Component (A2)

[0308] In the resist composition of the present embodiment, a base material component (hereinafter, also referred to as “component (A2)”) which does not correspond to the component (A1) and whose solubility in a developing solution is changed by the action of an acid may be used in combination as the component (A).

[0309] The component (A2) is not particularly limited and may be optionally selected from a plurality of components of the related art which have been known as base material components for a chemically amplified resist composition and used.

[0310] As the component (A2), a polymer compound or a low-molecular-weight compound may be used alone or in combination of two or more kinds thereof.

[0311] The proportion of the component (A1) in the component (A) is preferably 25% by mass or greater, more

preferably 50% by mass or greater, and still more preferably 75% by mass or greater, and may be 100% by mass with respect to the total mass of the component (A). In a case where the proportion thereof is 25% by mass or greater, a resist pattern having excellent various lithography characteristics such as high sensitivity, high resolution, and improved roughness is likely to be formed.

[0312] In the resist composition of the present embodiment, the content of the component (A) may be adjusted according to the thickness of the resist film intended to be formed.

#### <<Acid Generator Component (B)>>

[0313] It is preferable that the resist composition according to the present embodiment further contains an acid generator component (B) that generates an acid upon light exposure.

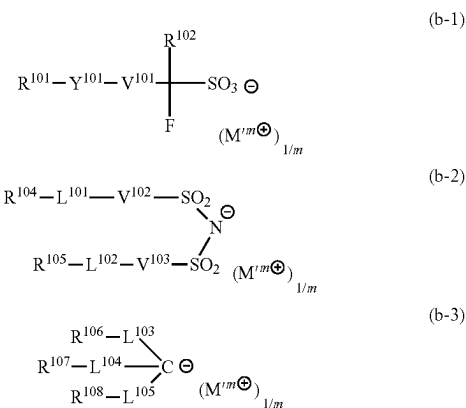
[0314] The component (B) is not particularly limited, and those which have been suggested so far as an acid generator for a chemically amplified resist composition in the related art can be used.

[0315] Examples of the acid generator include various acid generators, for example, onium salt-based acid generators such as iodonium salts and sulfonium salts; oxime sulfonate-based acid generators; diazomethane-based acid generators such as bisalkyl or bisaryl sulfonyl diazomethanes and poly(bis-sulfonyl)diazomethanes; nitrobenzyl sulfonate-based acid generators, iminosulfonate-based acid generators, and disulfone-based acid generators.

[0316] Examples of the onium salt-based acid generators include a compound represented by General Formula (b-1) (hereinafter, also referred to as “component (b-1)”), a compound represented by General Formula (b-2) (hereinafter, also referred to as “component (b-2)”), and a compound represented by General Formula (b-3) (hereinafter, also referred to as “component (b-3)”).

[0317] Examples of the onium salt-based acid generators include a compound represented by General Formula (b-1) (hereinafter, also referred to as “component (b-1)”), a compound represented by General Formula (b-2) (hereinafter, also referred to as “component (b-2)”), and a compound represented by General Formula (b-3) (hereinafter, also referred to as “component (b-3)”).

[Chemical Formula 48]



[0318] [In the formulae,  $R^{101}$  and  $R^{104}$  to  $R^{108}$  each independently represent a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent.  $R^{104}$  and  $R^{105}$  may be bonded to each other to form a ring structure.  $R^{102}$  represents a fluorinated alkyl group having 1 to 5 carbon atoms or a fluorine atom.  $Y^{101}$  represents a divalent linking group having an oxygen atom or a single bond.  $V^{101}$  to  $V^{105}$  each independently represent a single bond, an alkylene group, or a fluorinated alkylene group.  $L^{101}$  and  $L^{102}$  each independently represent a single bond or an oxygen atom.  $L^{103}$  to  $L^{105}$  each independently represent a single bond,  $-\text{CO}-$ , or  $-\text{SO}_2-$ .  $m$  represents an integer of 1 or greater, and  $M^{m+}$  represents an  $m$ -valent onium cation.]

#### {Anion Moiety}

##### Anions in Component (b-1)

[0319] In Formula (b-1),  $R^{101}$  represents a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent.

##### Cyclic Group which May have Substituent:

[0320] The cyclic group is preferably a cyclic hydrocarbon group, and the cyclic hydrocarbon group may be an aromatic hydrocarbon group or an aliphatic hydrocarbon group. The aliphatic hydrocarbon group indicates a hydrocarbon group that has no aromaticity. In addition, it is preferable that the aliphatic hydrocarbon group is saturated.

[0321] The aromatic hydrocarbon group as  $R^{101}$  is a hydrocarbon group having an aromatic ring. The aromatic hydrocarbon group has preferably 3 to 30 carbon atoms, more preferably 5 to 30 carbon atoms, still more preferably 5 to 20 carbon atoms, particularly preferably 6 to 15 carbon atoms, and most preferably 6 to 10 carbon atoms. Here, the number of carbon atoms in a substituent is not included in the number of carbon atoms.

[0322] Specific examples of the aromatic ring of the aromatic hydrocarbon group as  $R^{101}$  include benzene, fluorene, naphthalene, anthracene, phenanthrene, biphenyl, and an aromatic heterocyclic ring in which some carbon atoms constituting any of these aromatic rings are substituted with heteroatoms. Examples of the heteroatom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom.

[0323] Specific examples of the aromatic hydrocarbon group as  $R^{101}$  include a group obtained by removing one hydrogen atom from the aromatic ring (an aryl group such as a phenyl group or a naphthyl group) and a group in which one hydrogen atom of the aromatic ring is substituted with an alkylene group (for example, a benzyl group, a phenethyl group, or a 1-naphthylmethyl group). The alkylene group (alkyl chain in the arylalkyl group) has preferably 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and particularly preferably 1 carbon atom.

[0324] Examples of the cyclic aliphatic hydrocarbon group as  $R^{101}$  include an aliphatic hydrocarbon group having a ring in the structure thereof.

[0325] Examples of the aliphatic hydrocarbon group having a ring in the structure thereof include an alicyclic hydrocarbon group (a group in which one hydrogen atom has been removed from an aliphatic hydrocarbon ring), a group in which the alicyclic hydrocarbon group is bonded to the terminal of a linear or branched aliphatic hydrocarbon

group, and a group in which the alicyclic hydrocarbon group is interposed in the middle of a linear or branched aliphatic hydrocarbon group.

[0326] The alicyclic hydrocarbon group has preferably 3 to 20 carbon atoms and more preferably 3 to 12 carbon atoms.

[0327] The alicyclic hydrocarbon group may be a polycyclic group or a monocyclic group. As the monocyclic alicyclic hydrocarbon group, a group in which one or more hydrogen atoms have been removed from a monocycloalkane is preferable. The monocycloalkane has preferably 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane. As the polycyclic alicyclic hydrocarbon group, a group in which one or more hydrogen atoms have been removed from a polycycloalkane is preferable, and the number of carbon atoms of the polycycloalkane is preferably in a range of 7 to 30. Among these, a polycycloalkane having a crosslinked ring polycyclic skeleton such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane; and a polycycloalkane having a condensed ring polycyclic skeleton such as a cyclic group having a steroid skeleton are preferable as the polycycloalkane.

[0328] Among the examples, as the cyclic aliphatic hydrocarbon group as  $R^{101}$ , a group in which one or more hydrogen atoms have been removed from a monocycloalkane or a polycycloalkane is preferable, a group in which one hydrogen atom has been removed from a polycycloalkane is more preferable, an adamantyl group or a norbornyl group is still more preferable, and an adamantyl group is particularly preferable.

[0329] The linear aliphatic hydrocarbon group which may be bonded to the alicyclic hydrocarbon group has preferably 1 to 10 carbon atoms, more preferably 1 to 6 carbon atoms, still more preferably 1 to 4 carbon atoms, and most preferably 1 to 3 carbon atoms. As the linear aliphatic hydrocarbon group, a linear alkylene group is preferable, and specific examples thereof include a methylene group  $[-\text{CH}_2-]$ , an ethylene group  $[-(\text{CH}_2)_2-]$ , a trimethylene group  $[-(\text{CH}_2)_3-]$ , a tetramethylene group  $[-(\text{CH}_2)_4-]$ , and a pentamethylene group  $[-(\text{CH}_2)_5-]$ .

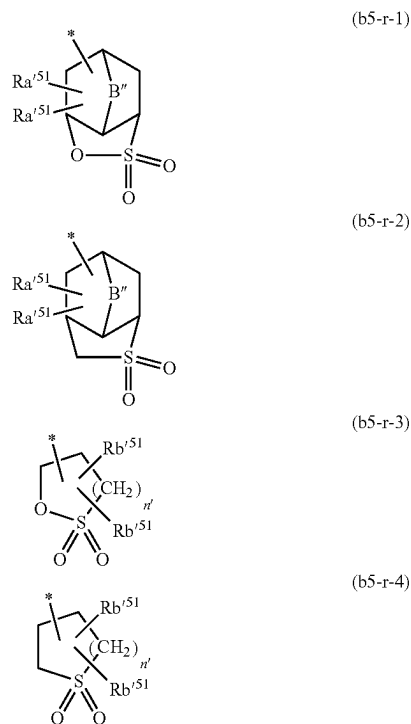
[0330] The branched aliphatic hydrocarbon group which may be bonded to the alicyclic hydrocarbon group has preferably 2 to 10 carbon atoms, more preferably 3 to 6 carbon atoms, still more preferably 3 or 4 carbon atoms, and most preferably 3 carbon atoms.

[0331] As the branched aliphatic hydrocarbon group, a branched alkylene group is preferable, and specific examples thereof include alkylalkylene groups, for example, alkylmethylene groups such as  $-\text{CH}(\text{CH}_3)-$ ,  $-\text{CH}(\text{CH}_2\text{CH}_3)-$ ,  $-\text{C}(\text{CH}_3)_2-$ ,  $-\text{C}(\text{CH}_3)(\text{CH}_2\text{CH}_3)-$ ,  $-\text{C}(\text{CH}_3)(\text{CH}_2\text{CH}_2\text{CH}_3)-$ , and  $-\text{C}(\text{CH}_2\text{CH}_3)_2-$ ; alkyl-ethylene groups such as  $-\text{CH}(\text{CH}_3)\text{CH}_2-$ ,  $-\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)-$ ,  $-\text{C}(\text{CH}_3)_2\text{CH}_2-$ ,  $-\text{CH}(\text{CH}_2\text{CH}_3)\text{CH}_2-$ , and  $-\text{C}(\text{CH}_2\text{CH}_3)_2\text{CH}_2-$ ; alkyltrimethylene groups such as  $-\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2-$  and  $-\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2-$ ; and alkyltetramethylene groups such as  $-\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_2-$  and  $-\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2-$ . As the alkyl group in the alkylalkylene group, a linear alkyl group having 1 to 5 carbon atoms is preferable.

[0332] Further, the cyclic hydrocarbon group as  $R^{101}$  may have a heteroatom such as a heterocyclic ring. Specific examples thereof include lactone-containing cyclic groups each represented by General Formulae (a2-r-1) to (a2-r-7),

—SO<sub>2</sub>—containing cyclic groups each represented by General Formulae (b5-r-1) to (b5-r-4), and other heterocyclic groups each represented by Chemical Formulae (r-hr-1) to (r-hr-16). In the formulae, \* represents a bonding site with respect to Y<sup>101</sup> in Formula (b-1).

[Chemical Formula 49]



**[0333]** [In the formulae, Rb<sup>51</sup>'s each independently represent a hydrogen atom, an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, —COOR'', —OC(=O)R'', a hydroxyalkyl group, or a cyano group, R'' represents a hydrogen atom, an alkyl group, a lactone-containing cyclic group, or a —SO<sub>2</sub>—containing cyclic group, B'' represents an alkylene group having 1 to 5 carbon atoms which may have an oxygen atom or a sulfur atom, an oxygen atom, or a sulfur atom, and n' represents an integer of 0 to 2. \* represents a bonding site.]

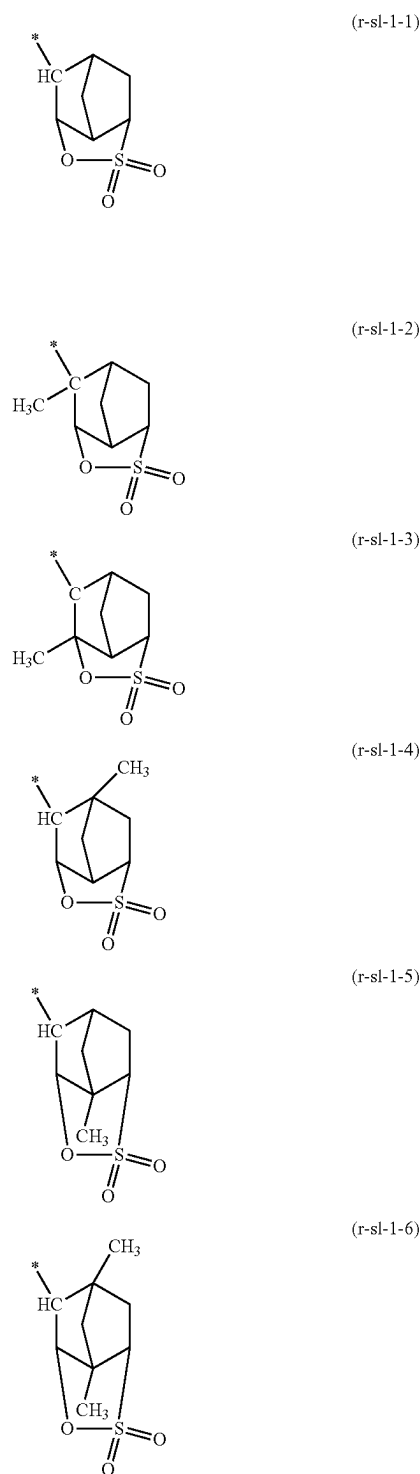
**[0334]** In General Formulae (b5-r-1) and (b5-r-2), B'' represents an alkylene group having 1 to 5 carbon atoms which may have an oxygen atom or a sulfur atom, an oxygen atom, or a sulfur atom.

**[0335]** B'' represents preferably an alkylene group having 1 to 5 carbon atoms or —O—, more preferably an alkylene group having 1 to 5 carbon atoms, and still more preferably a methylene group.

**[0336]** In General Formulae (b5-r-1) to (b5-r-4), Rb<sup>51</sup>'s each independently represent a hydrogen atom, an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, —COOR'', —OC(=O)R'', a hydroxyalkyl group, or a cyano group. Among these, it is preferable that Rb<sup>51</sup>'s each independently represent a hydrogen atom or a cyano group.

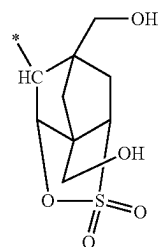
**[0337]** Specific examples of the groups each represented by General Formulae (b5-r-1) to (b5-r-4) are shown below. In the formulae shown below, “Ac” represents an acetyl group.

[Chemical Formula 50]

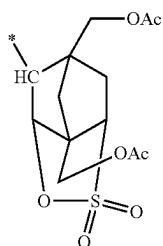




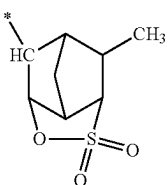
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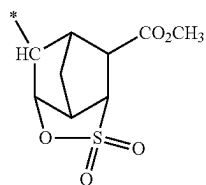
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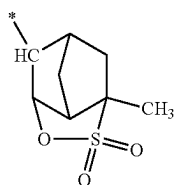
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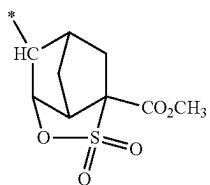
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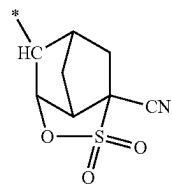
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(r-sl-1-11)

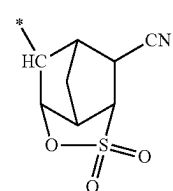


(r-sl-1-12)

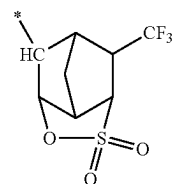


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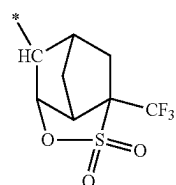
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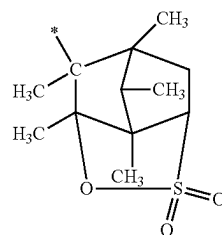
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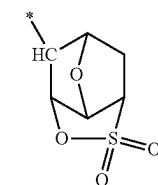
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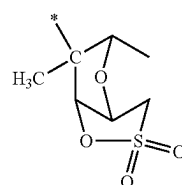
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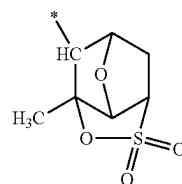
(r-sl-1-17)



(r-sl-1-18)

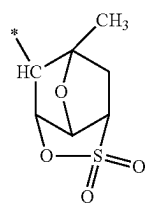


(r-sl-1-19)

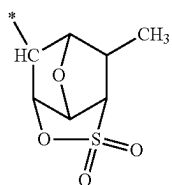
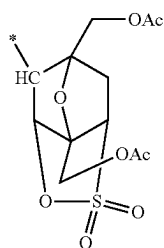
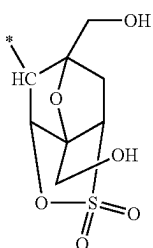
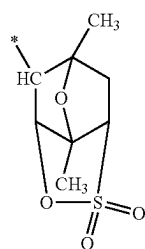
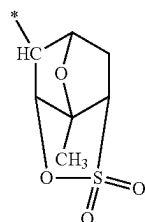


(r-sl-1-20)

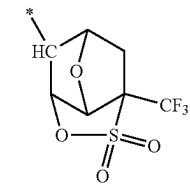
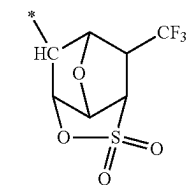
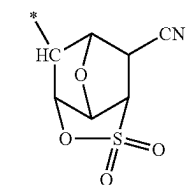
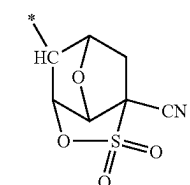
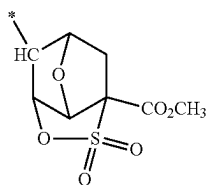
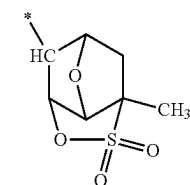
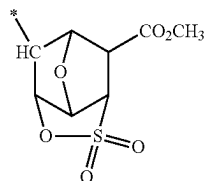
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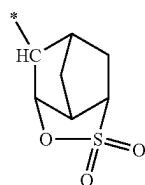
[Chemical Formula 51]



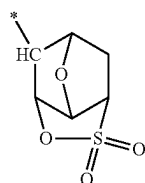
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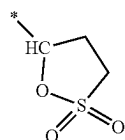
[Chemical Formula 52] -continued



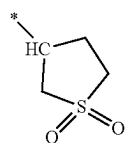
(r-sl-2-1)



(r-sl-2-2)

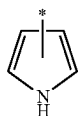


(r-sl-2-3)

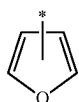


(r-sl-2-4)

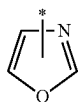
[Chemical Formula 53]



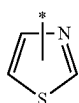
(r-hr-1)



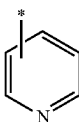
(r-hr-2)



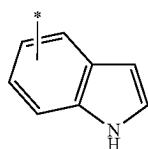
(r-hr-3)



(r-hr-4)

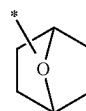


(r-hr-5)

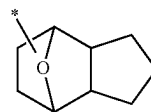


(r-hr-6)

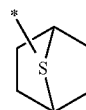
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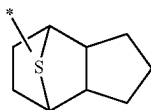
(r-hr-7)



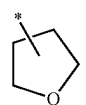
(r-hr-8)



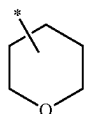
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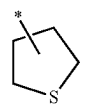
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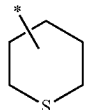
(r-hr-11)



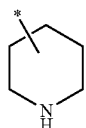
(r-hr-12)



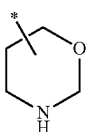
(r-hr-13)



(r-hr-14)



(r-hr-15)



(r-hr-16)

**[0338]** Examples of the substituent for the cyclic group as  $R^{101}$  include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, and a nitro group.

**[0339]** An alkyl group having 1 to 5 carbon atoms is preferable as the alkyl group serving as a substituent.

**[0340]** An alkoxy group having 1 to 5 carbon atoms is preferable as the alkoxy group serving as the substituent.

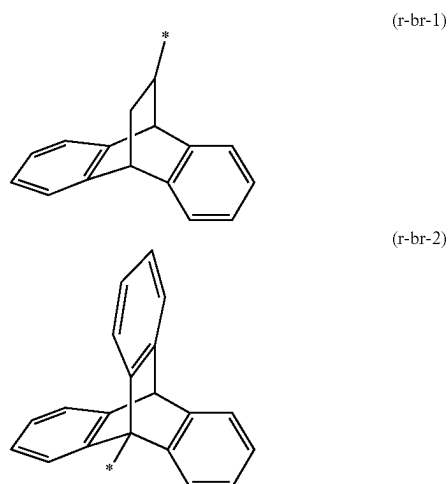
**[0341]** A fluorine atom, a bromine atom, or an iodine atom is preferable as the halogen atom serving as a substituent.

[0342] Example of the above-described halogenated alkyl group as the substituent includes a group in which some or all hydrogen atoms in an alkyl group having 1 to 5 carbon atoms such as a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group are substituted with the above-described halogen atoms.

[0343] The carbonyl group as the substituent is a group that substitutes a methylene group ( $-\text{CH}_2-$ ) constituting the cyclic hydrocarbon group.

[0344] The cyclic hydrocarbon group as  $\text{R}^{101}$  may be a condensed cyclic group having a condensed ring in which an aliphatic hydrocarbon ring and an aromatic ring are condensed. Examples of the condensed ring include those obtained by condensing one or more aromatic rings with a polycycloalkane having a crosslinked ring polycyclic skeleton. Specific examples of the crosslinked ring polycycloalkane include a bicycloalkane such as bicyclo[2.2.1]heptane (norbornane) and bicyclo[2.2.2]octane. As the condensed cyclic group, a group having a condensed ring in which two or three aromatic rings are condensed with a bicycloalkane is preferable, and a group having a condensed ring in which two or three aromatic rings are condensed with bicyclo[2.2.2]octane is more preferable. Specific examples of the condensed cyclic group as  $\text{R}^{101}$  include those represented by Formulae (r-br-1) and (r-br-2). In the formulae, \* represents a bonding site with respect to  $\text{Y}^{101}$  in Formula (b-1).

[Chemical Formula 54]



[0345] Examples of the substituent that the condensed cyclic group as  $\text{R}^{101}$  may have include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, a nitro group, an amino group, an aromatic hydrocarbon group, and an alicyclic hydrocarbon group.

[0346] Examples of the alkyl group, the alkoxy group, the halogen atom, and the halogenated alkyl group as the substituent of the condensed cyclic group include those exemplified as the substituent of the cyclic group as  $\text{R}^{101}$ .

[0347] Examples of the aromatic hydrocarbon group as the substituent of the condensed cyclic group include a group in which one hydrogen atom has been removed from the aromatic ring (an aryl group such as a phenyl group or a naphthyl group), a group in which one hydrogen atom in the aromatic ring has been substituted with an alkylene group

(for example, an arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, 1-naphthylethyl group, or a 2-naphthylethyl group), and a heterocyclic group represented by any of Formulae (r-hr-1) to (r-hr-6).

[0348] Examples of the alicyclic hydrocarbon group as the substituent of the condensed cyclic group include a group in which one hydrogen atom has been removed from a monocycloalkane such as cyclopentane or cyclohexane, a group in which one hydrogen atom has been removed from a polycycloalkane such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane, a lactone-containing cyclic group represented by any of General Formulae (a2-r-1) to (a2-r-7), a  $-\text{SO}_2-$ -containing cyclic group represented by any of General Formulae (b5-r-1) to (b5-r-4), and a heterocyclic group represented by any of Formulae (r-hr-7) to (r-hr-16).

Chain-Like Alkyl Group which May have Substituent:

[0349] The chain-like alkyl group as  $\text{R}^{101}$  may be linear or branched.

[0350] The linear alkyl group has preferably 1 to 20 carbon atoms, more preferably 1 to 15 carbon atoms, and most preferably 1 to 10 carbon atoms.

[0351] The branched alkyl group has preferably 3 to 20 carbon atoms, more preferably 3 to 15 carbon atoms, and most preferably 3 to 10 carbon atoms. Specific examples thereof include a 1-methylethyl group, a 1-methylpropyl group, a 2-methylpropyl group, a 1-methylbutyl group, a 2-methylbutyl group, a 3-methylbutyl group, a 1-ethylbutyl group, a 2-ethylbutyl group, a 1-methylpentyl group, a 2-methylpentyl group, a 3-methylpentyl group, and a 4-methylpentyl group.

Chain-Like Alkenyl Group which May have Substituent:

[0352] The chain-like alkenyl group as  $\text{R}^{101}$  may be linear or branched, and the number of carbon atoms thereof is preferably in a range of 2 to 10, more preferably in a range of 2 to 5, still more preferably in a range of 2 to 4, and particularly preferably 3. Examples of the linear alkenyl group include a vinyl group, a propenyl group (an allyl group), and a butenyl group. Examples of the branched alkenyl group include a 1-methylvinyl group, a 2-methylvinyl group, a 1-methylpropenyl group, and a 2-methylpropenyl group.

[0353] Among the examples, as the chain-like alkenyl group, a linear alkenyl group is preferable, a vinyl group or a propenyl group is more preferable, and a vinyl group is particularly preferable.

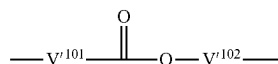
[0354] Examples of the substituent for the chain-like alkyl group or alkenyl group as  $\text{R}^{101}$  include an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, a nitro group, an amino group, and a cyclic group as  $\text{R}^{101}$ .

[0355] In Formula (b-1),  $\text{Y}^{101}$  represents a single bond or a divalent linking group having an oxygen atom.

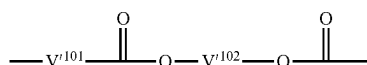
[0356] In a case where  $\text{Y}^{101}$  represents a divalent linking group containing an oxygen atom,  $\text{Y}^{101}$  may contain an atom other than the oxygen atom. Examples of atoms other than an oxygen atom include a carbon atom, a hydrogen atom, a sulfur atom, and a nitrogen atom.

[0357] Examples of the divalent linking group having an oxygen atom include linking groups each represented by General Formulae (y-al-1) to (y-al-7). Further, in General Formulae (y-al-1) to (y-al-7),  $\text{V}^{101}$  in General Formulae (y-al-1) to (y-al-7) is bonded to  $\text{R}^{101}$  in Formula (b-1).

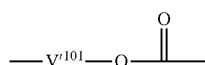
[Chemical Formula 55]



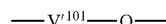
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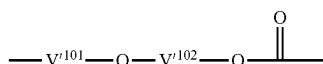
(y-al-2)



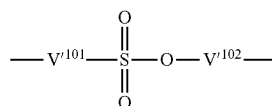
(y-al-3)



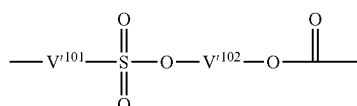
(y-al-4)



(y-al-5)



(y-al-6)



(y-al-7)

[0358] [In the formulae,  $V^{101}$  represents a single bond or an alkylene group having 1 to 5 carbon atoms, and  $V^{102}$  represents a divalent saturated hydrocarbon group having 1 to 30 carbon atoms.]

[0359] As the divalent saturated hydrocarbon group as  $V^{102}$ , an alkylene group having 1 to 30 carbon atoms is preferable, an alkylene group having 1 to 10 carbon atoms is more preferable, and an alkylene group having 1 to 5 carbon atoms is still more preferable.

[0360] The alkylene group as  $V^{101}$  and  $V^{102}$  may be a linear alkylene group or a branched alkylene group, and a linear alkylene group is preferable.

[0361] Specific examples of the alkylene group as  $V^{101}$  and  $V^{102}$  include a methylene group [ $\text{---CH}_2\text{---}$ ]; an alkylmethylene group such as  $\text{---CH(CH}_3\text{)---}$ ,  $\text{---CH(CH}_2\text{CH}_3\text{)---}$ ,  $\text{---C(CH}_3\text{)}_2\text{---}$ ,  $\text{---C(CH}_3\text{)(CH}_2\text{CH}_3\text{)---}$ ,  $\text{---C(CH}_3\text{)(CH(CH}_3\text{)CH}_2\text{CH}_3\text{)---}$ , or  $\text{---C(CH}_2\text{CH}_3\text{)}_2\text{---}$ ; an ethylene group [ $\text{---CH}_2\text{CH}_2\text{---}$ ]; an alkylethylene group such as  $\text{---CH(CH}_3\text{)CH}_2\text{---}$ ,  $\text{---CH(CH}_3\text{)CH(CH}_3\text{)---}$ ,  $\text{---C(CH}_3\text{)}_2\text{CH}_2\text{---}$ , or  $\text{---CH(CH}_2\text{CH}_3\text{)CH}_2\text{---}$ ; a trimethylene group (n-propylene group) [ $\text{---CH}_2\text{CH}_2\text{CH}_2\text{---}$ ]; an alkyltrimethylene group such as  $\text{---CH(CH}_3\text{)CH}_2\text{CH}_2\text{---}$  or  $\text{---CH}_2\text{CH}$

$(\text{CH}_3)\text{CH}_2\text{---}$ ; a tetramethylene group [ $\text{---CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{---}$ ]; an alkyltetramethylene group such as  $\text{---CH(CH}_3\text{)CH}_2\text{CH}_2\text{CH}_2\text{---}$  or  $\text{---CH}_2\text{CH(CH}_3\text{)CH}_2\text{CH}_2\text{---}$ ; and a pentamethylene group [ $\text{---CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{---}$ ].

[0362] Further, a part of methylene group in the alkylene group as  $V^{101}$  and  $V^{102}$  may be substituted with a divalent aliphatic cyclic group having 5 to 10 carbon atoms. As the aliphatic cyclic group, a divalent group in which one hydrogen atom has been further removed from the cyclic aliphatic hydrocarbon group (a monocyclic aliphatic hydrocarbon group or a polycyclic aliphatic hydrocarbon group) as  $\text{Ra}^{13}$  in Formula (a1-r-1) is preferable, and a cyclohexylene group, a 1,5-adamantylene group, or a 2,6-adamantylene group is more preferable.

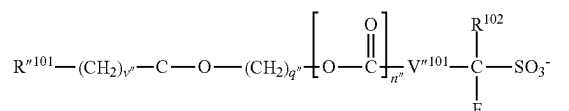
[0363] In Formula (b-1),  $V^{101}$  represents a single bond, an alkylene group, or a fluorinated alkylene group. It is preferable that the alkylene group and the fluorinated alkylene group as  $V^{101}$  have 1 to 4 carbon atoms. Among the examples, it is preferable that  $V^{101}$  represents a single bond or a linear fluorinated alkylene group having 1 to 4 carbon atoms.

[0364] In Formula (b-1),  $\text{R}^{102}$  represents a fluorine atom or a fluorinated alkyl group having 1 to 5 carbon atoms.  $\text{R}^{102}$  represents preferably a fluorine atom or a perfluoroalkyl group having 1 to 5 carbon atoms and more preferably a fluorine atom.

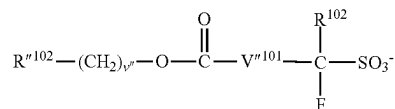
[0365] In a case where  $Y^{101}$  represents a single bond, specific example of the anion moiety represented by Formula (b-1) include a fluorinated alkylsulfonate anion such as a trifluoromethanesulfonate anion or a perfluorobutanesulfonate anion. Further, in a case where  $Y^{101}$  represents a divalent linking group having an oxygen atom, specific examples thereof include an anion represented by any of Formulae (an-1) to (an-3).

[Chemical Formula 56]

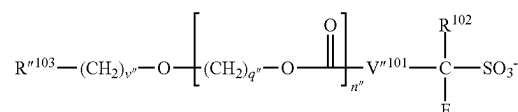
(an-1)



(an-2)



(an-3)



[0366] [In the formulae,  $\text{R}^{101}$  represents an aromatic cyclic group which may have a substituent, a monovalent heterocyclic group represented by any of Chemical Formulae (r-hr-1) to (r-hr-6), a condensed cyclic group represented by Formula (r-br-1) or (r-br-2), a chain-like alkyl group which may have a substituent, or an aromatic cyclic group which may have a substituent.  $\text{R}^{102}$  represents an aliphatic cyclic group which may have a substituent, a condensed

cyclic group represented by Formula (r-br-1) or (r-br-2), a lactone-containing cyclic group represented by any of General Formulae (a2-r-1) and (a2-r-3) to (a2-r-7), or a  $\text{—SO}_2\text{—}$ -containing cyclic group represented by any of General Formulae (b5-r-1) to (b5-r-4).  $R^{n103}$  represents an aromatic cyclic group which may have a substituent, an aliphatic cyclic group which may have a substituent, or a chain-like alkenyl group which may have a substituent.  $V^{n101}$  represents a single bond, an alkylene group having 1 to 4 carbon atoms, or a fluorinated alkylene group having 1 to 4 carbon atoms.  $R^{102}$  represents a fluorine atom or a fluorinated alkyl group having 1 to 5 carbon atoms. Each  $v''$  independently represents an integer of 0 to 3, each  $q''$  independently represents an integer of 0 to 20, and  $n''$  represents 0 or 1.]

[0367] As the aliphatic cyclic group as  $R^{n101}$ ,  $R^{n102}$ , and  $R^{n103}$  which may have a substituent, the same groups as those for the cyclic aliphatic hydrocarbon group as  $R^{101}$  in Formula (b-1) are preferable. Examples of the substituent include the same groups as those for the substituent which may substitute the cyclic aliphatic hydrocarbon group as  $R^{101}$  in Formula (b-1).

[0368] As the aromatic cyclic group which may have a substituent as  $R^{n101}$  and  $R^{n103}$  the same groups as those for the aromatic hydrocarbon group in the cyclic hydrocarbon group as  $R^{101}$  in Formula (b-1) are preferable. Examples of the substituent include the same groups as those for the substituent which may substitute the aromatic hydrocarbon group as  $R^{101}$  in Formula (b-1).

[0369] As the chain-like alkyl group as  $R^{n101}$  which may have a substituent, the same groups as those for the chain-like alkyl group as  $R^{101}$  in Formula (b-1) are preferable.

[0370] As the chain-like alkenyl group as  $R^{n103}$  which may have a substituent, the same groups as those for the chain-like alkenyl group as  $R^{101}$  in Formula (b-1) are preferable.

Anions in Component (b-2)

[0371] In Formula (b-2),  $R^{104}$  and  $R^{105}$  each independently represent a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof include the same groups as those for  $R^{101}$  in Formula (b-1). Here,  $R^{104}$  and  $R^{105}$  may be bonded to each other to form a ring.

[0372]  $R^{104}$  and  $R^{105}$  represent preferably a chain-like alkyl group which may have a substituent and more preferably a linear or branched alkyl group or a linear or branched fluorinated alkyl group.

[0373] The chain-like alkyl group has preferably 1 to 10 carbon atoms, more preferably 1 to 7 carbon atoms, and still more preferably 1 to 3 carbon atoms. It is preferable that the number of carbon atoms in the chain-like alkyl group as  $R^{104}$  and  $R^{105}$  decreases within the range of the number of carbon atoms from the viewpoint that the solubility in a solvent for a resist is also satisfactory. Further, in the chain-like alkyl group as  $R^{104}$  and  $R^{105}$ , it is preferable that the number of hydrogen atoms substituted with fluorine atoms is as large as possible from the viewpoint that the acid strength increases and the transparency to high energy light or electron beams having a wavelength of 250 nm or less is improved. The proportion of fluorine atoms in the chain-like alkyl group, that is, the fluorination ratio is preferably in a range of 70% to 100% and more preferably in a range of 90% to 100%,

and it is most preferable that the chain-like alkyl group is a perfluoroalkyl group in which all hydrogen atoms are substituted with fluorine atoms.

[0374] In Formula (b-2),  $V^{102}$  and  $V^{103}$  each independently represent a single bond, an alkylene group, or a fluorinated alkylene group, and examples thereof include the same groups as those for  $V^{101}$  in Formula (b-1).

[0375] In Formula (b-2),  $L^{101}$  and  $L^{102}$  each independently represent a single bond or an oxygen atom.

Anions in Component (b-3)

[0376] In Formula (b-3),  $R^{106}$  to  $R^{108}$  each independently represent a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof include the same groups as those for  $R^{101}$  in Formula (b-1).

[0377] In Formula (b-3),  $L^{103}$  to  $L^{105}$  each independently represent a single bond,  $\text{—CO—}$ , or  $\text{—SO}_2\text{—}$ .

[0378] Among the examples, as the anion moiety of the component (B), an anion in the component (b-1) is preferable.

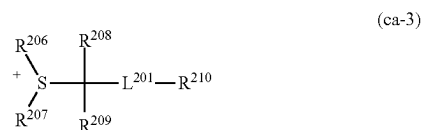
{Cation Moiety}

[0379] In Formulae (b-1), (b-2), and (b-3),  $M^{m+}$  represents an m-valent onium cation. Among these, a sulfonium cation and an iodonium cation are preferable.

[0380] m represents an integer of 1 or greater.

[0381] Preferred examples of the cation moiety ( $(M^{m+})_m$ ) include organic cations each represented by General Formulae (ca-1) to (ca-3).

[Chemical Formula 57]



[0382] [In the formulae,  $R^{201}$  to  $R^{207}$  each independently represent an aryl group, an alkyl group, or an alkenyl group, which may have a substituent.  $R^{201}$  to  $R^{203}$ , and  $R^{206}$  and  $R^{207}$  may be bonded to each other to form a ring with the sulfur atoms in the formulae.  $R^{208}$  and  $R^{209}$  each independently represent a hydrogen atom or an alkyl group having 1 to 5 carbon atoms.  $R^{210}$  represents an aryl group which may have a substituent, an alkyl group which may have a substituent, an alkenyl group which may have a substituent, or a  $\text{—SO}_2\text{—}$ -containing cyclic group which may have a substituent.  $L^{201}$  represents  $\text{—C(=O)—}$  or  $\text{—C(=O)—O—}$ .]

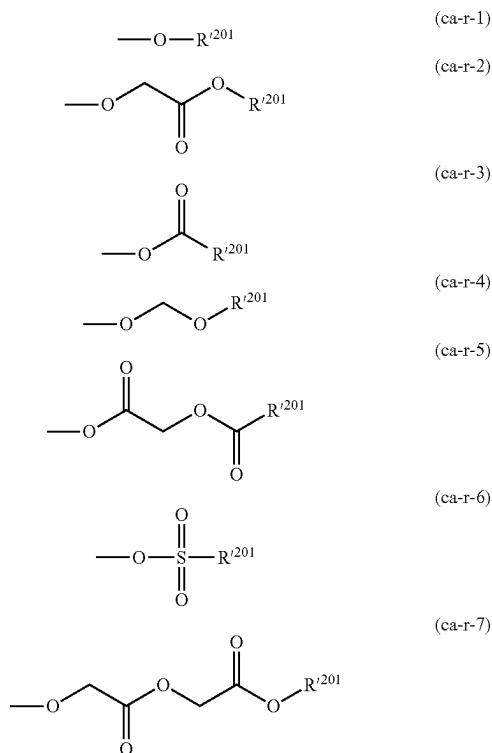
[0383] In General Formulae (ca-1) to (ca-3), examples of the aryl group as  $R^{201}$  to  $R^{207}$  include an unsubstituted aryl group having 6 to 20 carbon atoms. Among these, a phenyl group or a naphthyl group is preferable.

[0384] As the alkyl group as  $R^{201}$  to  $R^{207}$ , a chain-like or cyclic alkyl group having 1 to 30 carbon atoms is preferable.

[0385] It is preferable that the alkenyl group as  $R^{201}$  to  $R^{207}$  has 2 to 10 carbon atoms.

[0386] Examples of the substituent that  $R^{201}$  to  $R^{207}$  and  $R^{210}$  may have include an alkyl group, a halogen atom, a halogenated alkyl group, a carbonyl group, a cyano group, an amino group, an aryl group, and groups each represented by General Formulae (ca-r-1) to (ca-r-7).

[Chemical Formula 58]



[0387] [In the formulae,  $R^{201}$ 's each independently represent a hydrogen atom, a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent.]

Cyclic Group which May have Substituent:

[0388] The cyclic group is preferably a cyclic hydrocarbon group, and the cyclic hydrocarbon group may be an aromatic hydrocarbon group or an aliphatic hydrocarbon group. The aliphatic hydrocarbon group indicates a hydrocarbon group that has no aromaticity. Further, the aliphatic hydrocarbon group may be saturated or unsaturated. In general, it is preferable that the aliphatic hydrocarbon group is saturated.

[0389] The aromatic hydrocarbon group as  $R^{201}$  is a hydrocarbon group having an aromatic ring. The aromatic hydrocarbon group has preferably 3 to 30 carbon atoms, more preferably 5 to 30 carbon atoms, still more preferably 5 to 20 carbon atoms, particularly preferably 6 to 15 carbon atoms, and most preferably 6 to 10 carbon atoms. Here, the number of carbon atoms in a substituent is not included in the number of carbon atoms.

[0390] Specific examples of the aromatic ring contained in the aromatic hydrocarbon group as  $R^{201}$  include benzene, fluorene, naphthalene, anthracene, phenanthrene, biphenyl, or an aromatic heterocyclic ring in which some carbon atoms constituting any of these aromatic rings are substituted with heteroatoms. Examples of the heteroatom in the aromatic heterocyclic rings include an oxygen atom, a sulfur atom, and a nitrogen atom.

[0391] Specific examples of the aromatic hydrocarbon group as  $R^{201}$  include a group in which one hydrogen atom has been removed from the aromatic ring (an aryl group such as a phenyl group or a naphthyl group), and a group in which one hydrogen atom in the aromatic ring has been substituted with an alkylene group (for example, an arylalkyl group such as a benzyl group, a phenethyl group, a 1-naphthylmethyl group, a 2-naphthylmethyl group, 1-naphthylethyl group, or a 2-naphthylethyl group). The alkylene group (alkyl chain in the arylalkyl group) has preferably 1 to 4 carbon atoms, more preferably 1 or 2 carbon atoms, and particularly preferably 1 carbon atom.

[0392] Examples of the cyclic aliphatic hydrocarbon group as  $R^{201}$  include an aliphatic hydrocarbon group having a ring in the structure thereof.

[0393] Examples of the aliphatic hydrocarbon group having a ring in the structure thereof include an alicyclic hydrocarbon group (a group in which one hydrogen atom has been removed from an aliphatic hydrocarbon ring), a group in which the alicyclic hydrocarbon group is bonded to the terminal of a linear or branched aliphatic hydrocarbon group, and a group in which the alicyclic hydrocarbon group is interposed in the middle of a linear or branched aliphatic hydrocarbon group.

[0394] The alicyclic hydrocarbon group has preferably 3 to 20 carbon atoms and more preferably 3 to 12 carbon atoms.

[0395] The alicyclic hydrocarbon group may be a polycyclic group or a monocyclic group. As the monocyclic alicyclic hydrocarbon group, a group in which one or more hydrogen atoms have been removed from a monocycloalkane is preferable. The monocycloalkane has preferably 3 to 6 carbon atoms, and specific examples thereof include cyclopentane and cyclohexane. As the polycyclic alicyclic hydrocarbon group, a group in which one or more hydrogen atoms have been removed from a polycycloalkane is preferable, and the number of carbon atoms of the polycycloalkane is preferably in a range of 7 to 30. Among these, a polycycloalkane having a crosslinked ring polycyclic skeleton such as adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane; and a polycycloalkane having a condensed ring polycyclic skeleton such as a cyclic group having a steroid skeleton are preferable as the polycycloalkane.

[0396] Among these examples, as the cyclic aliphatic hydrocarbon group as  $R^{201}$ , a group in which one or more hydrogen atoms have been removed from a monocycloalkane or a polycycloalkane is preferable, a group in which one hydrogen atom has been removed from a polycycloalkane is more preferable, an adamantyl group or a norbornyl group is particularly preferable, and an adamantyl group is most preferable.

[0397] The linear or branched aliphatic hydrocarbon group which may be bonded to the alicyclic hydrocarbon group has preferably 1 to 10 carbon atoms, more preferably

1 to 6 carbon atoms, still more preferably 1 to 4 carbon atoms, and particularly preferably 1 to 3 carbon atoms.

**[0398]** As the linear aliphatic hydrocarbon group, a linear alkylene group is preferable, and specific examples thereof include a methylene group  $[-CH_2-]$ , an ethylene group  $[-(CH_2)_2-]$ , a trimethylene group  $[-(CH_2)_3-]$ , a tetramethylene group  $[-(CH_2)_4-]$ , and a pentamethylene group  $[-(CH_2)_5-]$ .

**[0399]** As the branched aliphatic hydrocarbon group, a branched alkylene group is preferable, and specific examples thereof include alkylalkylene groups, for example, alkylmethylene groups such as  $-CH(CH_3)-$ ,  $-CH(CH_2CH_3)-$ ,  $-C(CH_3)_2-$ ,  $-C(CH_3)(CH_2CH_3)-$ ,  $-C(CH_3)(CH_2CH_2CH_3)-$ , and  $-C(CH_2CH_3)_2-$ ; alkyl-ethylene groups such as  $-CH(CH_3)CH_2-$ ,  $-CH(CH_3)CH(CH_3)-$ ,  $-C(CH_3)_2CH_2-$ ,  $-CH(CH_2CH_3)CH_2-$ , and  $-C(CH_2CH_3)_2CH_2-$ ; alkyltrimethylene groups such as  $-CH(CH_3)CH_2CH_2-$  and  $-CH_2CH(CH_3)CH_2-$ ; and alkyltetramethylene groups such as  $-CH(CH_3)CH_2CH_2CH_2-$  and  $-CH_2CH(CH_3)CH_2CH_2-$ . As the alkyl group in the alkylalkylene group, a linear alkyl group having 1 to 5 carbon atoms is preferable.

**[0400]** Further, the cyclic hydrocarbon group as  $R^{201}$  may have a heteroatom such as a heterocyclic ring. Specific examples thereof include lactone-containing cyclic groups each represented by General Formulae (a2-r-1) to (a2-r-7),  $-SO_2-$ -containing cyclic group each represented by General Formulae (b5-r-1) to (b5-r-4), and other heterocyclic groups each represented by Chemical Formulae (r-lr-1) to (r-lr-16).

**[0401]** Examples of the substituent for the cyclic group as  $R^{201}$  include an alkyl group, an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, and a nitro group.

**[0402]** As the alkyl group as the substituent, an alkyl group having 1 to 5 carbon atoms is preferable, and a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group is most preferable.

**[0403]** As the alkoxy group as the substituent, an alkoxy group having 1 to 5 carbon atoms is preferable, a methoxy group, an ethoxy group, an n-propoxy group, an iso-propoxy group, an n-butoxy group, or a tert-butoxy group is more preferable, and a methoxy group or an ethoxy group is most preferable.

**[0404]** As the halogen atom as a substituent, a fluorine atom is preferable.

**[0405]** Example of the above-described halogenated alkyl group as the substituent includes a group in which some or all hydrogen atoms in an alkyl group having 1 to 5 carbon atoms such as a methyl group, an ethyl group, a propyl group, an n-butyl group, or a tert-butyl group are substituted with the above-described halogen atoms.

**[0406]** The carbonyl group as the substituent is a group that substitutes a methylene group  $(-CH_2-)$  constituting the cyclic hydrocarbon group.

Chain-Like Alkyl Group which May have Substituent:

**[0407]** The chain-like alkyl group as  $R^{201}$  may be linear or branched.

**[0408]** The linear alkyl group has preferably 1 to 20 carbon atoms, more preferably 1 to 15 carbon atoms, and most preferably 1 to 10 carbon atoms.

**[0409]** The branched alkyl group has preferably 3 to 20 carbon atoms, more preferably 3 to 15 carbon atoms, and most preferably 3 to 10 carbon atoms. Specific examples thereof include a 1-methylethyl group, a 1-methylpropyl

group, a 2-methylpropyl group, a 1-methylbutyl group, a 2-methylbutyl group, a 3-methylbutyl group, a 1-ethylbutyl group, a 2-ethylbutyl group, a 1-methylpentyl group, a 2-methylpentyl group, a 3-methylpentyl group, and a 4-methylpentyl group.

Chain-Like Alkenyl Group which May have Substituent:

**[0410]** The chain-like alkenyl group as  $R^{201}$  may be linear or branched, and the number of carbon atoms thereof is preferably in a range of 2 to 10, more preferably in a range of 2 to 5, still more preferably in a range of 2 to 4, and particularly preferably 3. Examples of the linear alkenyl group include a vinyl group, a propenyl group (an allyl group), and a butynyl group. Examples of the branched alkenyl group include a 1-methylvinyl group, a 2-methylvinyl group, a 1-methylpropenyl group, and a 2-methylpropenyl group.

**[0411]** Among the examples, as the chain-like alkenyl group, a linear alkenyl group is preferable, a vinyl group or a propenyl group is more preferable, and a vinyl group is particularly preferable.

**[0412]** Examples of the substituent for the chain-like alkyl group or alkenyl group as  $R^{201}$  include an alkoxy group, a halogen atom, a halogenated alkyl group, a hydroxyl group, a carbonyl group, a nitro group, an amino group, and a cyclic group as  $R^{201}$ .

**[0413]** Examples of the cyclic group which may have a substituent, the chain-like alkyl group which may have a substituent, and the chain-like alkenyl group which may have a substituent as  $R^{201}$  include those for the acid dissociable group represented by Formula (a1-r-2) which are the exemplary examples of the cyclic group which may have a substituent and the chain-like alkyl group which may have a substituent, in addition to those described above.

**[0414]** Among the examples,  $R^{201}$  represents preferably a cyclic group which may have a substituent and more preferably a cyclic hydrocarbon group which may have a substituent. More specific preferred examples thereof include a phenyl group, a naphthyl group, a group in which one or more hydrogen atoms have been removed from a polycycloalkane, a lactone-containing cyclic group represented by any of General Formulae (a2-r-1) to (a2-r-7), and a  $-SO_2-$ -containing cyclic group represented by any of General Formulae (b5-r-1) to (b5-r-4).

**[0415]** In General Formulae (ca-1) to (ca-3), in a case where  $R^{201}$  to  $R^{203}$  and  $R^{206}$  and  $R^{207}$  are bonded to each other to form a ring with a sulfur atom in the formula, these groups may be bonded to each other via a heteroatom such as a sulfur atom, an oxygen atom, or a nitrogen atom, or a functional group such as a carbonyl group,  $-SO-$ ,  $-SO_2-$ ,  $-SO_3-$ ,  $-COO-$ ,  $-CONH-$  or  $-N(R_N)-$  ( $R_N$  represents an alkyl group having 1 to 5 carbon atoms). As a ring to be formed, one ring containing the sulfur atom in the formula in the ring skeleton thereof is preferably a 3- to 10-membered ring and particularly preferably a 5- to 7-membered ring containing the sulfur atom. Specific examples of the ring to be formed include a thiophene ring, a thiazole ring, a benzothiophene ring, a dibenzothiophene ring, a 9H-thioxanthene ring, a thioxanthone ring, a thianthrene ring, a phenoxathiin ring, a tetrahydrothiophenium ring, and a tetrahydrothiopyranium ring.

**[0416]**  $R^{208}$  and  $R^{209}$  each independently represent a hydrogen atom or an alkyl group having 1 to 5 carbon atoms and preferably a hydrogen atom or an alkyl group having 1



to 3 carbon atoms. In a case where  $R^{208}$  and  $R^{209}$  represent an alkyl group,  $R^{208}$  and  $R^{209}$  may be bonded to each other to form a ring.

[0417]  $R^{210}$  represents an aryl group which may have a substituent, an alkyl group which may have a substituent, an alkenyl group which may have a substituent, or a  $-\text{SO}_2-$ -containing cyclic group which may have a substituent.

[0418] Examples of the aryl group as  $R^{210}$  include an unsubstituted aryl group having 6 to 20 carbon atoms. Among these, a phenyl group or a naphthyl group is preferable.

[0419] As the alkyl group as  $R^{210}$ , a chain-like or cyclic alkyl group having 1 to 30 carbon atoms is preferable.

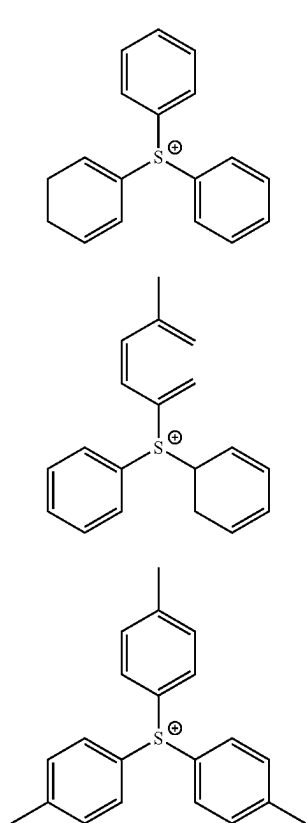
[0420] It is preferable that the alkenyl group as  $R^{210}$  has 2 to 10 carbon atoms.

[0421] The  $-\text{SO}_2-$ -containing cyclic group which may have a substituent as  $R^{210}$  is preferably “ $-\text{SO}_2-$ -containing polycyclic group” and more preferably a group represented by General Formula (b5-r-1).

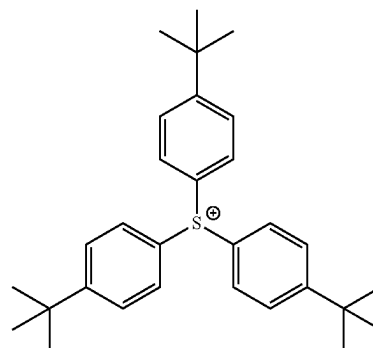
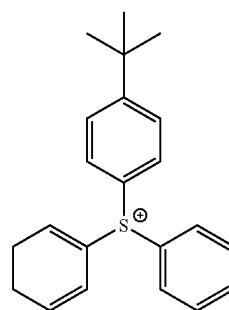
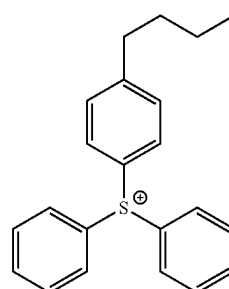
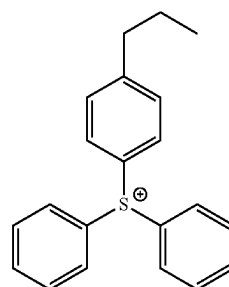
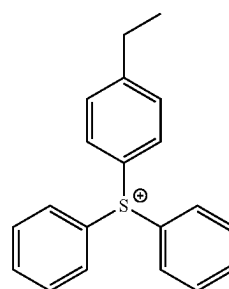
[0422] Specific examples of the cation represented by Formula (ca-1) are shown below.

[0423] Specific examples of the suitable cation represented by General Formula (ca-1) include cations each represented by the following chemical formulae.

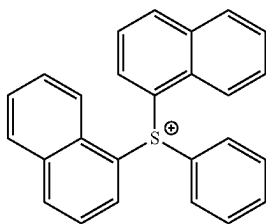
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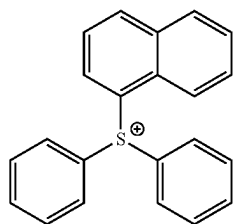
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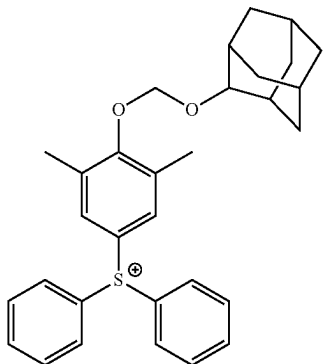
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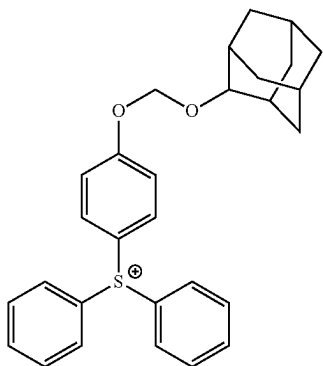
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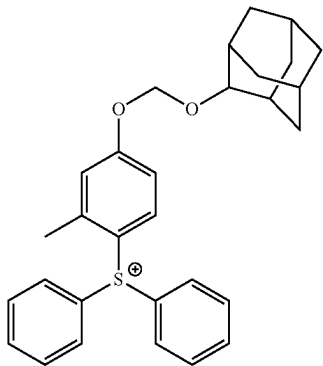
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(ca-1-11)

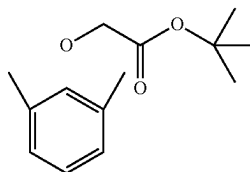


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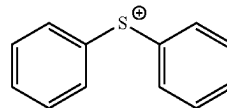


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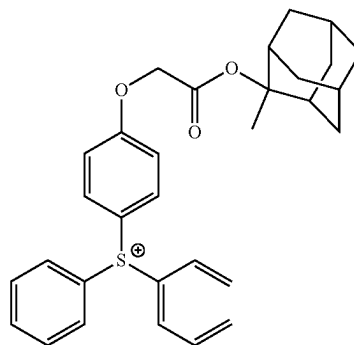
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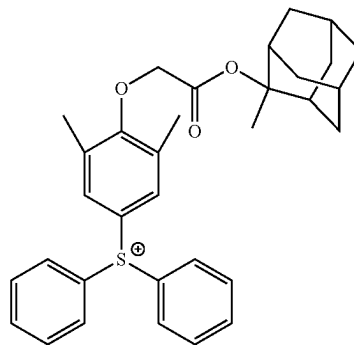
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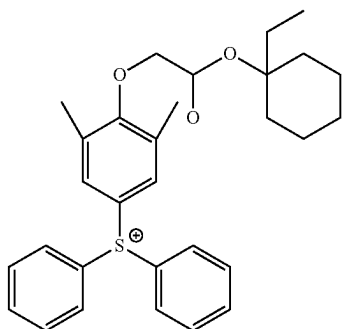
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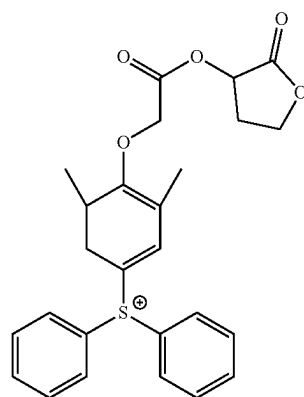
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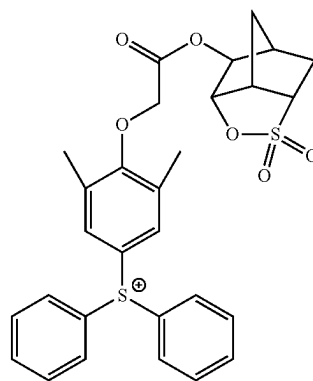
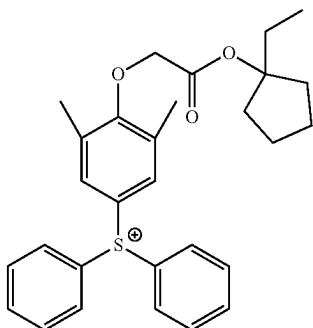
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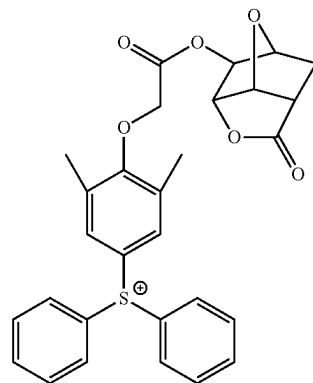
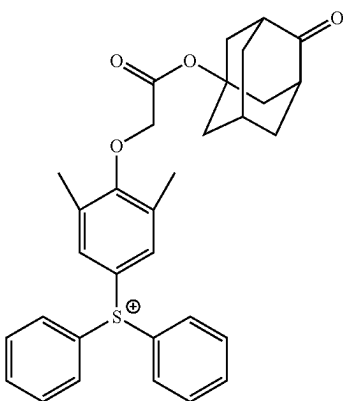
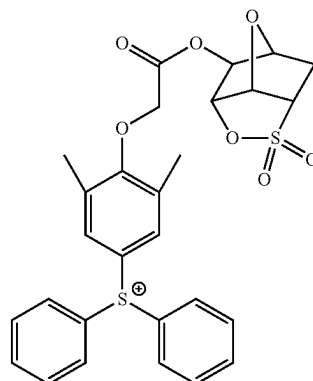
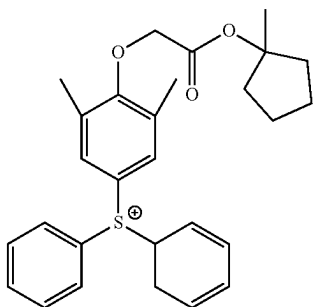
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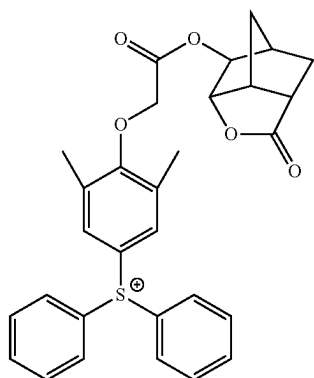
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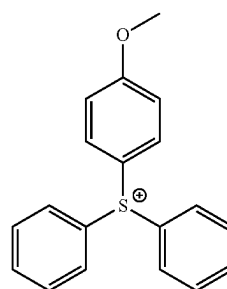
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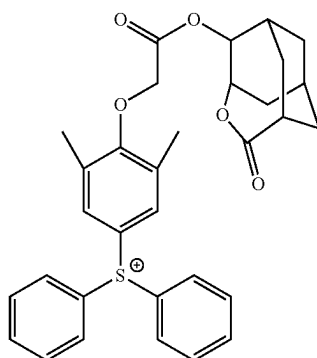
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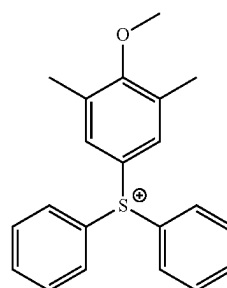
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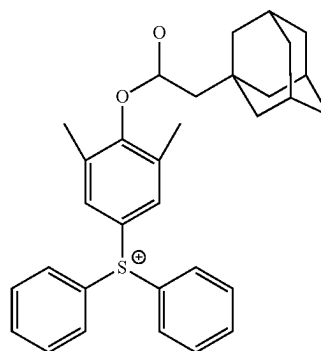
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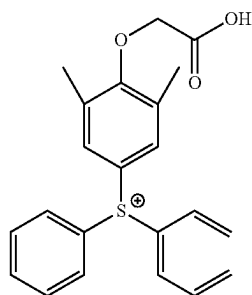
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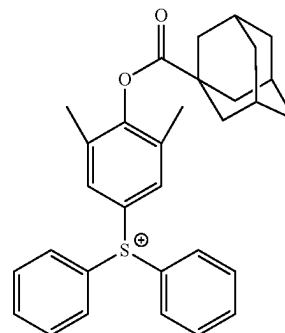
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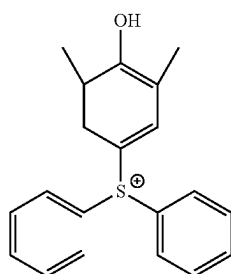
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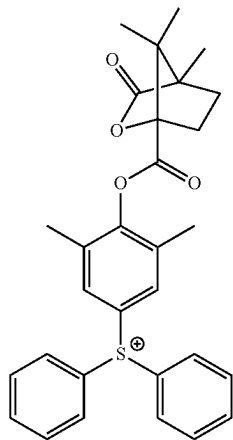


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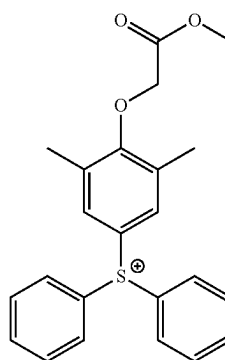
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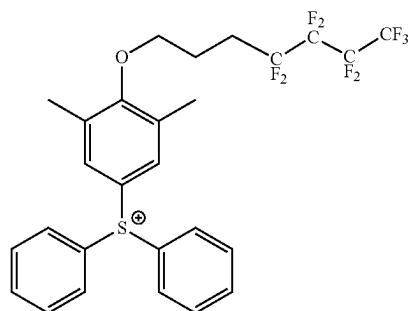
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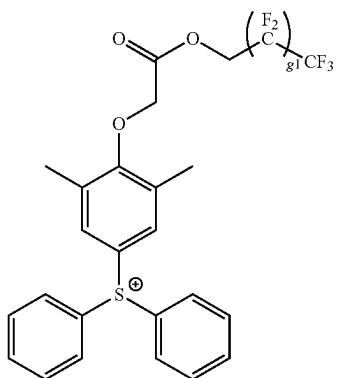
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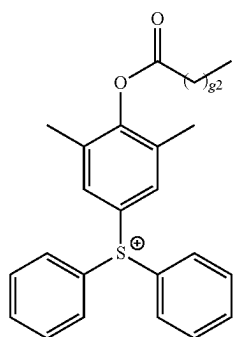
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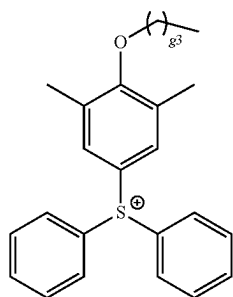
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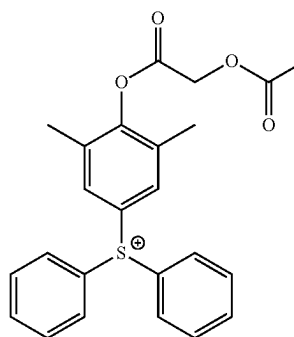
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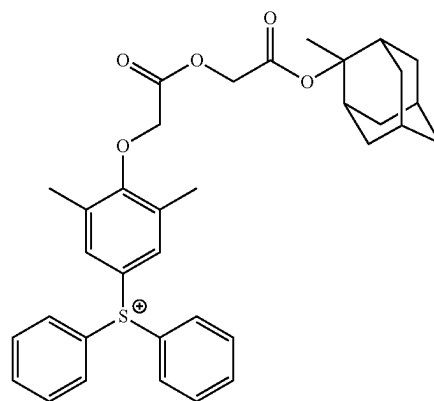
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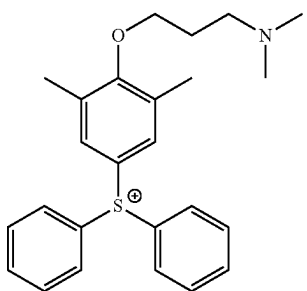
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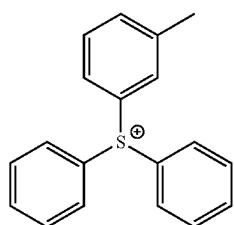
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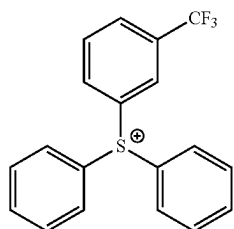
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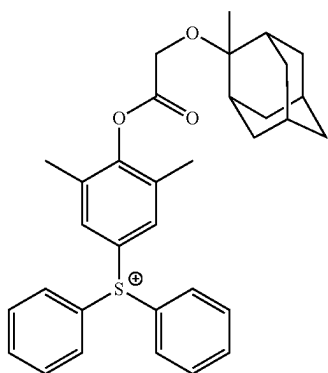
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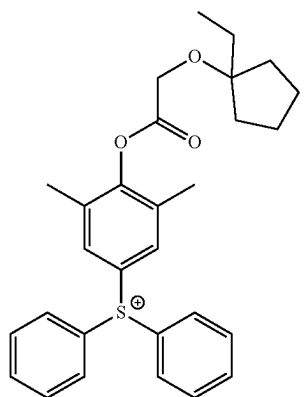
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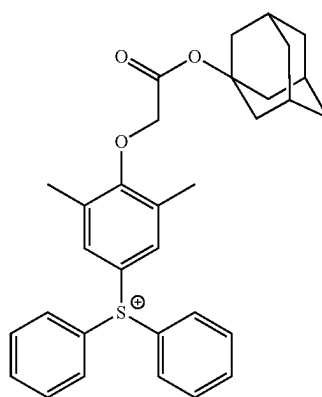


(ca-1-45)



(ca-1-46)

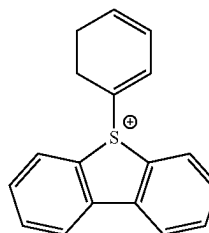
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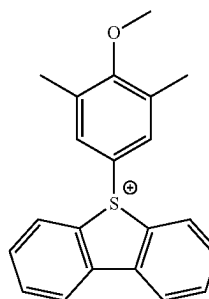
(ca-1-47)

**[0424]** [In the formulae, g1, g2, and g3 represent a repeating number, g1 represents an integer of 1 to 5, g2 represents an integer of 0 to 20, and g3 represents an integer of 0 to 20.

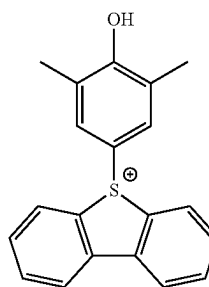
[Chemical Formula 62]



(ca-1-48)

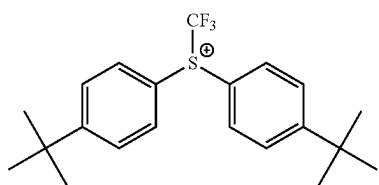
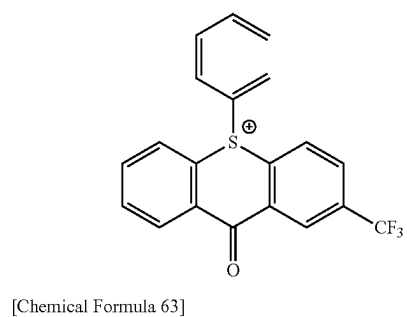
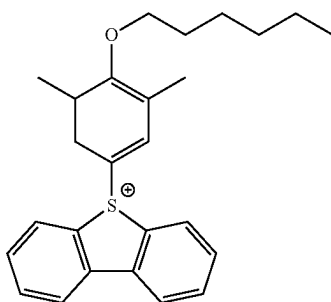
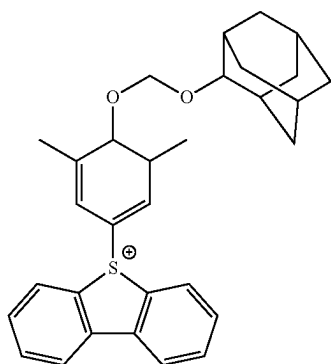
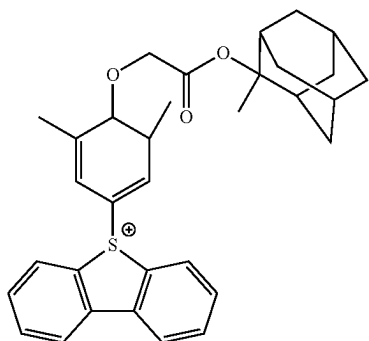


(ca-1-49)

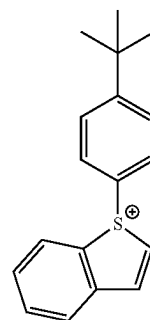
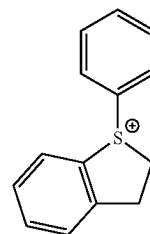
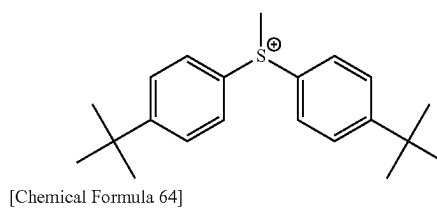
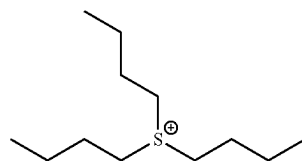
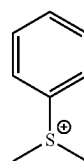
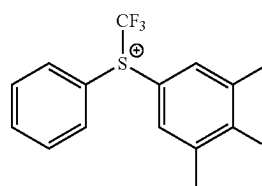
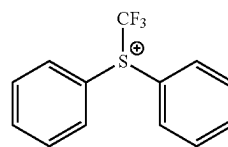


(ca-1-50)

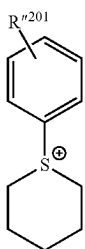
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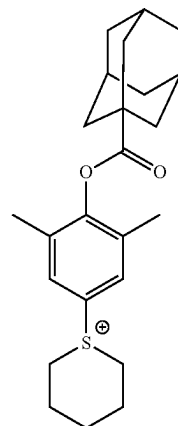


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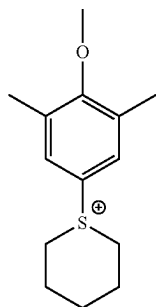


(ca-1-64)

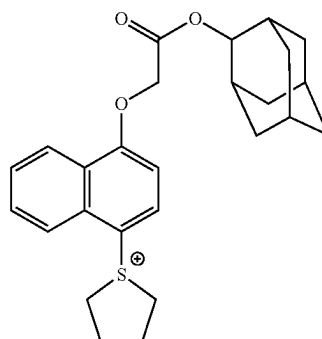
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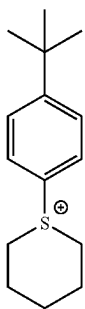
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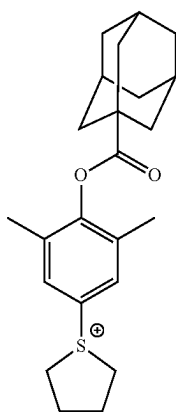
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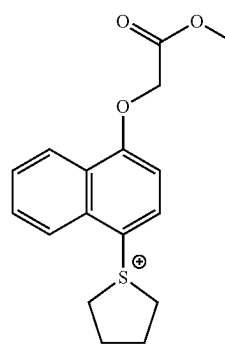
(ca-1-69)



(ca-1-66)



(ca-1-67)

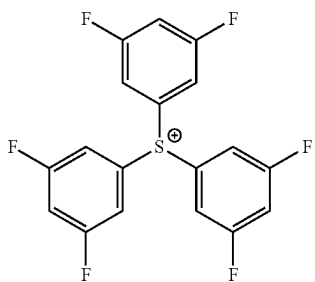
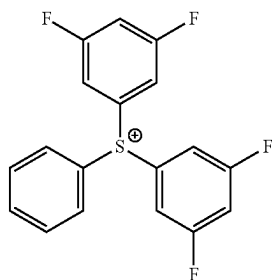
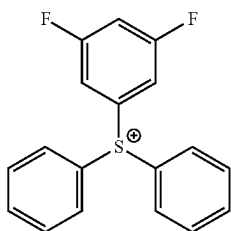
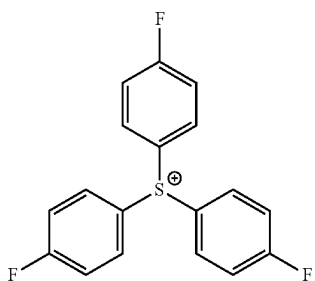
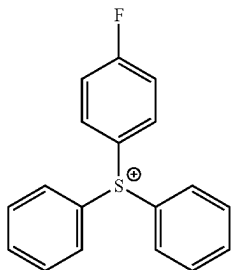


(ca-1-70)

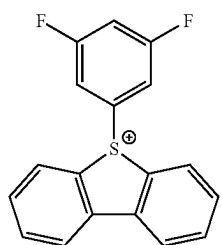
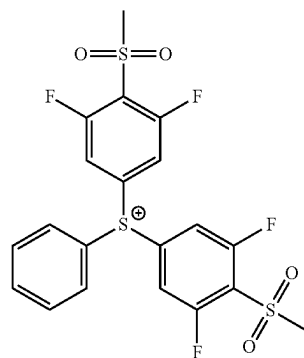
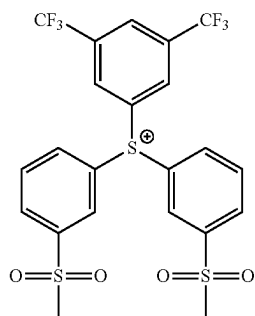
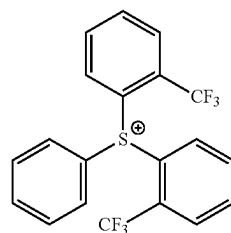
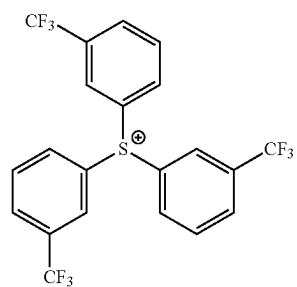
[0425] [In the formulae,  $R^{201}$  represents a hydrogen atom or a substituent, and examples of the substituent include the same groups as those for the substituents that  $R^{201}$  to  $R^{207}$  and  $R^{210}$  to  $R^{212}$  may have.]



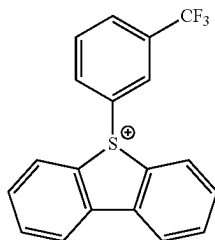
[Chemical Formula 65]



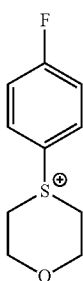
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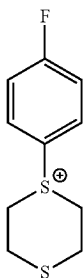
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(ca-1-81)

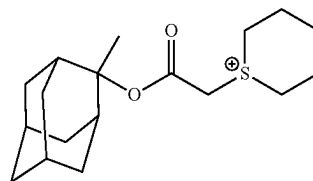


(ca-1-82)

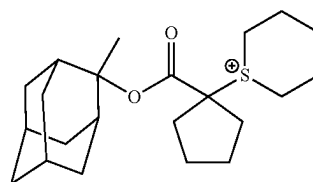


(ca-1-83)

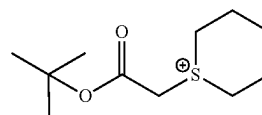
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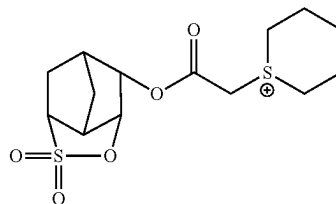
(ca-3-3)



(ca-3-4)



(ca-3-5)

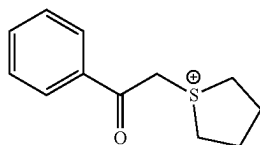


(ca-3-6)

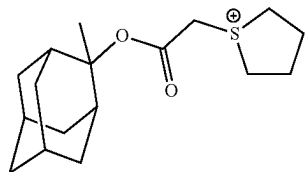
**[0426]** Specific examples of suitable cations represented by Formula (ca-2) include a diphenyliodonium cation and a bis(4-tert-butylphenyl) iodonium cation.

**[0427]** Specific examples of suitable cations represented by Formula (ca-3) include cations each represented by Formulae (ca-3-1) to (ca-3-6).

[Chemical Formula 66]



(ca-3-1)



(ca-3-2)

**[0428]** In the resist composition according to the present embodiment, the component (B) may be used alone or in combination of two or more kinds thereof.

**[0429]** In a case where the resist composition contains the component (B), the content of the component (B) in the resist composition is preferably less than 50 parts by mass, more preferably in a range of 5 to 40 parts by mass, and still more preferably in a range of 10 to 40 parts by mass with respect to 100 parts by mass of the component (A).

**[0430]** In a case where the content of the component (B) is set to be in the above-described preferable ranges, pattern formation can be sufficiently carried out. Further, it is preferable that each component of the resist composition is dissolved in an organic solvent from the viewpoint that a uniform solution is easily obtained and the storage stability of the resist composition is improved.

#### <Other Components>

**[0431]** The resist composition of the present embodiment may further contain other components in addition to the component (A) and the component (B) described above. Examples of other components include a component (D), a component (E), a component (F), and a component (S) described below.

## &lt;&lt;Base Component (D)&gt;&gt;

[0432] The resist composition of the present embodiment may further contain, in addition to the component (A), a base component (component (D)) that traps an acid generated upon light exposure (that is, controls diffusion of an acid). The component (D) acts as a quencher (an acid diffusion control agent) which traps the acid generated in the resist composition upon light exposure.

[0433] Examples of the component (D) include a photodecomposable base (D1) having an acid diffusion controllability (hereinafter, referred to as “component (D1)”) which is lost by the decomposition upon light exposure and a nitrogen-containing organic compound (D2) (hereinafter, referred to as “component (D2)”) which does not correspond to the component (D1). Among these, the photodecomposable base (component (D1)) is preferable from the viewpoint of easily increasing the sensitivity, reducing the roughness, and improving the characteristic of suppressing occurrence of coating defects.

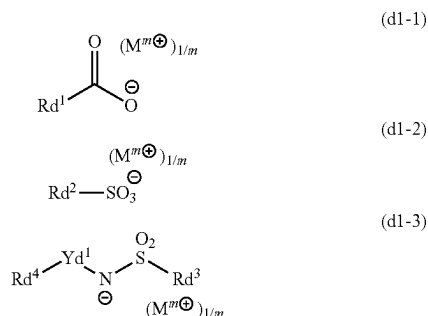
## In Regard to Component (D1)

[0434] In a case where a resist composition containing the component (D1) is obtained, the contrast between an exposed portion and an unexposed portion of the resist film can be further improved in a case of forming a resist pattern.

[0435] The component (D1) is not particularly limited as long as the component is decomposed upon light exposure and loses acid diffusion controllability, and one or more compounds selected from the group consisting of a compound represented by General Formula (d1-1) (hereinafter, referred to as “component (d1-1)”), a compound represented by General Formula (d1-2) (hereinafter, referred to as “component (d1-2)”), and a compound represented by General Formula (d1-3) (hereinafter, referred to as “component (d1-3)”) are preferable.

[0436] Since the components (d1-1) to (d1-3) are decomposed and lose the acid diffusion controllability (basicity), the components (d1-1) to (d1-3) do not function as a quencher at the exposed portion of the resist film, but function as a quencher at the unexposed portion of the resist film.

[Chemical Formula 67]



[0437] [In the formulae, Rd<sup>1</sup> to Rd<sup>4</sup> represent a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent. Here, no fluorine atom is bonded to the carbon atom adjacent to the S atom in Rd<sup>2</sup> of

Formula (d1-2). Yd<sup>1</sup> represents a single bond or a divalent linking group. m represents an integer of 1 or greater, and M<sup>m+</sup>s each independently represent an m-valent organic cation.]

{Component (d1-1)}

## Anion Moiety

[0438] In Formula (d1-1), Rd<sup>1</sup> represents a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof include the same groups as those for R<sup>201</sup>.

[0439] Among these, it is preferable that Rd<sup>1</sup> represents an aromatic hydrocarbon group which may have a substituent, an aliphatic cyclic group which may have a substituent, or a chain-like alkyl group which may have a substituent. Examples of the substituent that may be included in these groups include a hydroxyl group, an oxo group, an alkyl group, an aryl group, a fluorine atom, a fluorinated alkyl group, a lactone-containing cyclic group represented by any of General Formulae (a2-r-1) to (a2-r-7), an ether bond, an ester bond, and a combination thereof. In a case where an ether bond or an ester bond is included as the substituent, the substituent may be bonded through an alkylene group, and a linking group represented by any of Formulae (y-al-1) to (y-al-5) is preferable as the substituent. Further, in a case where the aromatic hydrocarbon group, the aliphatic cyclic group, or the chain-like alkyl group as Rd<sup>1</sup> contain a linking group represented by any of General Formulae (y-al-1) to (y-al-7) as a substituent, V<sup>101</sup> in General Formulae (y-al-1) to (y-al-7) is bonded to the carbon atom constituting the aromatic hydrocarbon group, the aliphatic cyclic group, or the chain-like alkyl group as Rd<sup>1</sup> in Formula (d3-1), in General Formulae (y-al-1) to (y-al-7).

[0440] Suitable examples of the aromatic hydrocarbon group include a phenyl group, a naphthyl group, and a polycyclic structure having a bicyclooctane skeleton (for example, a polycyclic structure formed of a bicyclooctane skeleton and a ring structure other than the bicyclooctane skeleton).

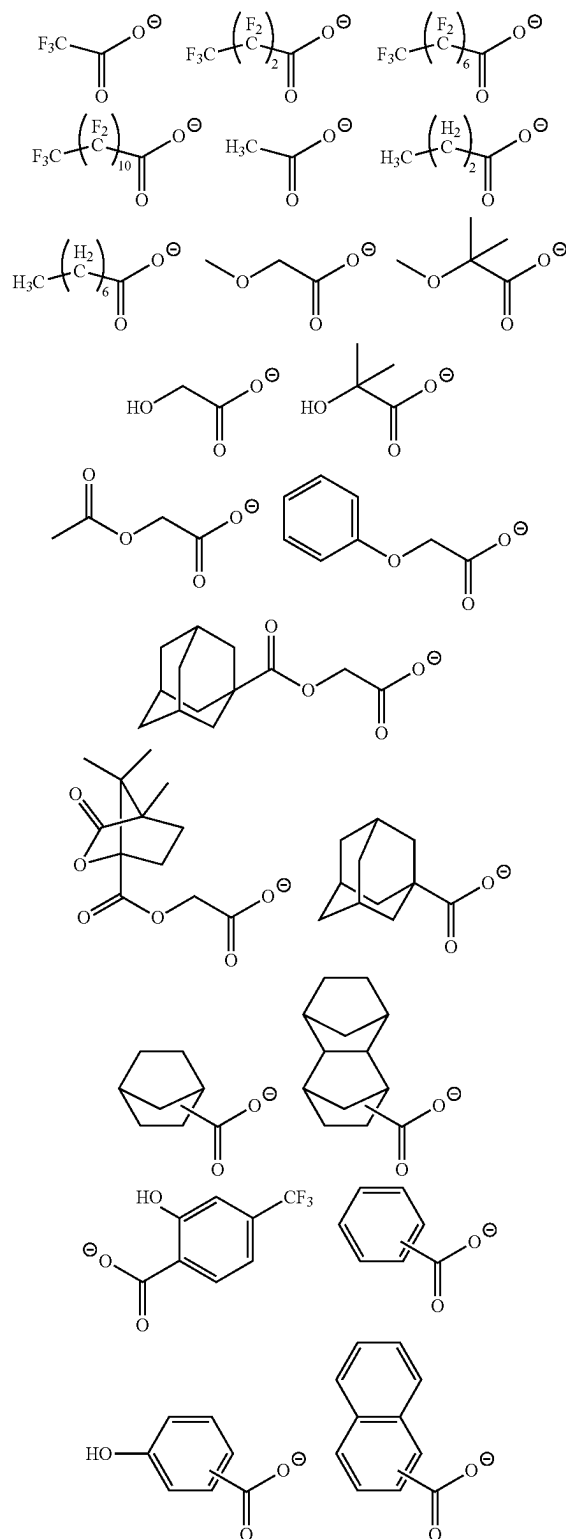
[0441] As the aliphatic cyclic group, a group in which one or more hydrogen atoms have been removed from a polycycloalkane such as adamantane, norbornane, isobornane, tricyclodecane or tetracyclodecane is more preferable.

[0442] It is preferable that the chain-like alkyl group has 1 to 10 carbon atoms, and specific examples thereof include a linear alkyl group such as a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a hexyl group, a heptyl group, an octyl group, a nonyl group, or a decyl group; and a branched alkyl group such as a 1-methylethyl group, a 1-methylpropyl group, a 2-methylpropyl group, a 1-methylbutyl group, a 2-methylbutyl group, a 3-methylbutyl group, a 1-ethylbutyl group, a 2-ethylbutyl group, a 1-methylpentyl group, a 2-methylpentyl group, a 3-methylpentyl group, or a 4-methylpentyl group.

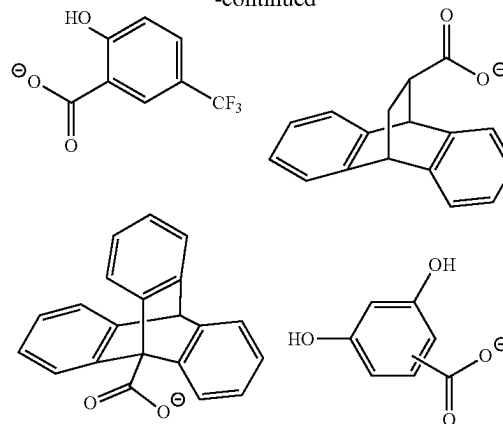
[0443] In a case where the chain-like alkyl group is a fluorinated alkyl group having a fluorine atom or a fluorinated alkyl group as a substituent, the fluorinated alkyl group has preferably 1 to 11 carbon atoms, more preferably 1 to 8 carbon atoms, and still more preferably 1 to 4 carbon atoms. The fluorinated alkyl group may have an atom other than a fluorine atom. Examples of the atom other than a fluorine atom include an oxygen atom, a sulfur atom, and a nitrogen atom.

[0444] Specific preferred examples of the anion moiety in the component (d1-1) are described below.

[Chemical Formula 68]



-continued



#### Cation Moiety

[0445] In Formula (d1-1),  $M^{m+}$  represents an m-valent organic cation.

[0446] Suitable examples of the organic cation as  $M^{m+}$  include the same cations as the cations each represented by General Formulae (ca-1) to (ca-5). Among these, a cation represented by General Formula (ca-1) is more preferable, and cations each represented by Formulae (ca-1-1) to (ca-1-70) are still more preferable.

[0447] The component (d1-1) may be used alone or in combination of two or more kinds thereof.

{Component (d1-2)}

#### Anion Moiety

[0448] In Formula (d1-2),  $Rd^2$  represents a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof include the same groups as those for  $R^{201}$ .

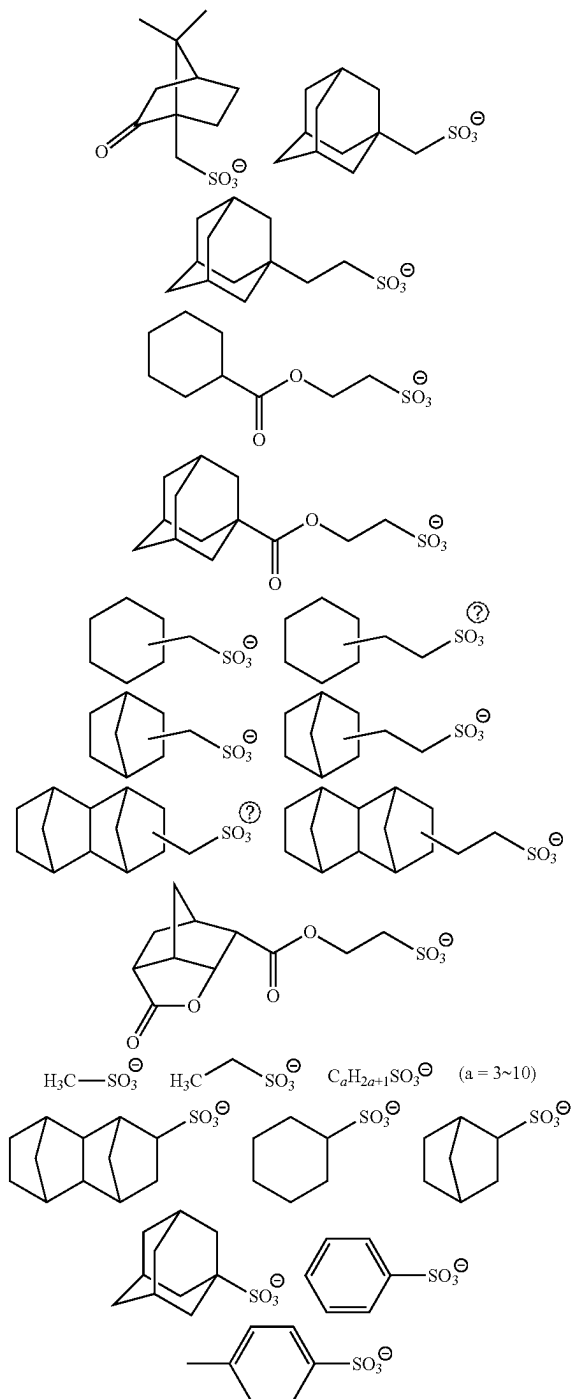
[0449] Here, no fluorine atom is bonded to the carbon atom adjacent to the S atom in  $Rd^2$  (the carbon atom is not substituted with fluorine). In this manner, the anion of the component (d1-2) is an appropriately weak acid anion, thereby improving the quenching ability of the component (D).

[0450] It is preferable that  $Rd^2$  represents a chain-like alkyl group which may have a substituent or an aliphatic cyclic group which may have a substituent. The chain-like alkyl group has preferably 1 to 10 carbon atoms and more preferably 3 to 10 carbon atoms. As the aliphatic cyclic group, a group in which one or more hydrogen atoms have been removed from adamantane, norbornane, isobornane, tricyclodecane, or tetracyclododecane (a group which may have a substituent); and a group in which one or more hydrogen atoms have been removed from camphor are more preferable.

[0451] The hydrocarbon group as  $Rd^2$  may have a substituent, and examples of the substituent include the same groups as those for the substituent that the hydrocarbon group (such as an aromatic hydrocarbon group, an aliphatic cyclic group, or a chain-like alkyl group) as  $Rd^1$  in Formula (d1-1) may have.

[0452] Specific preferred examples of the anion moiety in the component (d1-2) are described below.

[Chemical Formula 69]



② indicates text missing or illegible when filed

## Cation Moiety

**[0453]** In Formula (d1-2),  $M^{m+}$  represents an m-valent organic cation and has the same definition as that for  $M^{m+}$  in Formula (d1-1).

**[0454]** The component (d1-2) may be used alone or in combination of two or more kinds thereof.  
{Component (d1-3))}

## Anion Moiety

**[0455]** In Formula (d1-3),  $Rd^2$  represents a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof include the same groups as those for  $R^{1201}$ . Among these, a cyclic group having a fluorine atom, a chain-like alkyl group, or a chain-like alkenyl group is preferable. Among these, a fluorinated alkyl group is preferable, and the same groups as those for the fluorinated alkyl group represented by  $Rd^1$  are more preferable.

**[0456]** In Formula (d1-3),  $Rd^4$  represents a cyclic group which may have a substituent, a chain-like alkyl group which may have a substituent, or a chain-like alkenyl group which may have a substituent, and examples thereof include the same groups as those for  $R^{1201}$ .

[0457] Among these, an alkyl group which may have a substituent, an alkoxy group which may have a substituent, an alkenyl group which may have a substituent, or a cyclic group which may have a substituent is preferable.

[0458] It is preferable that the alkyl group as  $Rd^4$  is a linear or branched alkyl group having 1 to 5 carbon atoms, and specific examples thereof include a methyl group, an ethyl group, a propyl group, an isopropyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, and a neopentyl group. Some hydrogen atoms in the alkyl group as  $Rd^4$  may be substituted with a hydroxyl group, a cyano group, or the like.

**[0459]** It is preferable that the alkoxy group as  $Rd^4$  is an alkoxy group having 1 to 5 carbon atoms, and specific examples of the alkoxy group having 1 to 5 carbon atoms include a methoxy group, an ethoxy group, an n-propoxy group, an iso-propoxy group, an n-butoxy group, and a tert-butoxy group. Among these, a methoxy group and an ethoxy group are preferable.

**[0460]** Examples of the alkenyl group as  $Rd^4$  include the same groups as those for the alkenyl group as  $R^{1201}$ . Among these, a vinyl group, a propenyl group (an allyl group), a 1-methylpropenyl group, and a 2-methylpropenyl group are preferable. These groups may have an alkyl group having 1 to 5 carbon atoms or a halogenated alkyl group having 1 to 5 carbon atoms as a substituent.

**[0461]** Examples of the cyclic group as  $Rd^4$  include the same groups as those for the cyclic group as  $R^{1201}$ . Among these, an alicyclic group in which one or more hydrogen atoms have been removed from a cycloalkane such as cyclopentane, cyclohexane, adamantane, norbornane, isobornane, tricyclodecane or tetracyclododecane or an aromatic group such as a phenyl group or a naphthyl group is preferable. In a case where  $Rd^4$  represents an alicyclic group, the resist composition is satisfactorily dissolved in an organic solvent so that the lithography characteristics are enhanced. Further, in a case where  $Rd^4$  represents an aromatic group, the resist composition has excellent light absorption efficiency in lithography using EUV or the like as an exposure light source, and thus the sensitivity and lithography characteristics are enhanced.

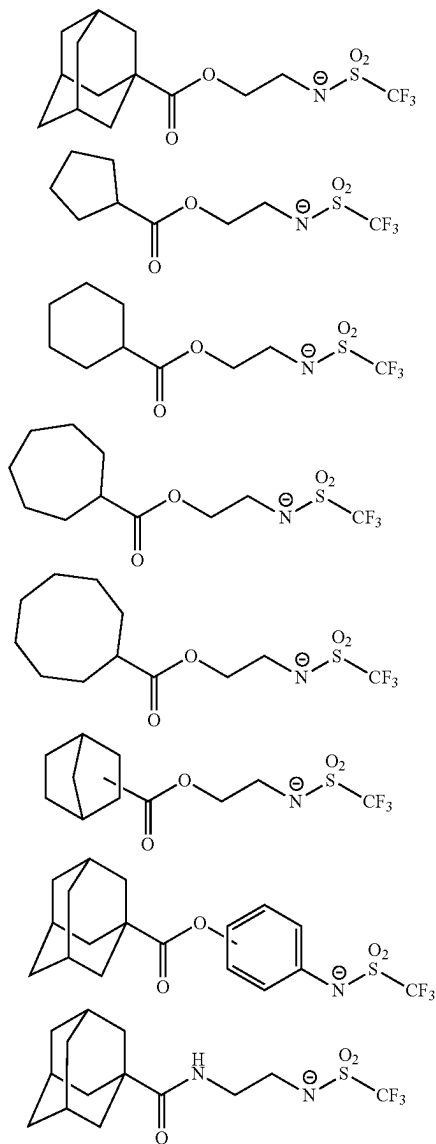
**[0462]** In Formula (d1-3), Yd<sup>1</sup> represents a single bond or a divalent linking group.

[0463] The divalent linking group as Yd<sup>1</sup> is not particularly limited, and examples thereof include a divalent hydrocarbon group (an aliphatic hydrocarbon group or an aromatic hydrocarbon group) which may have a substituent and a divalent linking group having a heteroatom. Examples of the divalent linking groups are the same as those for the divalent hydrocarbon group which may have a substituent and the divalent linking group having a heteroatom described in the section of the divalent linking group as Ya<sup>21</sup> in Formula (a2-1).

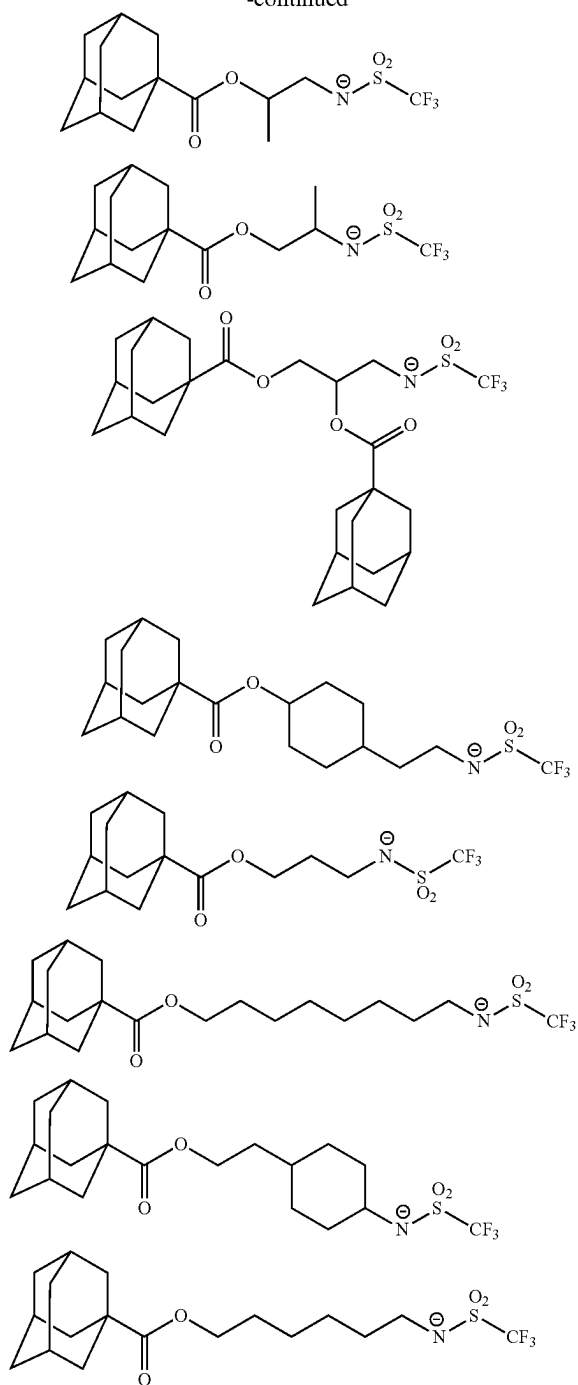
[0464] It is preferable that Yd<sup>1</sup> represents a carbonyl group, an ester bond, an amide bond, an alkylene group, or a combination thereof. As the alkylene group, a linear or branched alkylene group is more preferable, and a methylene group or an ethylene group is still more preferable.

[0465] Specific preferred examples of the anion moiety in the component (d1-3) are described below.

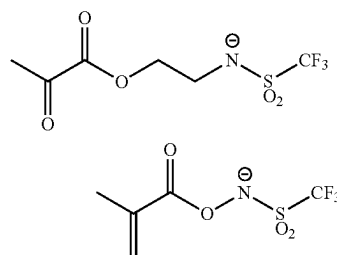
[Chemical Formula 70]

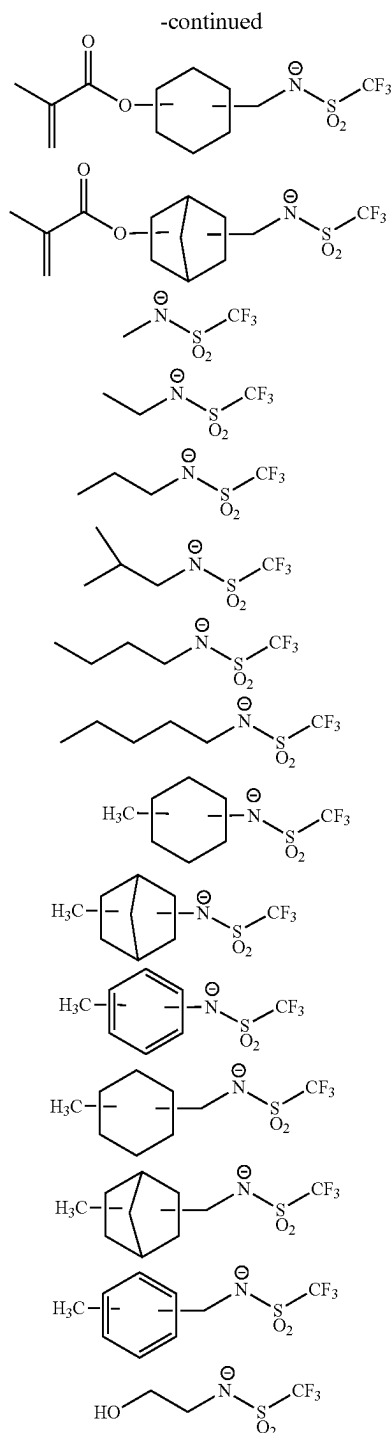


-continued



[Chemical Formula 71]





#### Cation Moiety

**[0466]** In Formula (d1-3),  $M^{m+}$  represents an m-valent organic cation and has the same definition as that for  $M^{m+}$  in Formula (d1-1).

**[0467]** The component (d1-3) may be used alone or in combination of two or more kinds thereof.

**[0468]** As the component (D1), only one of the above-described components (d1-1) to (d1-3) or a combination of two or more kinds thereof may be used.

**[0469]** In a case where the resist composition contains the component (D1), the content of the component (D1) in the resist composition is preferably in a range of 0.5 to 20 parts by mass, more preferably in a range of 1 to 15 parts by mass, and still more preferably in a range of 2 to 8 parts by mass with respect to 100 parts by mass of the component (A1).

**[0470]** In a case where the content of the component (D1) is greater than or equal to the lower limits of the above-described ranges, particularly excellent lithography characteristics and an excellent resist pattern shape are easily obtained. On the contrary, in a case where the content is less than or equal to the upper limits of the above-described ranges, the sensitivity can be satisfactorily maintained and the throughput is also excellent.

#### Method of Producing Component (D1):

**[0471]** The methods of producing the component (d1-1) and the component (d1-2) are not particularly limited, and these components can be produced by known methods.

**[0472]** Further, the method of producing the component (d1-3) is not particularly limited, and the component is produced by the same method as disclosed in United States Patent Application, Publication No. 2012-0149916.

#### In Regard to Component (D2)

**[0473]** The component (D) may contain a nitrogen-containing organic compound component (hereinafter, referred to as "component (D2)") that does not correspond to the component (D1) described above.

**[0474]** The component (D2) is not particularly limited as long as the component functions as an acid diffusion control agent and does not correspond to the component (D1), and an optional component may be selected from known components and then used. Among the examples, an aliphatic amine is preferable, and particularly a secondary aliphatic amine and a tertiary aliphatic amine are more preferable.

**[0475]** The aliphatic amine is an amine containing one or more aliphatic groups, and the number of carbon atoms in the aliphatic group is preferably in a range of 1 to 12.

**[0476]** Examples of the aliphatic amine include amines in which at least one hydrogen atom of ammonia  $NH_3$  has been substituted with an alkyl group or hydroxyalkyl group having 12 or less carbon atoms (alkylamines or alkylalcoholamines), and cyclic amines.

**[0477]** Specific examples of the alkylamines and the alkylalcoholamines include monoalkylamines such as n-hexylamine, n-heptylamine, n-octylamine, n-nonylamine, and n-decylamine; dialkylamines such as diethylamine, di-n-propylamine, di-n-heptylamine, di-n-octylamine, and dicyclohexylamine; trialkylamines such as trimethylamine, triethylamine, tri-n-propylamine, tri-n-butylamine, tri-n-pentylamine, tri-n-hexylamine, tri-n-heptylamine, tri-n-octylamine, tri-n-nonylamine, tri-n-decylamine, and tri-n-dodecylamine; and alkylalcoholamines such as diethanolamine, triethanolamine, diisopropanolamine, triisopropanolamine, di-n-octanolamine, and tri-n-octanolamine. Among these, a trialkylamine having 5 to 10 carbon atoms is preferable, and tri-n-pentylamine and tri-n-octylamine are particularly preferable.

[0478] Examples of the cyclic amine include a heterocyclic compound having a nitrogen atom as a heteroatom. The heterocyclic compound may be a monocyclic compound (aliphatic monocyclic amine) or a polycyclic compound (aliphatic polycyclic amine).

[0479] Specific examples of the aliphatic monocyclic amine include piperidine and piperazine.

[0480] The aliphatic polycyclic amine preferably has 6 to 10 carbon atoms, and specific examples thereof include 1,5-diazabicyclo[4.3.0]-5-nonene, 1,8-diazabicyclo[5.4.0]-7-undecene, hexamethylenetetramine, and 1,4-diazabicyclo[2.2.2]octane.

[0481] Examples of other aliphatic amines include tris(2-methoxymethoxyethyl)amine, tris{2-(2-methoxyethoxy)ethyl}amine, tris{2-(2-methoxyethoxymethoxy)ethyl}amine, tris{2-(1-methoxyethoxy)ethyl}amine, tris{2-(1-ethoxyethoxy)ethyl}amine, tris{2-(1-ethoxypropoxy)ethyl}amine, tris[2-{2-(2-hydroxyethoxy)ethoxy}ethyl]amine, and triethanolamine triacetate. Among these, triethanolamine triacetate is preferable.

[0482] As the component (D2), an aromatic amine may be used.

[0483] Examples of aromatic amines include 4-dimethylaminopyridine, pyrrole, indole, pyrazole, imidazole, and derivatives thereof, tribenzylamine, 2,6-diisopropylaniline, N-tert-butoxycarbonylpyrrolidine, and 2,6-di-tert-butylpyridine.

[0484] The component (D2) may be used alone or in combination of two or more kinds thereof.

[0485] In a case where the resist composition contains the component (D2), the content of the component (D2) in the resist composition is typically in a range of 0.01 to 5 parts by mass with respect to 100 parts by mass of the component (A1). In a case where the content thereof is set to be in the above-described range, the resist pattern shape, the post exposure temporal stability, and the like are improved.

<<At Least One Compound (E) Selected from Group Consisting of Organic Carboxylic Acids, Phosphorus Oxo Acids, and Derivatives Thereof>>

[0486] For the purpose of preventing any deterioration in sensitivity and improving the resist pattern shape and the post-exposure temporal stability, the resist composition according to the present embodiment may contain, as an optional component, at least one compound (E) (hereinafter referred to as “component (E)”) selected from the group consisting of an organic carboxylic acid, and a phosphorus oxo acid and a derivative thereof.

[0487] Specific examples of the organic carboxylic acid include acetic acid, malonic acid, citric acid, malic acid, succinic acid, benzoic acid, and salicylic acid. Among these, salicylic acid is preferable.

[0488] Examples of the phosphorus oxo acid include phosphoric acid, phosphonic acid, and phosphinic acid. Among these, phosphonic acid is particularly preferable.

[0489] Examples of the phosphorus oxo acid derivative include an ester obtained by substituting a hydrogen atom in the above-described oxo acid with a hydrocarbon group. Examples of the hydrocarbon group include an alkyl group having 1 to 5 carbon atoms and an aryl group having 6 to 15 carbon atoms.

[0490] Examples of the phosphoric acid derivatives include phosphoric acid esters such as phosphoric acid di-n-butyl ester and phosphoric acid diphenyl ester.

[0491] Examples of the phosphonic acid derivatives include phosphonic acid esters such as phosphonic acid dimethyl ester, phosphonic acid di-n-butyl ester, phenylphosphonic acid, phosphonic acid diphenyl ester, and phosphonic acid dibenzyl ester.

[0492] Examples of the phosphinic acid derivatives include phosphinic acid ester and phenylphosphinic acid.

[0493] In the resist composition of the present embodiment, the component (E) may be used alone or in combination of two or more kinds thereof.

[0494] In a case where the resist composition contains the component (E), the content of the component (E) is preferably in a range of 0.01 to 5 parts by mass and more preferably in a range of 0.05 to 3 parts by mass with respect to 100 parts by mass of the component (A). In a case where the content thereof is in the above-described range, the sensitivity, lithography characteristics, and the like are improved.

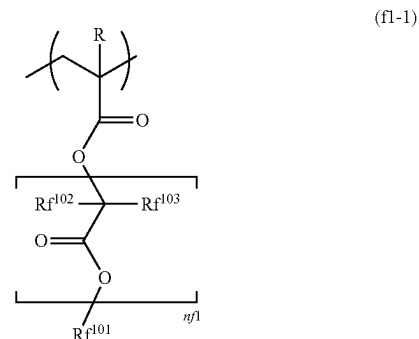
<<Fluorine Additive Component (F)>>

[0495] The resist composition according to the present embodiment may further contain a fluorine additive component (hereinafter, referred to as “component (F)”) as a hydrophobic resin. The component (F) is used to impart water repellency to the resist film and used as a resin different from the component (A), whereby the lithography characteristics can be improved.

[0496] As the component (F), for example, the fluorine-containing polymer compounds described in Japanese Unexamined Patent Application, First Publication Nos. 2010-002870, 2010-032994, 2010-277043, 2011-13569, and 2011-128226 can be used.

[0497] Specific examples of the component (F) include a polymer having a constitutional unit (f1) represented by General Formula (f1-1). As the polymer, a polymer (homopolymer) formed of only the constitutional unit (f1) represented by Formula (f1-1); a copolymer of the constitutional unit (f1) and the constitutional unit (a1); or a copolymer of the constitutional unit (f1), a constitutional unit derived from acrylic acid or methacrylic acid, and the constitutional unit (a1) is preferable, and a copolymer of the constitutional unit (f1) and the constitutional unit (a1) is more preferable. Here, as the constitutional unit (a1) copolymerized with the constitutional unit (f1), a constitutional unit derived from 1-ethyl-1-cyclooctyl (meth)acrylate or a constitutional unit derived from 1-methyl-1-adamantyl (meth)acrylate is preferable, and a constitutional unit derived from 1-ethyl-1-cyclooctyl (meth)acrylate is more preferable.

[Chemical Formula 72]





[0498] [In the formula, R has the same definition as described above, R<sup>f102</sup> and R<sup>f103</sup> each independently represent a hydrogen atom, a halogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, and R<sup>f102</sup> and R<sup>f103</sup> may be the same as or different from each other. n<sup>f1</sup> represents an integer of 0 to 5, and R<sup>f101</sup> represents an organic group having a fluorine atom.]

[0499] In Formula (f1-1), R bonded to the carbon atom at the  $\alpha$ -position has the same definition as described above. It is preferable that R represents a hydrogen atom or a methyl group.

[0500] In Formula (f1-1), a fluorine atom is preferable as the halogen atom as R<sup>f102</sup> and R<sup>f103</sup>. Examples of the alkyl group having 1 to 5 carbon atoms as R<sup>f102</sup> and R<sup>f103</sup> include the same groups as those for the alkyl group having 1 to 5 carbon atoms as R. Among the examples, a methyl group or an ethyl group is preferable. Specific examples of the halogenated alkyl group having 1 to 5 carbon atoms as R<sup>f102</sup> and R<sup>f103</sup> include groups in which some or all hydrogen atoms of an alkyl group having 1 to 5 carbon atoms are substituted with halogen atoms. Among these, a fluorine atom is preferable as the halogen atom. Among these, R<sup>f102</sup> and R<sup>f103</sup> represent preferably a hydrogen atom, a fluorine atom, or an alkyl group having 1 to 5 carbon atoms, more preferably a hydrogen atom, a fluorine atom, a methyl group, or an ethyl group, and still more preferably a hydrogen atom.

[0501] In Formula (f1-1), n<sup>f1</sup> represents an integer of 0 to 5, preferably an integer of 0 to 3, and more preferably 1 or 2.

[0502] In Formula (f1-1), R<sup>f101</sup> represents an organic group having a fluorine atom and preferably a hydrocarbon group having a fluorine atom.

[0503] The hydrocarbon group having a fluorine atom may be linear, branched, or cyclic, and the number of carbon atoms thereof is preferably in a range of 1 to 20, more preferably in a range of 1 to 15, and particularly preferably in a range of 1 to 10.

[0504] In the hydrocarbon group having a fluorine atom, preferably 25% or more of the hydrogen atoms in the hydrocarbon group are fluorinated, more preferably 50% or more thereof are fluorinated, and particularly preferably 60% or more thereof are fluorinated from the viewpoint of increasing the hydrophobicity of the resist film during immersion exposure.

[0505] Among the examples, R<sup>f101</sup> represents more preferably a fluorinated hydrocarbon group having 1 to 6 carbon atoms and particularly preferably a trifluoromethyl group, —CH<sub>2</sub>—CF<sub>3</sub>, —CH<sub>2</sub>—CF<sub>2</sub>—CF<sub>3</sub>, —CH(CF<sub>3</sub>)<sub>2</sub>, —CH<sub>2</sub>—CH<sub>2</sub>—CF<sub>3</sub>, or —CH<sub>2</sub>—CH<sub>2</sub>—CF<sub>2</sub>—CF<sub>2</sub>—CF<sub>2</sub>—CF<sub>3</sub>.

[0506] The weight-average molecular weight (Mw) (in terms of polystyrene according to gel permeation chromatography) of the component (F) is preferably in a range of 1000 to 50000, more preferably in a range of 5000 to 40000, and most preferably in a range of 10000 to 30000. In a case where the weight-average molecular weight thereof is less than or equal to the upper limits of the above-described ranges, the resist composition exhibits a satisfactory solubility in a solvent for a resist enough to be used as a resist. Meanwhile, in a case where the weight-average molecular weight thereof is greater than or equal to the lower limits of the above-described ranges, water repellency of the resist film is improved.

[0507] Further, the dispersity (Mw/Mn) of the component (F) is preferably in a range of 1.0 to 5.0, more preferably in a range of 1.0 to 3.0, and most preferably in a range of 1.0 to 2.5.

[0508] In the resist composition according to the present embodiment, the component (F) may be used alone or in combination of two or more kinds thereof.

[0509] In a case where the resist composition contains the component (F), the content of the component (F) is preferably in a range of 0.5 to 10 parts by mass and more preferably in a range of 1 to 10 parts by mass with respect to 100 parts by mass of the component (A).

#### <<Organic Solvent Component(S)>>

[0510] The resist composition of the present embodiment can be produced by dissolving the resist materials in an organic solvent component (hereinafter, referred to as "component(S)").

[0511] The component(S) may be any organic solvent which can dissolve each component to be used to obtain a uniform solution, and an optional organic solvent can be appropriately selected from those which have been known as solvents of a chemically amplified resist composition and then used.

[0512] In the resist composition of the present embodiment, the component(S) may be used alone or in the form of a mixed solvent of two or more kinds thereof. Among these, PGMEA, PGME,  $\gamma$ -butyrolactone, EL, or cyclohexanone is preferable.

[0513] Further, a mixed solvent obtained by mixing PGMEA with a polar solvent is also preferable as the component(S). The blending ratio (mass ratio) may be appropriately determined in consideration of the compatibility between PGMEA and the polar solvent.

[0514] Further, a mixed solvent of  $\gamma$ -butyrolactone and at least one selected from PGMEA and EL is also preferable as the component(S). In this case, as the mixing ratio, the mass ratio between the former and the latter is preferably in a range of 70:30 to 95:5.

[0515] The amount of the component(S) to be used is not particularly limited and is appropriately set to have a concentration which enables coating a substrate or the like depending on the thickness of the coated film. The component(S) is typically used in an amount such that the solid content concentration of the resist composition is set to be in a range of 0.1% to 20% by mass and preferably in a range of 0.2% to 15% by mass.

[0516] As desired, miscible additives such as additive resins, dissolution inhibitors, plasticizers, stabilizers, colorants, halation prevention agents, and dyes for improving the performance of the resist film can be added to the resist composition of the present embodiment, as appropriate.

[0517] After the resist material is dissolved in the component(S), impurities may be removed from the resist composition of the present embodiment using a porous polyimide film, a porous polyamideimide film, or the like. For example, the resist composition may be filtered using a filter formed of a porous polyimide film, a filter formed of a porous polyamideimide film, a filter formed of a porous polyimide film and a porous polyamideimide film, or the like. Examples of the porous polyimide film and the porous polyamideimide film include those described in Japanese Unexamined Patent Application, First Publication No. 2016-155121.

[0518] The resist composition according to the present embodiment described above contains the resin component (A1) having the constitutional unit (a0) derived from the compound represented by General Formula (a0-1).

[0519] In the constitutional unit (a0), the cyclic alkene A is bonded to the carbonyloxy group in Formula (a0-1) as an acid dissociable group to form an acid decomposable group. At least one iodine atom (I) is bonded to the cyclic alkene A. The iodine atom has a large absorption of EUV having a wavelength of 13.5 nm.

[0520] In addition, the cyclic alkene A has a carbon-carbon unsaturated bond in the ring skeleton. Therefore, the stability of the carbocation formed by the dissociation of the cyclic alkene A by the action of an acid is likely to increase. In particular, in a case where the carbocation is formed between the carbon atom (C<sup>a</sup>) at a position ( $\alpha$ -position of the carbon atom (C<sup>d</sup>)) adjacent to the carbon atom (C<sup>d</sup>) which forms the ring skeleton of the cyclic alkene A bonded to the carbonyloxy group in Formula (a0-1) and the carbon atom (C<sup>b</sup>) at a position ( $\beta$ -position of the carbon atom (C<sup>d</sup>)) adjacent to the carbon atom (C<sup>d</sup>), the stability of the above-described carbocation is likely to increase.

[0521] In conjunction with the above-described effects, the resist composition according to the present embodiment is assumed to achieve high sensitivity.

(Method for Forming Resist Pattern)

[0522] A resist pattern formation method according to the second aspect of the present invention is a method including a step of forming a resist film on a support using the resist composition according to the first aspect of the present invention described above, a step of exposing the resist film to light, and a step of developing the resist film exposed to light to form a resist pattern.

[0523] According to the embodiment of the method for forming a resist pattern, a method for forming a resist pattern, which is performed in the following manner is an exemplary example.

[0524] First, a support is coated with the resist composition of the present embodiment using a spinner or the like, and a bake (post applied bake (PAB)) treatment is performed under a temperature condition of 80° C. to 150° C. for 40 to 120 seconds and preferably 60 to 90 seconds to form a resist film.

[0525] Next, the selective exposure is performed on the resist film by, for example, light exposure through a mask (mask pattern) having a predetermined pattern formed thereon using an exposure apparatus such as an electron beam lithography apparatus or an ArF exposure apparatus, or direct irradiation with an electron beam for drawing without using a mask pattern, and a bake treatment (post-exposure bake (PEB)) is carried out, for example, under a temperature condition of 80° C. to 150° C. for 40 to 120 seconds and preferably 60 to 90 seconds.

[0526] Next, the resist film is subjected to a developing treatment. The developing treatment is conducted using an alkali developing solution in a case of an alkali developing process and using a developing solution containing an organic solvent (organic developing solution) in a case of a solvent developing process.

[0527] After the developing treatment, it is preferable to conduct a rinse treatment. As the rinse treatment, water rinsing using pure water is preferable in a case of the alkali

developing process, and rinsing using a rinse solution containing an organic solvent is preferable in a case of the solvent developing process.

[0528] In a case of the solvent developing process, after the developing treatment or the rinse treatment, the developing solution or the rinse solution attached onto the pattern may be removed by a treatment using a supercritical fluid.

[0529] After the developing treatment or the rinse treatment, drying is conducted. As desired, a bake treatment (post-bake) may be conducted after the developing treatment.

[0530] The support is not particularly limited and a known support of the related art can be used, and examples thereof include a substrate for an electronic component and a substrate on which a predetermined wiring pattern has been formed. Specific examples thereof include a metal substrate such as a silicon wafer, copper, chromium, iron, or aluminum; and a glass substrate. As the materials of the wiring pattern, copper, aluminum, nickel, or gold can be used.

[0531] The wavelength used for light exposure is not particularly limited and the light exposure can be conducted using radiation such as an ArF excimer laser, a KrF excimer laser, an F<sub>2</sub> excimer laser, extreme ultraviolet rays (EUV), vacuum ultraviolet rays (VUV), electron beams (EB), X-rays, and soft X-rays.

[0532] The method of exposing the resist film to light can be general exposure (dry exposure) conducted in air or an inert gas such as nitrogen, or liquid immersion exposure (liquid immersion lithography).

[0533] The liquid immersion exposure is an exposure method in which the region between the resist film and the lens at the lowermost position of the exposure apparatus is filled with a solvent (liquid immersion medium) in advance that has a refractive index greater than the refractive index of air, and the exposure (immersion exposure) is conducted in this state.

[0534] As the liquid immersion medium, a solvent having a refractive index greater than the refractive index of air but less than the refractive index of the resist film to be exposed is preferable, and examples thereof include water, a fluorine-based inert liquid, a silicon-based solvent, and a hydrocarbon-based solvent.

[0535] As the liquid immersion medium, water is preferably used.

[0536] As the alkali developing solution used for the developing treatment in the alkali developing process, a 0.1 to 10 mass % tetramethylammonium hydroxide (TMAH) aqueous solution is an exemplary example.

[0537] The organic solvent contained in the organic developing solution used for the developing treatment in the solvent developing process may be any solvent that is capable of dissolving the component (A) (the component (A) before light exposure) and can be appropriately selected from known organic solvents. Specific examples thereof include a polar solvent such as a ketone-based solvent, an ester-based solvent, an alcohol-based solvent, a nitrile-based solvent, an amide-based solvent, and an ether-based solvent, and a hydrocarbon-based solvent.

[0538] Examples of the ester-based solvent include methyl acetate, butyl acetate, ethyl acetate, isopropyl acetate, pentyl acetate, isopentyl acetate, amyl acetate, propylene glycol monomethyl ether acetate, ethylene glycol monoethyl ether acetate, diethylene glycol monobutyl ether acetate, diethylene glycol monoethyl ether acetate, ethyl-3-ethoxypropyl

onate, 3-methoxybutyl acetate, 3-methyl-3-methoxybutyl acetate, methyl formate, ethyl formate, butyl formate, propyl formate, ethyl lactate, butyl lactate, propyl lactate, butyl butanoate, methyl 2-hydroxyisobutyrate, isoamyl acetate, isobutyl isobutyrate, and butyl propionate.

[0539] Examples of the nitrile-based solvent include acetonitrile, propionitrile, valeronitrile, and butyronitrile.

[0540] Known additives can be blended into the organic developing solution as necessary. Examples of the additive include a surfactant. The surfactant is not particularly limited, and for example, an ionic or non-ionic fluorine-based and/or silicon-based surfactant can be used.

[0541] The developing treatment can be performed according to a known developing method, and examples thereof include a method of immersing a support in a developing solution for a certain time (a dip method), a method of raising a developing solution on the surface of a support using the surface tension and maintaining the state for a certain time (a puddle method), a method of spraying a developing solution to the surface of a support (spray method), and a method of continuously ejecting a developing solution onto a support rotating at a certain rate while scanning a developing solution ejection nozzle at a certain rate (dynamic dispense method).

[0542] As the organic solvent contained in the rinse solution used for the rinse treatment after the developing treatment in the solvent developing process, a solvent that is unlikely to dissolve a resist pattern can be appropriately selected from the organic solvents described as the organic solvent used in the organic developing solution and then used. Typically, at least one solvent selected from a hydrocarbon-based solvent, a ketone-based solvent, an ester-based solvent, an alcohol-based solvent, an amide-based solvent, and an ether-based solvent is used.

[0543] These organic solvents may be used alone or in combination of two or more kinds thereof. Further, an organic solvent other than the above-described solvents and water may be mixed and used.

[0544] The rinse treatment carried out using a rinse solution (washing treatment) can be performed according to a known rinse method. Examples of the method of performing the rinse treatment include a method of continuously ejecting a rinse solution onto a support rotating at a certain rate (rotary coating method), a method of immersing a support in a rinse solution for a certain time (dip method), and a method of spraying a rinse solution to the surface of a support (spray method).

[0545] According to the resist pattern formation method of the present embodiment described above, it is presumed that high sensitivity can be achieved during formation of a resist pattern because the above-described resist composition is used.

[0546] Various materials that are used in the resist composition according to the above-described embodiment and the resist pattern formation method according to the above-described embodiment (for example, a resist solvent, a developing solution, a rinse liquid, a composition for forming an antireflection film, and a composition for forming a top coat) preferably do not contain impurities such as a metal, a metal salt containing halogen, an acid, an alkali, and a component containing a sulfur atom or phosphorus atom.

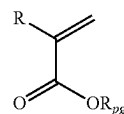
[0547] Here, examples of the impurities containing metal atoms include Na, K, Ca, Fe, Cu, Mn, Mg, Al, Cr, Ni, Zn, Ag, Sn, Pb, Li, and salts thereof. The content of the

impurities contained in these materials is preferably 200 ppb or less, more preferably 1 ppb or less, still more preferably 100 parts per trillion (ppt) or less, particularly preferably 10 ppt or less, and most preferably substantially zero (less than or equal to the detection limit of the measuring device).

(Compound)

[0548] The compound of the present embodiment is represented by General Formula (a0-1).

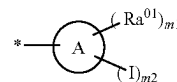
[Chemical Formula 73]



(a0-1)

[0549] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. Rpg represents an acid dissociable group represented by General Formula (a0-pg).]

[Chemical Formula 74]



(a0-pg)

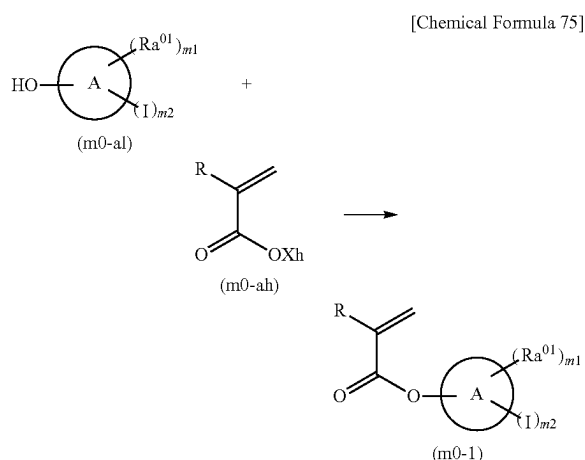
[0550] [In the formula, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton.  $\text{Ra}^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms.  $m_1$  represents an integer of 0 to 20.  $m_2$  represents an integer of 1 to 20. Here,  $m_1 + m_2 \leq$  (a total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied. In a case where  $m_1$  represents an integer of 2 or greater, a plurality of  $\text{Ra}^{01}$ 's may be the same as or different from each other.]

[0551] The compound according to the present embodiment is the same as the compound from which the constitutional unit (a0) of the component (A1) of the resist composition according to the first aspect is derived.

[0552] The compound of the present embodiment is preferably at least one selected from the group consisting of a compound represented by General Formula (a0-t1) and a compound represented by General Formula (a0-s1), more preferably at least one selected from the group consisting of a compound represented by General Formula (a0-t1-1), a compound represented by General Formula (a0-s1-1), a compound represented by General Formula (a0-s1-2), and a compound represented by General Formula (a0-s1-3), still more preferably at least one selected from the group consisting of a compound represented by General Formula (a0-t1-1), a compound represented by General Formula (a0-s1-1), and a compound represented by General Formula (a0-s1-2), and particularly preferably at least one selected from the group consisting of a compound represented by General Formula (a0-t1-1) and a compound represented by General Formula (a0-s1-1).

(Method of Producing Compound)

[0553] A method of producing the compound according to the present embodiment is not particularly limited. For example, a compound represented by General Formula (m0-1) (hereinafter, also referred to as “compound (m0-1)”) can be obtained by reacting a compound represented by General Formula (m0-al) (hereinafter, also referred to as “compound (m0-al)”) with a compound represented by General Formula (m0-ah) (hereinafter, also referred to as “compound (m0-ah)”).



[0554] [In the formula, R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms. A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton.  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms. m1 represents an integer of 0 to 20. m2 represents an integer of 1 to 20. Here,  $m1+m2$  (a total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied. In a case where m1 represents an integer of 2 or greater, a plurality of  $Ra^{01}$ 's may be the same as or different from each other. Xh represents Cl or Br.]

[0555] In Formulae (m0-ah) and (m0-1), R has the same definition as that for R in Formula (a0-1).

[0556] In Formulae (m0-al) and (m0-1), A,  $Ra^{01}$ , m1, and m2 each have the same definition as that for A,  $Ra^{01}$ , m1, and m2 in Formula (a0-pg).

[0557] In Formula (m0-ah), it is preferable that Xh represents Cl.

[0558] The compound (m0-1) contains an acid decomposable group whose polarity is increased by the action of an acid. Specifically, the cyclic alkene A is bonded to the carbonyloxy group in Formula (m0-1) as an acid dissociable group (Rpg in Formula (a0-1)) to form an acid decomposable group. Therefore, in a case where the carbon atom ( $C^d$ ) forming the ring skeleton of the cyclic alkene A, which is bonded to the carbonyloxy group in Formula (m0-1), is a secondary carbon atom, the carbon-carbon unsaturated bond of the ring skeleton of the cyclic alkene A is formed between the carbon atom ( $C^a$ ) at a position ( $\alpha$ -position of the carbon atom ( $C^d$ )) adjacent to the carbon atoms ( $C^d$ ) and the carbon atom ( $C^b$ ) at a position ( $\beta$ -position of the carbon atom ( $C^d$ )) adjacent to the carbon atoms ( $C^a$ ). In a case where the

carbon atom ( $C^d$ ) is a tertiary carbon atom, the position of the carbon-carbon unsaturated bond included in the carbocyclic skeleton of the cyclic alkene A is not particularly limited. From the viewpoint of improving acid dissociation properties, in a case where the carbon atom ( $C^d$ ) is a tertiary carbon atom, it is preferable that the carbon-carbon unsaturated bond included in the ring skeleton of the cyclic alkene A is formed between the carbon atom ( $C^a$ ) and the carbon atom ( $C^b$ ).

[0559] As the compound (m0-al) and the compound (m0-ah), commercially available products or compounds synthesized by a known production method may be used.

[0560] The reaction solvent for reacting the above-described compound (m0-al) and the above-described compound (m0-ah) may be a solvent which can dissolve the above-described compound (m0-al) and the above-described compound (m0-ah) and does not react with the compounds, and examples of such a solvent include dichloromethane, dichloroethane, chloroform, tetrahydrofuran, N,N-dimethylformamide, acetonitrile, propionitrile, N,N'-dimethylacetamide, and dimethyl sulfoxide.

[0561] The reaction of the compound (m0-al) and the compound (m0-ah) described above may be carried out in the presence of a base such as triethylamine, pyridine, and 4-dimethylaminopyridine.

[0562] In the reaction of the compound (m0-al) and the compound (m0-ah), the amount of the compound (m0-ah) to be added is preferably in a range of 1.00 to 1.70 equivalents and more preferably in a range of 1.20 to 1.50 equivalents with respect to the compound (m0-al).

[0563] The reaction time of the above-described reaction depends on the kinds of the compound (m-al) and the compound (m0-ah), but is preferably in a range of 1 to 10 hours and more preferably in a range of 3 to 5 hours.

[0564] The reaction temperature of the above-described reaction depends on the reaction temperature between the compound (m0-al) and the compound (m0-ah) and the kinds of the compound (m0-al) and the compound (m0-ah), but is preferably in a range of  $-10^\circ\text{C}$ . to  $50^\circ\text{C}$ . and more preferably in a range of  $20^\circ\text{C}$ . to  $30^\circ\text{C}$ .

[0565] After completion of the above-described reaction, the compound (m0-1) in the reaction solution may be isolated and purified. A known method in the related art can be used for isolation and purification, and for example, concentration, solvent extraction, distillation, crystallization, recrystallization, or chromatography can be appropriately combined and used.

[0566] The structure of the compound (m0-1) obtained as described above can be identified by typical organic analysis methods such as  $^1\text{H}$ -nuclear magnetic resonance (NMR) spectroscopy,  $^{13}\text{C}$ -NMR spectroscopy,  $^{19}\text{F}$ -NMR spectroscopy, infrared (IR) absorption spectroscopy, mass spectrometry (MS), an elemental analysis method, and an X-ray crystal diffraction method.

[0567] The compound according to the present embodiment is useful for producing a polymer compound having the constitutional unit (a0) (resin component (A1) according to the first aspect).

<Polymer Compound>

[0568] The polymer compound according to the present embodiment has a constitutional unit (a0) derived from a compound represented by the compound (a0-1). That is, the

polymer compound of the present embodiment is the same as the resin component (A1) according to the first aspect.

#### EXAMPLES

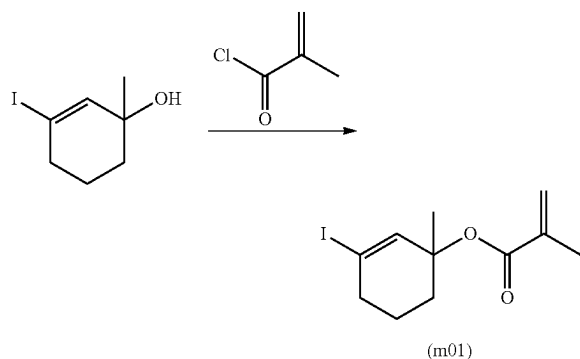
[0569] Hereinafter, the present invention will be described in more detail based on examples, but the present invention is not limited to these examples.

#### <Synthesis of Compounds>

##### Synthesis Example 1: Synthesis of Monomer (m01)

[0570] 20.0 g of 1-methyl-3-iodo-2-cyclohexene-1-ol, 12.8 g of triethylamine, 1.0 g of 4-dimethylaminopyridine, and 200.0 g of dichloromethane were placed in a three-neck flask and stirred, 11.4 g of methacrylic acid chloride was added thereto, and the mixture was stirred for 2 hours. 200.0 g of water was added thereto to complete the reaction, and the organic layer was concentrated using a rotary evaporator, thereby obtaining 16.2 g of a monomer (m01).

[Chemical Formula 76]



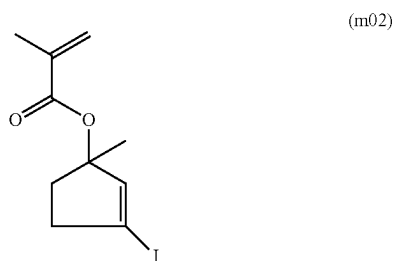
[0571] The obtained compound was subjected to NMR measurement, and the structure thereof was identified based on the following results.

[0572]  $^1\text{H-NMR}$  (dms $\text{o-d}_6$ , 400 MHz):  $\delta$  (ppm)=1.24 (3H, s), 1.65 (2H, m), 1.83 (2H, m), 1.96 (2H, m), 2.01 (3H, s), 6.40 (2H, m), 6.48 (1H, s)

##### Synthesis Example 2: Synthesis of Monomer (m02)

[0573] A monomer (m02) was obtained in the same manner as in Synthesis Example 1 except that 1-methyl-3-iodo-2-cyclopentene-1-ol was used instead of 1-methyl-3-iodo-2-cyclohexene-1-ol.

[Chemical Formula 77]



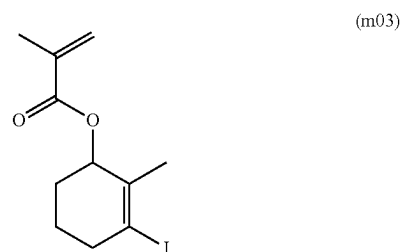
[0574] The obtained compound was subjected to NMR measurement, and the structure thereof was identified based on the following results.

[0575]  $^1\text{H-NMR}$  (dms $\text{o-d}_6$ , 400 MHz):  $\delta$  (ppm)=1.24 (3H, s), 2.09 (2H, m), 2.31 (2H, m), 2.01 (3H, s), 6.40 (2H, m), 6.48 (1H, m)

##### Synthesis Example 3: Synthesis of Monomer (m03)

[0576] A monomer (m03) was obtained in the same manner as in Synthesis Example 1 except that 2-methyl-3-iodo-2-cyclohexene-1-ol was used instead of 1-methyl-3-iodo-2-cyclohexene-1-ol.

[Chemical Formula 78]



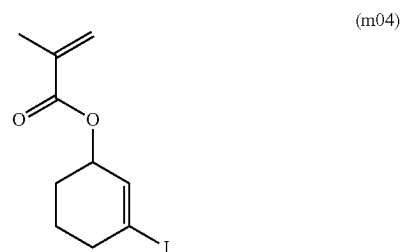
[0577] The obtained compound was subjected to NMR measurement, and the structure thereof was identified based on the following results.

[0578]  $^1\text{H-NMR}$  (dms $\text{o-d}_6$ , 400 MHz):  $\delta$  (ppm)=1.65 (2H, m), 1.79 (3H, s), 1.89 (2H, m), 1.96 (2H, m), 2.01 (3H, s), 5.13 (1H, t), 6.40 (1H, m), 6.48 (1H, m)

##### Synthesis Example 4: Synthesis of Monomer (m04)

[0579] A monomer (m04) was obtained in the same manner as in Synthesis Example 1 except that 3-iodo-2-cyclohexene-1-ol was used instead of 1-methyl-3-iodo-2-cyclohexene-1-ol.

[Chemical Formula 79]



[0580] The obtained compound was subjected to NMR measurement, and the structure thereof was identified based on the following results.

[0581]  $^1\text{H-NMR}$  (dms $\text{o-d}_6$ , 400 MHz):  $\delta$  (ppm)=1.65 (2H, m), 1.89 (2H, m), 1.96 (2H, m), 2.01 (3H, s), 5.13 (1H, dd), 6.40 (2H, m), 6.48 (1H, m)

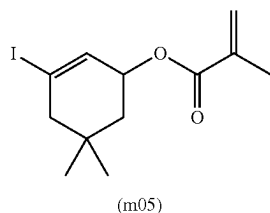
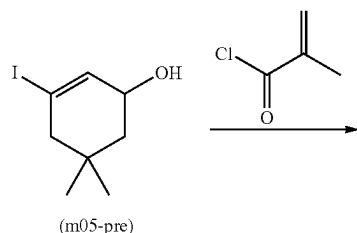
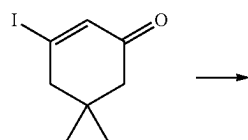
##### Synthesis Example 5: Synthesis of Monomer (m05)

[0582] 20.0 g of 3-iodo-5,5-dimethyl-2-cyclohexene-1-one, 35.8 g of a cerium chloride heptahydrate, and 200 g of

methanol were placed in a three-neck flask and stirred, and 3.9 g of sodium borohydride was added thereto in an ice bath. The mixture was stirred for 2 hours, 200 g of a saturated ammonium chloride aqueous solution and 200 g of dichloromethane were added thereto, and only the organic layer was collected. The organic layer was concentrated using a rotary evaporator, thereby obtaining 15.2 g of the compound (m05-pre) (3-iodo-5,5-dimethyl-2-cyclohexene-1-ol).

**[0583]** 15.0 g of the compound (m05-pre), 8.6 g of triethylamine, 0.7 g of 4-dimethylaminopyridine, and 150.0 g of dichloromethane were placed in a three-neck flask and stirred, 7.7 g of methacrylic acid chloride was added thereto, and the mixture was stirred for 2 hours. 200.0 g of water was added thereto to complete the reaction, and the organic layer was concentrated using a rotary evaporator, thereby obtaining 15.0 g of a compound (m05).

[Chemical Formula 80]



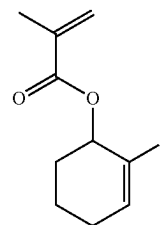
**[0584]** The obtained compound was subjected to NMR measurement, and the structure thereof was identified based on the following results.

**[0585]**  $^1\text{H-NMR}$  (dms $\text{-d}_6$ , 400 MHz):  $\delta$  (ppm)=0.94 (6H, s), 1.78 (2H, m), 1.85 (2H, m), 2.01 (3H, s), 5.13 (1H, dd), 6.40 (2H, m), 6.48 (1H, m)

#### Synthesis Example 6: Synthesis of Monomer (m06)

**[0586]** A monomer (m06) was obtained in the same manner as in Synthesis Example 1 except that 2-iodo-2-cyclohexene-1-ol was used instead of 1-methyl-3-iodo-2-cyclohexene-1-ol.

[Chemical Formula 81]



(m06)

**[0587]** The obtained compound was subjected to NMR measurement, and the structure thereof was identified based on the following results.

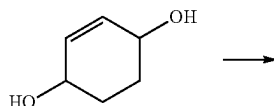
**[0588]**  $^1\text{H-NMR}$  (dms $\text{-d}_6$ , 400 MHz):  $\delta$  (ppm)=1.65 (2H, m), 1.89 (2H, m), 2.01 (3H, s), 2.06 (2H, m), 5.13 (1H, t), 6.40 (2H, m), 6.48 (1H, m)

#### Synthesis Example 7: Synthesis of Monomer (m07)

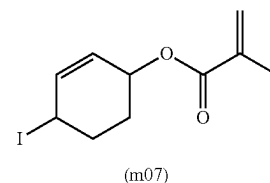
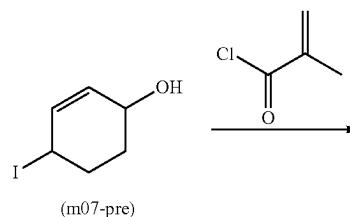
**[0589]** 50.0 g of 4-hydroxy-2-cyclohexene-1-one, 35.8 g of imidazole, 137.9 g of triphenylphosphine, and 500.0 g of dichloromethane were placed in a three-neck flask and stirred, and 133.4 g of iodine was added thereto in an ice bath. After the mixture was stirred for 8 hours, 500.0 g of water was added thereto, and only the organic layer was collected. The organic layer was dewatered with magnesium sulfate, concentrated using a rotary evaporator, and purified by silica gel column chromatography, thereby obtaining 23.6 g of a compound (m07-pre) (4-iodo-2-cyclohexene-1-ol).

**[0590]** 20.0 g of the compound (m07-pre), 12.6 g of triethylamine, 1.1 g of 4-dimethylaminopyridine, and 200.0 g of dichloromethane were placed in a three-neck flask and stirred, 12.1 g of methacrylic acid chloride was added thereto, and the mixture was stirred for 2 hours. 200.0 g of water was added thereto to complete the reaction, and the organic layer was concentrated using a rotary evaporator, thereby obtaining 19.6 g of a monomer (m07).

[Chemical Formula 82]



(m07-pre)



[0591] The obtained compound was subjected to NMR measurement, and the structure thereof was identified based on the following results.

[0592]  $^1\text{H-NMR}$  (dms $o$ -d $_6$ , 400 MHz):  $\delta$  (ppm)=1.89 (2H, m), 2.01 (3H, s), 2.19 (2H, m), 4.52 (1H, m), 5.13 (1H, dd), 5.59 (2H, m), 6.40 (1H, m), 6.48 (1H, m)

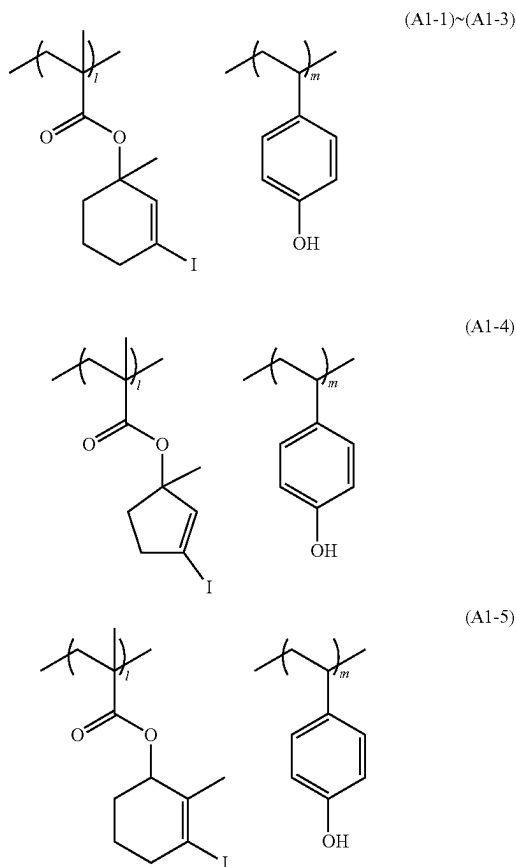
#### <Production of Polymer Compound>

[0593] Each of polymer compounds (A1-1) to (A1-9) used in present examples was obtained by carrying out radical polymerization using monomers from which constitutional units constituting the polymer compounds were derived at a predetermined molar ratio and performing a deprotection reaction.

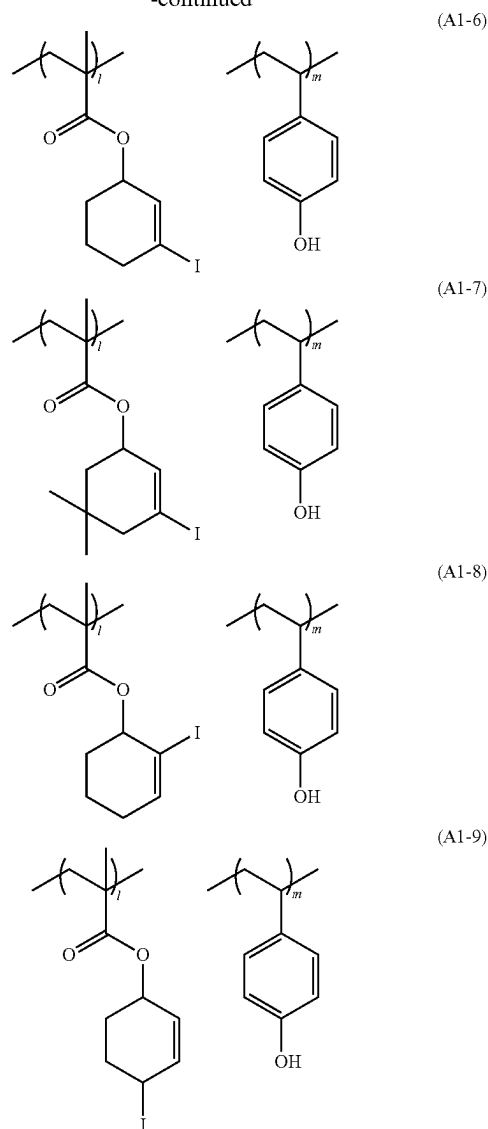
[0594] The weight-average molecular weight ( $M_w$ ) and the polydispersity ( $M_w/M_n$ ) of each of the obtained polymer compounds were determined by GPC measurement (in terms of the standard polystyrene).

[0595] In addition, the copolymerization composition ratio (the proportion (molar ratio) of each constitutional unit in the structural formula) of each of the obtained polymer compounds was determined from the carbon 13 nuclear magnetic resonance spectrum (600 MHz  $^{13}\text{C-NMR}$ ).

[Chemical Formula 83]



-continued



[0596] Polymer compound (A1-1): weight-average molecular weight ( $M_w$ ) of 7300, polydispersity ( $M_w/M_n$ ) of 1.73,  $l/m=70/30$

[0597] Polymer compound (A1-2): weight-average molecular weight ( $M_w$ ) of 7200, polydispersity ( $M_w/M_n$ ) of 1.71,  $l/m=60/40$

[0598] Polymer compound (A1-3): weight-average molecular weight ( $M_w$ ) of 6900, polydispersity ( $M_w/M_n$ ) of 1.68,  $l/m=50/50$

[0599] Polymer compound (A1-4): weight-average molecular weight ( $M_w$ ) of 7200, polydispersity ( $M_w/M_n$ ) of 1.69,  $l/m=60/40$

[0600] Polymer compound (A1-5): weight-average molecular weight ( $M_w$ ) of 7000, polydispersity ( $M_w/M_n$ ) of 1.71,  $l/m=60/40$

[0601] Polymer compound (A1-6): weight-average molecular weight ( $M_w$ ) of 6900, polydispersity ( $M_w/M_n$ ) of 1.70,  $l/m=60/40$

[0602] Polymer compound (A1-7): weight-average molecular weight (Mw) of 7500, polydispersity (Mw/Mn) of 1.67, l/m=60/40

[0603] Polymer compound (A1-8): weight-average molecular weight (Mw) of 6800, polydispersity (Mw/Mn) of 1.70, l/m=60/40

[0604] Polymer compound (A1-9): weight-average molecular weight (Mw) of 7300, polydispersity (Mw/Mn) of 1.72, l/m=60/40

#### <Preparation of Resist Composition>

Examples 1 to 13 and Comparative Examples 1 to 6

[0605] Each of the components shown in Table 1 was mixed and dissolved to prepare a resist composition of each Example.

TABLE 1

|                          | Component<br>(A) | Component<br>(B) | Component<br>(C) | Component<br>(S) |
|--------------------------|------------------|------------------|------------------|------------------|
| Example 1                | (A)-1<br>[100]   | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 2                | (A)-2<br>[100]   | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 3                | (A)-3<br>[100]   | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>8000]   |
| Example 4                | (A)-4<br>[100]   | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 5                | (A)-5<br>[100]   | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 6                | (A)-6<br>[100]   | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 7                | (A)-7<br>[100]   | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 8                | (A)-8<br>[100]   | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 9                | (A)-9<br>[100]   | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 10               | (A)-2<br>[100]   | (B)-2<br>[17.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 11               | (A)-2<br>[100]   | (B)-3<br>[19.0]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 12               | (A)-2<br>[100]   | (B)-4<br>[21.6]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Example 13               | (A)-2<br>[100]   | (B)-5<br>[20.3]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Comparative<br>Example 1 | (A)-10<br>[100]  | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Comparative<br>Example 2 | (A)-11<br>[100]  | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Comparative<br>Example 3 | (A)-12<br>[100]  | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Comparative<br>Example 4 | (A)-13<br>[100]  | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Comparative<br>Example 5 | (A)-14<br>[100]  | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |
| Comparative<br>Example 6 | (A)-15<br>[100]  | (B)-1<br>[16.2]  | (D)-1<br>[5]     | (S)-1<br>[8000]  |

[0606] In Table 1, each abbreviation has the following meaning. The numerical values in the brackets are blending amounts (parts by mass).

[0607] (A)-1 to (A)-9: polymer compounds (A1-D to (A1-9) shown above

[0608] (A)-10: polymer compound represented by Chemical Formula (A1-10) The weight-average molecular weight (Mw) of the polymer compound (A1-10) in terms of standard polystyrene determined by GPC measurement was 6900, and the polydispersity

(Mw/Mn) thereof was 1.69. The copolymerization composition ratio (the ratio (molar ratio) of each constitutional unit in the structural formula) determined by <sup>13</sup>C-NMR was l/m=60/40.

[0609] (A)-11: polymer compound represented by Chemical Formula (A1-11) The weight-average molecular weight (Mw) of the polymer compound (A1-11) in terms of standard polystyrene determined by GPC measurement was 7200, and the polydispersity (Mw/Mn) thereof was 1.67. The copolymerization composition ratio (the ratio (molar ratio) of each constitutional unit in the structural formula) determined by <sup>13</sup>C-NMR was l/m=60/40.

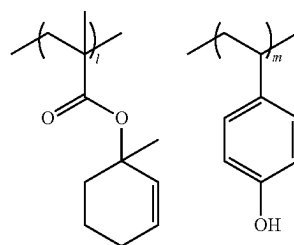
[0610] (A)-12: polymer compound represented by Chemical Formula (A1-12) The weight-average molecular weight (Mw) of the polymer compound (A1-12) in terms of standard polystyrene determined by GPC measurement was 6800, and the polydispersity (Mw/Mn) thereof was 1.73. The copolymerization composition ratio (the ratio (molar ratio) of each constitutional unit in the structural formula) determined by <sup>13</sup>C-NMR was l/m=60/40.

[0611] (A)-13: polymer compound represented by Chemical Formula (A1-13) The weight-average molecular weight (Mw) of the polymer compound (A1-13) in terms of standard polystyrene determined by GPC measurement was 6900, and the polydispersity (Mw/Mn) thereof was 1.74. The copolymerization composition ratio (the ratio (molar ratio) of each constitutional unit in the structural formula) determined by <sup>13</sup>C-NMR was l/m=60/40.

[0612] (A)-14: polymer compound represented by Chemical Formula (A1-14) The weight-average molecular weight (Mw) of the polymer compound (A1-14) in terms of standard polystyrene determined by GPC measurement was 6700, and the polydispersity (Mw/Mn) thereof was 1.74. The copolymerization composition ratio (the ratio (molar ratio) of each constitutional unit in the structural formula) determined by <sup>13</sup>C-NMR was l/m=60/40.

[0613] (A)-15: polymer compound represented by Chemical Formula (A1-15) The weight-average molecular weight (Mw) of the polymer compound (A1-15) in terms of standard polystyrene determined by GPC measurement was 6700, and the polydispersity (Mw/Mn) thereof was 1.77. The copolymerization composition ratio (the ratio (molar ratio) of each constitutional unit in the structural formula) determined by <sup>13</sup>C-NMR was l/m=60/40.

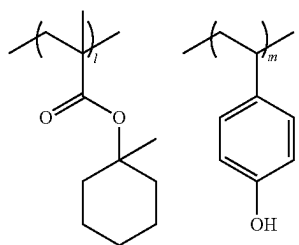
[Chemical Formula 84]



(A1-10)



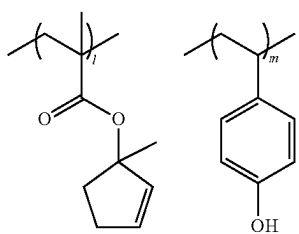
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(A1-11)

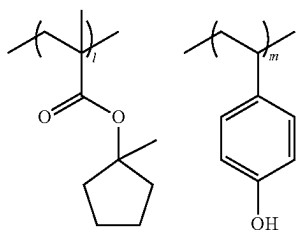
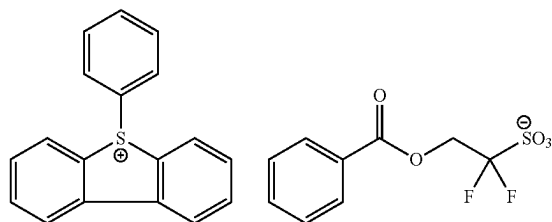
**[0614]** (B)-1: acid generator formed of compound (B1-1) shown below**[0615]** (B)-2: acid generator formed of compound (B1-2) shown below**[0616]** (B)-3: acid generator formed of compound (B1-3) shown below**[0617]** (B)-4: acid generator formed of compound (B1-4) shown below**[0618]** (B)-5: acid generator formed of compound (B1-5) shown below

[Chemical Formula 85]



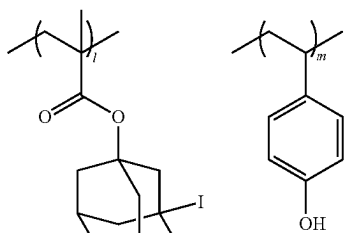
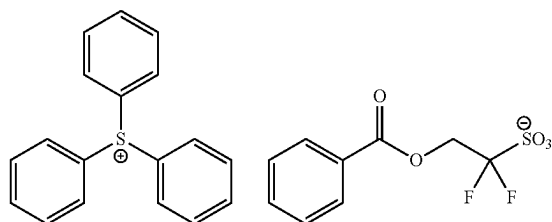
(A1-12)

(B1-1)



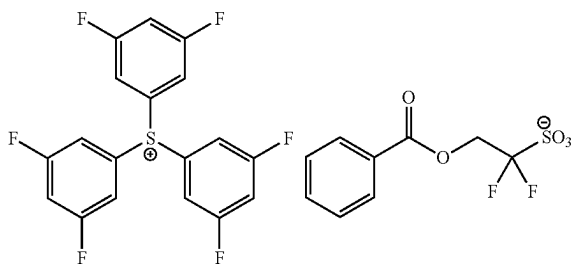
(A1-13)

(B1-2)



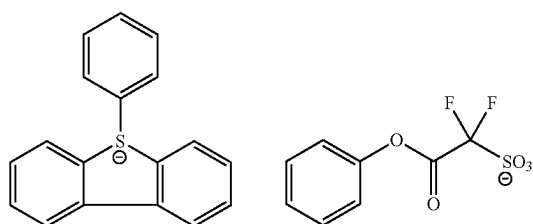
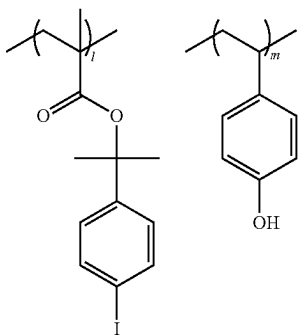
(A1-14)

(B1-3)



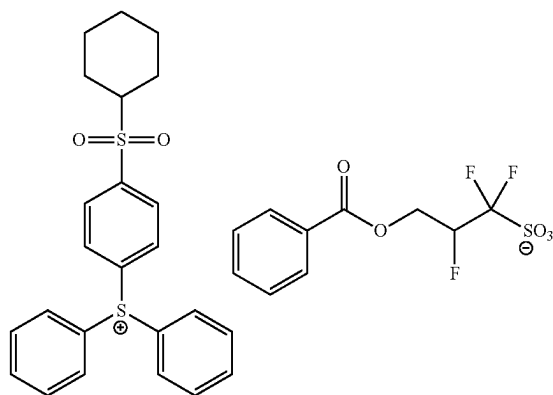
(A1-15)

(B1-4)



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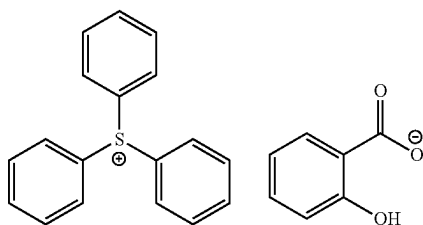
(B1-5)



[0619] (D)-1: acid diffusion control agent formed of compound (D1-D) shown below

[Chemical Formula 86]

(D1-1)



[0620] (S)-1: mixed solvent of propylene glycol monomethyl ether acetate/propylene glycol monomethyl ether=60/40 (mass ratio)

&lt;Resist Pattern Formation&gt;

[0621] The resist composition of each Example was applied onto an 8-inch silicon substrate which had been subjected to a hexamethyldisilazane (HMDS) treatment using a spinner, the coated wafer was subjected to a pre-bake (PAB) treatment on a hot plate at a temperature of 110° C. for 60 seconds so that the coated wafer was dried to form a resist film having a film thickness of 50 nm. Next, drawing (exposure) was carried out on the resist film by using an electron beam lithography device JEOL-JBX-9300FS (manufactured by JEOL Ltd.), with the target size set to a 1:1 line and space pattern (hereinafter, referred to as “LS pattern”) of a line width of 50 nm, at an acceleration voltage of 100 kV. Thereafter, a post exposure bake (PEB) treatment was performed thereon at 100° C. for 60 seconds. Subsequently, alkali development was performed at 23° C. for 60 seconds using a 2.38 mass % tetramethylammonium hydroxide (TMAH) aqueous solution “NMD-3” (trade name, manufactured by TOKYO OHKA KOGYO CO., LTD.). Thereafter, water rinsing was performed for 15 seconds using pure water. As a result, a 1:1 LS pattern having a line width of 50 nm was formed.

[Evaluation of Optimum Exposure Amount (Eop)]

[0622] The optimum exposure amount Eop ( $\mu\text{C}/\text{cm}^2$ ) at which an LS pattern having a target size was formed by the

above-described method for forming a resist pattern was determined. The results are listed in the columns of “Eop ( $\mu\text{C}/\text{cm}^2$ )” in Table 2.

[Evaluation of Reduction of Film Thickness (DL)]

[0623] The film thickness of the resist film immediately after the PAB treatment and the film thickness of the unexposed portion of the resist film after the water rinsing were measured in <Resist pattern formation> described above, and the amount of a decrease in the film thickness was determined. The results are listed in the columns of “DL (nm)” in Table 2.

TABLE 2

|                       | Eop ( $\mu\text{C}/\text{cm}^2$ ) | DL (nm) |
|-----------------------|-----------------------------------|---------|
| Example 1             | 79                                | 4.8     |
| Example 2             | 82                                | 3.2     |
| Example 3             | 91                                | 2.9     |
| Example 4             | 82                                | 4.2     |
| Example 5             | 83                                | 3.5     |
| Example 6             | 87                                | 3.5     |
| Example 7             | 92                                | 3.2     |
| Example 8             | 89                                | 3.5     |
| Example 9             | 95                                | 3.7     |
| Example 10            | 81                                | 3.2     |
| Example 11            | 78                                | 3.2     |
| Example 12            | 88                                | 3.4     |
| Example 13            | 81                                | 3.2     |
| Comparative Example 1 | 107                               | 8.9     |
| Comparative Example 2 | 101                               | 9.7     |
| Comparative Example 3 | 109                               | 8.3     |
| Comparative Example 4 | 111                               | 9.1     |
| Comparative Example 5 | 109                               | 4.5     |
| Comparative Example 6 | 101                               | 4.7     |

[0624] As shown in the results listed in Table 2, it was confirmed that the resist compositions of Examples 1 to 13 achieved high sensitivity as compared with the resist compositions of Comparative Examples 1 to 6. Further, it was confirmed that in the resist compositions of Examples 1 to 13, reduction of the film thickness was improved as compared with the resist compositions of Comparative Examples 1 to 4.

[0625] While preferred embodiments of the present invention have been described and illustrated above, it should be understood that these are exemplary of the present invention and are not to be considered as limiting. Additions, omissions, substitutions, and other modifications can be made without departing from the scope of the invention. Accordingly, the present invention is not to be considered as being limited by the foregoing description and is only limited by the scope of the appended claims.

What is claimed is:

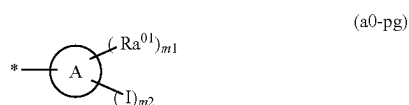
1. A resist composition which generates an acid upon light exposure and whose solubility in a developing solution is changed by an action of the acid, the resist composition comprising:

a resin component (A1) whose solubility in a developing solution is changed by the action of the acid,

wherein the resin component (A1) has a constitutional unit (a0) derived from a compound represented by General Formula (a0-1),

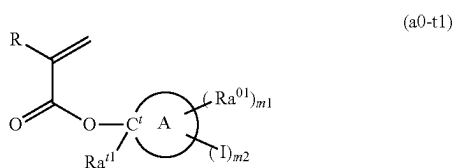


wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, and Rpg represents an acid dissociable group represented by General Formula (a0-pg),



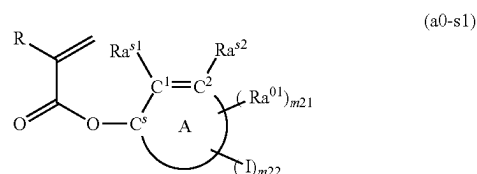
wherein A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m1 represents an integer of 0 to 20, m2 represents an integer of 1 to 20, where  $m1+m2 \leq$  (a total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied, a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m1 represents an integer of 2 or greater, and \* represents a bonding site.

2. The resist composition according to claim 1, wherein the constitutional unit (a0) is a constitutional unit derived from a compound represented by General Formula (a0-t1),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C'$  represents a tertiary carbon atom,  $Ra^{t1}$  represents a linear or branched alkyl group having 1 to 12 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m1 represents an integer of 0 to 20, m2 represents an integer of 1 to 20, where  $m1+m2 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m1 represents an integer of 2 or greater.

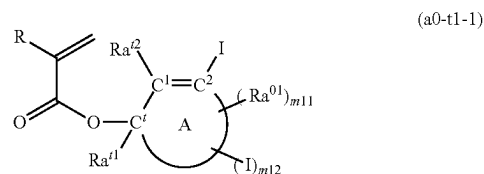
3. The resist composition according to claim 1, wherein the constitutional unit (a0) is a constitutional unit derived from a compound represented by General Formula (a0-s1),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C^s$  represents a secondary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{s1}$  and  $Ra^{s2}$  each independently represent a hydrogen atom, an iodine atom, or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m21 represents an integer of 0 to 20, m22 represents an integer of 0 to 20, where  $m21+m22 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, at least one of  $Ra^{s1}$  or  $Ra^{s2}$  represents an iodine atom in a case where m22 represents 0, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m21 represents an integer of 2 or greater.

4. The resist composition according to claim 1, wherein, in General Formula (a0-1), an iodine atom is bonded to at least one of carbon atoms constituting the carbon-carbon unsaturated bond in the ring skeleton of the cyclic alkene as A.

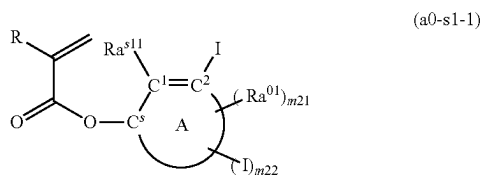
5. The resist composition according to claim 2, wherein the constitutional unit (a0) is a constitutional unit derived from a compound represented by General Formula (a0-t1-1),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C'$  represents a tertiary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{t1}$  represents a linear or branched alkyl group having 1 to 12 carbon atoms,  $Ra^{t2}$  represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m11 represents an integer of 0 to 20, m12 represents an integer of 0 to 19, where  $m11+m12 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, and a

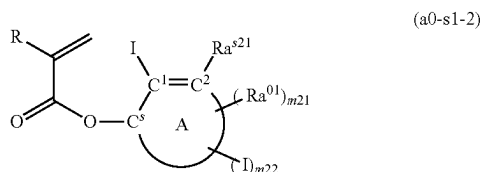
plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m11 represents an integer of 2 or greater.

6. The resist composition according to claim 3, wherein the constitutional unit (a0) is a constitutional unit derived from a compound represented by General Formula (a0-s1-1),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C^s$  represents a secondary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{s11}$  represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m21 represents an integer of 0 to 20, m22 represents an integer of 0 to 19, where  $m21+m22 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m21 represents an integer of 2 or greater.

7. The resist composition according to claim 3, wherein the constitutional unit (a0) is a constitutional unit derived from a compound represented by General Formula (a0-s1-2),

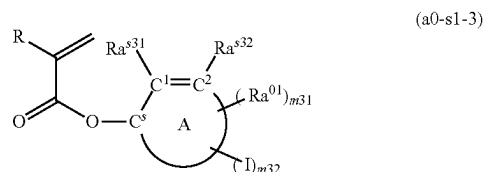


wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C^s$  represents a secondary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{s21}$  represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m21 represents an integer of 0 to 20, m22 represents an integer of 0 to 19, where  $m21+m22 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, and a

plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m21 represents an integer of 2 or greater.

8. The resist composition according to claim 3,

wherein the constitutional unit (a0) is a constitutional unit derived from a compound represented by General Formula (a0-s1-3),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C^s$  represents a secondary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{s31}$  and  $Ra^{s32}$  each independently represent a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m31 represents an integer of 0 to 20, m32 represents an integer of 1 to 20, where  $m31+m32 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m31 represents an integer of 2 or greater.

9. A resist pattern formation method comprising:

forming a resist film on a support using the resist composition according to claim 1;

exposing the resist film to light; and

developing the resist film exposed to light to form a resist pattern.

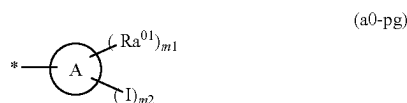
10. The resist pattern formation method according to claim 9,

wherein exposing the resist film comprises exposing the resist film to extreme ultraviolet rays or electron beams.

11. A compound which is represented by General Formula (a0-1),

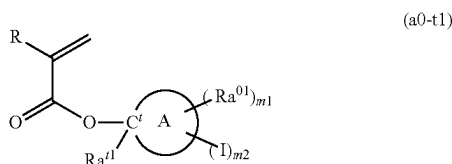


wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, and Rpg represents an acid dissociable group represented by General Formula (a0-pg),



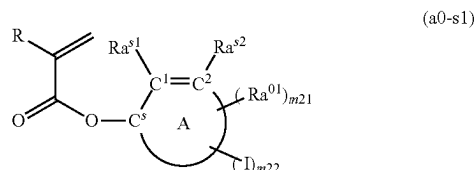
wherein A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m1 represents an integer of 0 to 20, m2 represents an integer of 1 to 20, where  $m1+m2 \leq$  (a total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied, a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m1 represents an integer of 2 or greater, and \* represents a bonding site.

12. The compound according to claim 11, which is represented by General formula (a0-t1),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C^t$  represents a tertiary carbon atom,  $Ra^{t1}$  represents a linear or branched alkyl group having 1 to 12 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m1 represents an integer of 0 to 20, m2 represents an integer of 1 to 20, where  $m1+m2 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-1) is satisfied, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m1 represents an integer of 2 or greater.

13. The compound according to claim 11, which is represented by General formula (a0-s1),



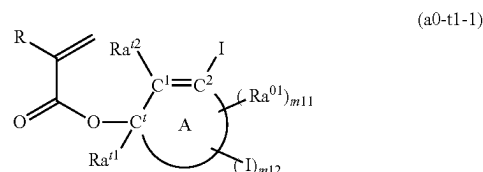
wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C^s$  represents a secondary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{s1}$  and  $Ra^{s2}$  each

independently represent a hydrogen atom, an iodine atom, or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m21 represents an integer of 0 to 20, m22 represents an integer of 0 to 20, where  $m21+m22 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, at least one of  $Ra^{s1}$  or  $Ra^{s2}$  represents an iodine atom in a case where m22 represents 0, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m21 represents an integer of 2 or greater.

14. The compound according to claim 11,

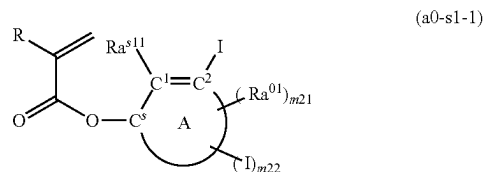
wherein, in General Formula (a0-1), an iodine atom is bonded to at least one of carbon atoms constituting the carbon-carbon unsaturated bond in the ring skeleton of the cyclic alkene as A.

15. The compound according to claim 12, which is represented by General Formula (a0-t1-1),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C^t$  represents a tertiary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{t1}$  represents a linear or branched alkyl group having 1 to 12 carbon atoms,  $Ra^{t2}$  represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms, m11 represents an integer of 0 to 20, m12 represents an integer of 0 to 19, where  $m11+m12 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where m11 represents an integer of 2 or greater.

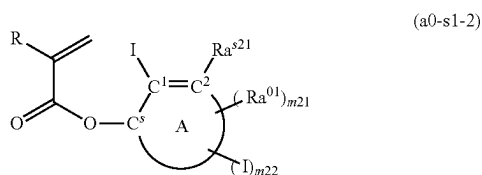
16. The compound according to claim 13, which is represented by General Formula (a0-s1-1),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated

bond in a ring skeleton,  $C^s$  represents a secondary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{s11}$  represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $m21$  represents an integer of 0 to 20,  $m22$  represents an integer of 0 to 19, where  $m21+m22 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where  $m21$  represents an integer of 2 or greater.

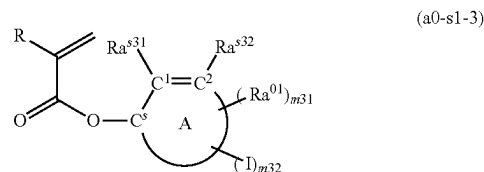
17. The compound according to claim 13, which is represented by General Formula (a0-s1-2),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C^s$  represents a secondary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{s21}$  represents a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $m21$  represents an integer of 0 to 20,  $m22$

represents an integer of 0 to 19, where  $m21+m22 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where  $m21$  represents an integer of 2 or greater.

18. The compound according to claim 13, which is represented by General Formula (a0-s1-3),



wherein R represents a hydrogen atom, an alkyl group having 1 to 5 carbon atoms, or a halogenated alkyl group having 1 to 5 carbon atoms, A represents a cyclic alkene having only one carbon-carbon unsaturated bond in a ring skeleton,  $C^s$  represents a secondary carbon atom,  $C^1$  and  $C^2$  each independently represent a carbon atom constituting the carbon-carbon unsaturated bond of the cyclic alkene as A,  $Ra^{s31}$  and  $Ra^{s32}$  each independently represent a hydrogen atom or a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $Ra^{01}$  represents a monovalent hydrocarbon group having 1 to 20 carbon atoms,  $m31$  represents an integer of 0 to 20,  $m32$  represents an integer of 1 to 20, where  $m31+m32 \leq$  (the total number of carbon atoms constituting the ring skeleton of the cyclic alkene as A-3) is satisfied, and a plurality of  $Ra^{01}$ 's may be the same as or different from each other in a case where  $m31$  represents an integer of 2 or greater.

\* \* \* \* \*